Elementary Analysis

Sections 21–38

All 214 exercises, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Sections 21–38, 214 exercises. Short descriptions identify the problems; consult the linked second-edition text for the original statements.

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21 More on metric spaces: continuity

A metric d measures distance. The open ball $B_r(s)$ consists of the points whose distance from s is less than r. A continuous function sends points sufficiently close to a *fixed* point to points close to its image. Uniform continuity uses one input tolerance for all pairs of points. An open set in a subspace is understood relative to that subspace; it need not be open in the surrounding Euclidean space.

21.1 Exercise 21.1: Uniform continuity of a vector-valued function

What we must show. Show that $\gamma(t) = (f_1(t), \ldots, f_k(t))$ is uniformly continuous when all its coordinate functions are uniformly continuous.

Plan. Make each coordinate error small enough that the sum of their squares is less than the desired squared distance.

Choose the input tolerance before choosing the points.

Fix $\varepsilon > 0$. For each j, uniform continuity of f_j gives $\delta_j > 0$ such that, for *all* $s, t \in [a, b]$,

$$|s - t| < \delta_j \implies |f_j(s) - f_j(t)| < \varepsilon/\sqrt{k}.$$

The minimum $\delta = \min(\delta_1, \ldots, \delta_k)$ is positive because there are only finitely many coordinates. If $|s - t| < \delta$, every coordinate estimate holds. Consequently

$$d(\gamma(s), \gamma(t)) = \left(\sum_{j=1}^{k} |f_j(s) - f_j(t)|^2 \right)^{1/2} < \left(k \frac{\varepsilon^2}{k} \right)^{1/2} = \varepsilon.$$

Conclusion. The same δ works for every pair s, t, which is precisely uniform continuity.

21.2 Exercise 21.2: Continuity at one point in terms of open sets

What we must show. Prove both directions of the open-neighborhood characterization at $s_0 \in S$.

Continuity implies the neighborhood condition.

Suppose f is continuous at s_0 and $U \subseteq S^*$ is open with $f(s_0) \in U$. Openness provides $\varepsilon > 0$ for which $B^*_\varepsilon(f(s_0)) \subseteq U$. Continuity supplies $\delta > 0$ with $d(s, s_0) < \delta$ implying $d^*(f(s), f(s_0)) < \varepsilon$. The open set $V = B_\delta(s_0)$ contains s_0 and satisfies $f(V) \subseteq U$.

The neighborhood condition implies continuity.

Conversely, fix $\varepsilon > 0$ and apply the assumed condition to the open ball $U = B^*_\varepsilon(f(s_0))$. We obtain an open V containing s_0 with $f(V) \subseteq U$. Since V is open, it contains a ball $B_\delta(s_0)$ for some $\delta > 0$. Hence

$$d(s, s_0) < \delta \implies s \in V \implies f(s) \in U \implies d^*(f(s), f(s_0)) < \varepsilon.$$

Conclusion. This is exactly the metric definition of continuity at s_0.

21.3 Exercise 21.3: Distance from a fixed point is uniformly continuous

What we must show. For fixed s_0, prove uniform continuity of $f(s) = d(s, s_0)$.

Derive the reverse triangle inequality.

The triangle inequality gives $d(s, s_0) \leq d(s, t) + d(t, s_0)$, so $d(s, s_0) - d(t, s_0) \leq d(s, t)$. Interchanging s, t bounds the negative of the same difference. Together these statements give

$$|f(s) - f(t)| = |d(s, s_0) - d(t, s_0)| \leq d(s, t).$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Whenever $d(s, t) < \delta$, the displayed inequality implies $|f(s) - f(t)| < \varepsilon$.

Conclusion. Distance from a fixed point is even Lipschitz with constant 1; this means its output distance never exceeds its input distance.

21.4 Exercise 21.4: Rational intervals suffice to test continuity

What we must show. Establish the equivalence of continuity, openness of preimages of all open intervals, and openness for intervals with rational endpoints.

The first two implications.

If f is continuous, Theorem 21.3 makes the inverse image of every open subset of $\mathbb{R}$ open in S. In particular, this holds for every interval (a, b). Restricting a, b to rationals gives the third condition.

Recover continuity using only rational endpoints.

Assume $f^{-1}((a, b))$ is open whenever $a < b$ are rational. Fix $s_0 \in S$ and $\varepsilon > 0$. Density of $\mathbb{Q}$ gives rationals with

$$f(s_0) - \varepsilon < a < f(s_0) < b < f(s_0) + \varepsilon.$$

The set $V = f^{-1}((a, b))$ is open and contains s_0, so some $B_\delta(s_0)$ is contained in V. If $d(s, s_0) < \delta$, then $a < f(s) < b$, and therefore $|f(s) - f(s_0)| < \varepsilon$.

Conclusion. Continuity at every s_0 follows. A dense set of possible interval endpoints gives arbitrarily small neighborhoods of every output.

21.5 Exercise 21.5: What can fail on a noncompact Euclidean set

What we must show. Construct (a) an unbounded continuous function and (b) a bounded continuous function with no maximum on a noncompact $E \subseteq \mathbb{R}^k$.

Plan. By Heine–Borel, a compact Euclidean set is exactly a closed bounded set. Failure of compactness gives either points escaping to infinity or a missing limit point.

(a) Construct an unbounded function.

If E is unbounded, set $h(x) = \|x\|$, the Euclidean distance from zero. Exercise 21.3 proves continuity. For each $M > 0$, unboundedness of E supplies $x \in E$ with $h(x) > M$.

If E is bounded, it cannot be closed. Choose $x_0 \in \overline{E} \setminus E$ and set $h(x) = 1/\|x - x_0\|$ on E. The denominator is positive on E, and the distance function and the reciprocal function on $(0, \infty)$ are

continuous, so h is continuous. Since $x_0 \in \overline{E}$, for every $M > 0$ some $x \in E$ satisfies $0 < \|x - x_0\| < 1/M$; thus $h(x) > M$.

(b) Compress the unbounded values into a bounded interval.

In either case $h \geq 0$ and h is unbounded above. Define

$$g(x) = \frac{h(x)}{1 + h(x)} = 1 - \frac{1}{1 + h(x)}.$$

This is continuous and $0 \leq g(x) < 1$. For any $x \in E$, choose $y \in E$ with $h(y) > h(x)$. Since $t/(1+t)$ is strictly increasing for $t \geq 0$ (as follows by cross multiplication), $g(y) > g(x)$. Thus no point can maximize g.

Conclusion. Compactness is essential both for boundedness of continuous functions and for attainment of their extreme values.

21.6 Exercise 21.6: Compositions of continuous metric-space functions

What we must show. Show that $g \circ f : S_1 \to S_3$ is continuous when both component functions are continuous.

Pull an open set back in two stages.

Let $U \subseteq S_3$ be open. Continuity of g says $g^{-1}(U)$ is open in S_2. Continuity of f then says $f^{-1}(g^{-1}(U))$ is open in S_1. The identity

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$$

holds because each side consists of exactly those s with $g(f(s)) \in U$. Theorem 21.3 therefore proves continuity of $g \circ f$.

Conclusion. The order of inverse images is the reverse of the order in which the functions are applied.

21.7 Exercise 21.7: Restricted metrics and continuity of a path

(a) Restricting a metric.

For $s, t, u \in E \subseteq S$, the numbers $d(s,t)$ are nonnegative, $d(s,t) = 0$ exactly when $s = t$, symmetry holds, and $d(s,u) \leq d(s,t) + d(t,u)$. These statements already hold in S, so they still hold when the available points are restricted to E. Thus (E, d) is a metric space. In particular $d(x,y) = |x - y|$ is a metric on every subset of $\mathbb{R}$.

(b) Spell out the definition for a path.

A function $\gamma : [a,b] \to \mathbb{R}^k$ is continuous if, for every $t_0 \in [a,b]$ and every $\varepsilon > 0$, there is $\delta > 0$ such that

$$t \in [a,b], \quad |t - t_0| < \delta \quad \Longrightarrow \quad \left(\sum_{j=1}^{k} |\gamma_j(t) - \gamma_j(t_0)|^2 \right)^{1/2} < \varepsilon.$$

Conclusion. At an endpoint, only points of $[a,b]$ are tested. The definition does not ask for values of the path outside its domain.

21.8 Exercise 21.8: Uniform continuity sends Cauchy sequences to Cauchy sequences

What we must show. Prove that $(f(s_n))$ is Cauchy in S^* when (s_n) is Cauchy in S and f is uniformly continuous.

Match the two definitions.

Fix $\varepsilon > 0$. Uniform continuity gives $\delta > 0$ such that $d(s,t) < \delta$ implies $d^*(f(s), f(t)) < \varepsilon$, for all $s, t \in S$. Since (s_n) is Cauchy, some N has $d(s_m, s_n) < \delta$ whenever $m, n > N$. Consequently

$$m, n > N \implies d^*(f(s_m), f(s_n)) < \varepsilon.$$

Conclusion. This is the Cauchy condition in the target. We did not assume either sequence has a limit. Mere continuity would not suffice: $1/x$ on $(0,1)$ sends the Cauchy sequence $1/(n+1)$ to the non-Cauchy sequence $n+1$.

21.9 Exercise 21.9: Continuous surjections involving a square

(a) Project the square onto an interval.

On $[0,1]^2$, put $p(x_1, x_2) = x_1$. The inequality $|p(x) - p(y)| \le \|x - y\|$ proves continuity. Every $u \in [0,1]$ is the image of $(u, 0)$, so the range is exactly $[0,1]$.

(b) Can an interval fill a square?

Yes: a continuous surjection $[0,1] \to [0,1]^2$ is called a *space-filling curve*. Dimension alone does not prevent one. The question asks for a judgment, rather than a formula; here is an existence argument explaining why the answer is not merely a guess.

Let C be the middle-thirds Cantor set. Each point of C has a unique expansion $t = \sum_{j \ge 1} 2b_j 3^{-j}$ with $b_j \in \{0, 1\}$. Define

$$H(t) = \left(\sum_{j \ge 1} b_{2j-1} 2^{-j},\ \sum_{j \ge 1} b_{2j} 2^{-j} \right).$$

Any two numbers in $[0,1]$ have binary expansions; interlacing their digits produces a point of C with precisely those two images. Thus $H(C) = [0,1]^2$. If two Cantor points have their first $2m$ ternary digits equal, their coordinate differences are at most $\sum_{j>m} 2^{-j} = 2^{-m}$. Conversely, points in distinct cylinders specified by the first $2m$ digits are separated by a positive gap. These finite-cylinder gaps prove continuity of H on C.

Extend H to each removed interval (u, v) by straight-line interpolation between $H(u)$ and $H(v)$. The resulting map stays in the square because line segments between square points stay in the square. It is continuous inside each gap. To check continuity at $t \in C$, consider any sequence $t_n \to t$. For points in C use continuity of H. For points in gaps, either a subsequence lies in a fixed gap whose endpoint is t (where linear interpolation is continuous), or the gap lengths tend to zero. In the latter case both endpoints tend to t, and their images tend to $H(t)$, so every convex combination does too. Only finitely many gaps have length at least any prescribed positive number; hence these cases cover every possible subsequence. The sequential criterion proves continuity.

Conclusion. The extension is a continuous map on $[0,1]$ and is already onto the square on its Cantor subset. A curve can therefore fill a square.

21.10 Exercise 21.10: Four questions about continuous ranges

(a) Map $(0,1)$ onto $[0,1]$.

Use $f(t) = (1 + \sin(4\pi t))/2$. This continuous function takes values only in $[0,1]$, attains 1 at $t = 1/8$, and attains 0 at $t = 3/8$. The intermediate value theorem on $[1/8, 3/8]$ supplies every value between them. Hence the map is onto.

(b) Map $(0,1)$ onto $\mathbb{R}$.

Use $g(t) = \tan(\pi(t - 1/2))$. Its argument lies in $(-\pi/2, \pi/2)$, where tangent is continuous and strictly increasing, with limits $-\infty$ and $+\infty$ at the two ends. For any real y, choose interior points where g is below and above y and apply the intermediate value theorem.

(c) Map $[0,1] \cup [2,3]$ onto $[0,1]$.

Define $h(t) = t$ on $[0,1]$ and $h(t) = t - 2$ on $[2,3]$. On each component this is continuous. A ball of radius less than 1 around any domain point cannot reach the other component, so continuity on each component proves continuity on the whole domain. Each component already maps onto $[0,1]$.

(d) Rule out two proposed ranges.

A continuous image of the compact interval $[0,1]$ is compact. Neither $(0,1)$ (not closed in $\mathbb{R}$) nor $\mathbb{R}$ (unbounded) is compact. If a continuous map were onto either, its image would be both compact and noncompact, a contradiction.

Conclusion. The formulas supply (a)–(c); preservation of compactness excludes both maps in (d).

21.11 Exercise 21.11: The rationals are not a countable intersection of open sets

What we must show. Use the Baire theorem to rule out $\mathbb{Q} = \bigcap_{n \geq 1} U_n$ with all U_n open in $\mathbb{R}$.

Plan. Such U_n would all be dense. Intersect them also with the complements of individual rationals and reach an empty dense set.

Construct dense open sets.

Suppose the representation exists. Since every U_n contains $\mathbb{Q}$ and $\mathbb{Q}$ is dense, U_n is dense. List the rationals as $\mathbb{Q} = \{q_1, q_2, \ldots\}$. Each $V_n = \mathbb{R} \setminus \{q_n\}$ is open and dense: every nonempty open interval contains points other than a single fixed point.

Apply completeness exactly where needed.

The real line is a complete metric space. Theorem 21.7(a), applied to the interlaced sequence $U_1, V_1, U_2, V_2, \ldots$, says their intersection is dense. But it equals

$$\left(\bigcap_n U_n\right) \cap \left(\bigcap_n V_n\right) = \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \varnothing,$$

which is not dense in the nonempty space $\mathbb{R}$.

Take complements for the equivalent formulation.

If $\mathbb{R} \setminus \mathbb{Q} = \bigcup_n F_n$ with each F_n closed, De Morgan's law would give $\mathbb{Q} = \bigcap_n (\mathbb{R} \setminus F_n)$ with open terms.

Conclusion. Neither proposed representation is possible.

21.12 Exercise 21.12: Infinitely many disjoint sets of second category

What we must show. Give pairwise disjoint subsets of $\mathbb{R}$, each not a countable union of nowhere dense subsets of $\mathbb{R}$.

Choose intervals and invoke the right Baire consequence.

For $n \geq 1$, let $E_n = (2n, 2n+1)$. These intervals are pairwise disjoint. Suppose some E_n were a countable union of nowhere dense sets. By Theorem 21.7(c), its complement would be dense in $\mathbb{R}$. But that complement misses the nonempty open interval E_n, contradicting density.

Conclusion. The sequence $((2n, 2n+1))_{n\geq 1}$ works. The category here is category in the ambient space $\mathbb{R}$, as the exercise requires.

21.13 Exercise 21.13: Oscillation detects continuity

What we must show. For

$$\omega_f(x) = \inf_{\delta>0} \sup\{|f(y) - f(z)| : y, z \in (x-\delta, x+\delta)\},$$

prove f is continuous at x exactly when $\omega_f(x) = 0$. The suprema may be infinite when f is locally unbounded; all quantities are nonnegative.

Continuity makes the oscillation zero.

Fix $\varepsilon > 0$. Continuity gives $\delta > 0$ such that $|f(y) - f(x)| < \varepsilon/4$ for $|y - x| < \delta$. For two such points, the triangle inequality gives $|f(y) - f(z)| < \varepsilon/2$. Taking the supremum can change a strict inequality into a nonstrict one, but still gives a bound at most $\varepsilon/2 < \varepsilon$. Therefore $0 \leq \omega_f(x) < \varepsilon$. Since this holds for every $\varepsilon > 0$, $\omega_f(x) = 0$.

Zero oscillation gives continuity.

If the nonnegative infimum is zero, then for every $\varepsilon > 0$ some $\delta > 0$ has the corresponding supremum less than ε. Set $z = x$, which is permitted because x belongs to the interval. Then

$$|y - x| < \delta \implies |f(y) - f(x)| \leq \sup_{u,v\in(x-\delta,x+\delta)} |f(u) - f(v)| < \varepsilon.$$

Conclusion. This is continuity at x. It is important that the interval in the oscillation definition includes its center.

21.14 Exercise 21.14: Reduce every interval to the bounded closed case

What we must show. Complete Theorem 21.11 after its proof for $[a, b]$.

Assume a forbidden decomposition of an arbitrary interval.

Suppose a nondegenerate interval I is a disjoint union of at least two nondegenerate closed intervals J_α. Choose two distinct members. Select an interior point x of one and an interior point y of the other; relabel so $x < y$. Because I is an interval, $[x, y] \subseteq I$.

Intersect every piece with $[x, y]$.

Discard empty intersections. The remaining $K_\alpha = J_\alpha \cap [x, y]$ are pairwise disjoint closed intervals whose union is $[x, y]$. The interval containing x has positive length to the right of x, and the interval containing y has positive length to the left of y, because x, y were chosen in interiors. Every other surviving interval lies strictly between x and y and retains its original positive length. Thus none of the surviving intersections is a singleton. There are at least two of them, since the intervals containing x and y are distinct.

Use the case already established.

We have written the bounded closed interval $[x, y]$ as a disjoint union of at least two nondegenerate closed intervals. This contradicts the bounded closed case of Theorem 21.11.

Conclusion. The theorem holds for all nondegenerate intervals, including open, half-open, and unbounded intervals. The interior-point choice prevents unwanted singleton pieces at the truncation endpoints.

22 More on metric spaces: connectedness

A separation of a set is a partition into two nonempty relatively open sets. A set is connected when it has no separation. A path is a continuous map from an interval; path-connectedness asks for a path between every two points of the set. Path-connected sets are connected, but the converse fails. In this section $C(S)$ means the *bounded* continuous real-valued functions on $S \subseteq \mathbb{R}$, and its distance is

$$d(f,g) = \|f-g\|_\infty = \sup_{x\in S} |f(x)-g(x)|.$$

Here $\|h\|_\infty$ denotes the supremum of $|h|$. When S is empty, there is just one function and its distance from itself is taken to be zero; the nontrivial arguments below concern nonempty S.

22.1 Exercise 22.1: Two impossible continuous surjections

(a) A disconnected union of intervals.

The interval $[0,1]$ is connected. By Theorem 22.2 a continuous image must be connected. But $[0,1]\cup[2,3]$ has a separation into its two nonempty pieces, obtained by intersecting with $(-\infty, 3/2)$ and $(3/2,\infty)$. Hence no continuous map from $[0,1]$ is onto that union.

(b) A rational range.

The interval $(0,1)$ is connected, whereas $\mathbb{Q}$ is not an interval: for example, it contains $1, 2$ but not $\sqrt{2}$ between them. Equivalently, $\mathbb{Q}\cap(-\infty,\sqrt{2})$ and $\mathbb{Q}\cap(\sqrt{2},\infty)$ separate $\mathbb{Q}$.

Conclusion. Neither surjection exists, because a continuous image of a connected set is connected.

22.2 Exercise 22.2: The unit circle is connected

What we must show. Show the set $x_1^2 + x_2^2 = 1$ is connected.

Give a continuous parametrization with exactly that image.

Define $\gamma : [0, 2\pi] \to \mathbb{R}^2$ by $\gamma(t) = (\cos t, \sin t)$. Both coordinates are continuous, so Proposition 21.2 proves continuity of γ. The identity $\cos^2 t + \sin^2 t = 1$ puts its image on the circle. Conversely every point of the unit circle has a polar angle $t \in [0, 2\pi]$, and hence is of this form.

Conclusion. The circle is the continuous image of the connected interval $[0, 2\pi]$, so Theorem 22.2 makes it connected.

22.3 Exercise 22.3: The closure of a connected set is connected

What we must show. Prove connectedness of $\overline{E}$ from connectedness of E.

Suppose there were a separation of the closure.

Let open ambient sets U, V give disjoint nonempty sets $\overline{E}\cap U$, $\overline{E}\cap V$ whose union is $\overline{E}$. Choose $u \in \overline{E}\cap U$. Since U is an open neighborhood of a closure point, $U \cap E \neq \varnothing$. The same argument gives $V \cap E \neq \varnothing$.

Restrict the separation.

The sets $E \cap U$ and $E \cap V$ are disjoint, nonempty and relatively open in E, and their union is E. They would separate E, contrary to its connectedness.

Conclusion. No separation of $\overline{E}$ exists. For $E = \varnothing$, the closure is empty and connected by definition.

22.4 Exercise 22.4: The closed topologist's sine curve

What we must show. Study $E = \{(x, \sin(1/x)) : 0 < x \leq 1\}$.

(a) Find all new limit points.

We claim

$$\overline{E} = E \cup (\{0\} \times [-1, 1]).$$

If $(x_n, \sin(1/x_n)) \to (a, b)$, then $a \in [0, 1]$ and $b \in [-1, 1]$. If $a > 0$, continuity of $\sin(1/x)$ at a forces $b = \sin(1/a)$, so the limit is already in E. If $a = 0$, it lies on the stated vertical segment. Conversely, given $b \in [-1, 1]$, choose an angle θ with $\sin\theta = b$. For all sufficiently large integers n, $x_n = 1/(\theta + 2\pi n)$ belongs to $(0, 1]$ and tends to zero, while $\sin(1/x_n) = b$. Thus $(0, b) \in \overline{E}$.

(b) Establish connectedness before dealing with paths.

The map $x \mapsto (x, \sin(1/x))$ is continuous on the connected interval $(0, 1]$. Its image E is connected by Theorem 22.2, and Exercise 22.3 makes its closure connected.

(c) Explain why no path can leave the vertical segment.

Suppose a path $\gamma(t) = (u(t), v(t))$, $0 \leq t \leq 1$, starts on the vertical segment and ends in E. The coordinate functions are continuous, $u(0) = 0$, and $u(1) > 0$. The zero set of u is a nonempty closed subset of $[0, 1]$, so it has a largest element $t_0 < 1$. We have $u(t_0) = 0$ and $u(t) > 0$ for $t > t_0$.

Choose a fixed $t_1 > t_0$. Because $u(t_1) > 0$, the intermediate value theorem applied to u on $[t_0, t_1]$ shows that it attains every sufficiently small positive number. In particular, it attains the numbers

$$a_n = \frac{1}{\pi/2 + 2\pi n}, \qquad b_n = \frac{1}{3\pi/2 + 2\pi n}.$$

Choose corresponding parameters $r_n, s_n \in (t_0, t_1]$. They tend to t_0: for each $\eta > 0$ with $t_0 + \eta \leq t_1$, the positive continuous u has a positive minimum on $[t_0 + \eta, t_1]$; eventually a_n, b_n are smaller than that minimum. Thus their preimages cannot lie in this latter interval.

For positive first coordinate, membership in $\overline{E}$ forces $v(t) = \sin(1/u(t))$. Hence $v(r_n) = 1$ and $v(s_n) = -1$. Both parameter sequences approach t_0, contradicting continuity of v there.

Conclusion. The closure is connected but is not path-connected. An accumulating curve can touch a segment in its closure without supplying any continuous path from that segment to the curve.

22.5 Exercise 22.5: Unions and intersections of connected sets

(a) A shared point joins two connected sets.

Suppose $E \cap F \neq \varnothing$ but $E \cup F$ has a separation A, B. Restricting the separation to E shows E lies entirely in one side, since otherwise it would be disconnected. Similarly, F lies entirely in one

side. A point of $E \cap F$ forces them to lie on the same side. The other side would then be empty, a contradiction.

(b) An intersection can be disconnected.

Let

$$E = \{(x, y) : x^2 + y^2 = 1,\ y \geq 0\}, \qquad F = \{(x, y) : x^2 + y^2 = 1,\ y \leq 0\}.$$

Each semicircle is a continuous image of a closed interval, hence connected. But $E \cap F = \{(-1, 0), (1, 0)\}$ has two isolated points and is disconnected: balls of radius $1/2$ about these points separate it.

Conclusion. A nonempty intersection guarantees a connected union; it does not guarantee a connected intersection.

22.6 Exercise 22.6: Why the supremum distance is a metric on bounded functions

(a) Verify finiteness and the metric axioms.

If $|f| \leq M$ and $|g| \leq N$ on S, then $0 \leq |f(x) - g(x)| \leq M + N$. Thus $d(f, g)$ is finite and nonnegative. It is symmetric because $|f(x) - g(x)| = |g(x) - f(x)|$. If $d(f, g) = 0$, then $|f(x) - g(x)| = 0$ at every x, so $f = g$; conversely identical functions have distance zero. Finally for every x,

$$|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)| \leq d(f, g) + d(g, h).$$

Taking the supremum over x proves the triangle inequality.

(b) Distinguish a finite metric from a convergence definition.

Without boundedness, a pair such as $f(x) = x$, $g(x) = 0$ on $\mathbb{R}$ would have $d(f, g) = \infty$, forbidden for an ordinary metric. Uniform convergence itself only asks whether the errors $|f_n(x) - f(x)|$ are eventually small for all x simultaneously. The functions can be unbounded while their differences are uniformly small: $f_n(x) = x + 1/n$ converges uniformly to $f(x) = x$ on $\mathbb{R}$.

Conclusion. Boundedness makes the supremum distance finite for every pair in $C(S)$; it is not needed to state uniform convergence.

22.7 Exercise 22.7: A bounded sequence is a bounded continuous function on $\mathbb{N}$

What we must show. Identify the metric space B from Exercise 13.3 with $C(\mathbb{N})$.

Use discreteness to understand continuity.

Under the usual distance inherited from $\mathbb{R}$, the ball of radius $1/2$ about $n \in \mathbb{N}$ contains only n. Hence *every* function $f : \mathbb{N} \to \mathbb{R}$ is continuous: choose $\delta = 1/2$ at each point, and all tested values are identical to the center value.

Match both the elements and their distances.

The sequence $(x_1, x_2, \ldots)$ corresponds to the function $f(n) = x_n$. A sequence is bounded exactly when its function is bounded. Moreover,

$$\sup_{n \in \mathbb{N}} |x_n - y_n| = \sup_{n \in \mathbb{N}} |f(n) - g(n)|.$$

Conclusion. This correspondence is a bijection preserving distance, so B and $C(\mathbb{N})$ are the same metric space after relabeling their elements.

22.8 Exercise 22.8: Evaluation at one point is uniformly continuous

What we must show. For fixed $s_0 \in S$, study $F(f) = f(s_0)$.

Bound one error by the largest error.

For $f, g \in C(S)$,

$$|F(f) - F(g)| = |f(s_0) - g(s_0)| \leq \sup_{x \in S} |f(x) - g(x)| = d(f, g).$$

Given $\varepsilon > 0$, take $\delta = \varepsilon$. If $d(f, g) < \delta$, the displayed inequality gives $|F(f) - F(g)| < \varepsilon$.

Conclusion. Evaluation is 1-Lipschitz and therefore uniformly continuous.

22.9 Exercise 22.9: A straight line in a function space

What we must show. Prove uniform continuity of $F(t) = tf + (1 - t)g$ as a map $\mathbb{R} \to C(S)$.

First check that the values belong to the claimed space.

For fixed t, the function $tf + (1 - t)g$ is continuous and is bounded by $|t|\|f\|_\infty + |1 - t|\|g\|_\infty$. Thus $F(t) \in C(S)$.

Compute the distance between two values.

The difference is $F(t) - F(s) = (t - s)(f - g)$, so

$$d(F(t), F(s)) = |t - s| \, \|f - g\|_\infty.$$

Let $M = \|f - g\|_\infty$. If $M = 0$, F is constant. Otherwise $\delta = \varepsilon/M$ makes $|t - s| < \delta$ imply $d(F(t), F(s)) < \varepsilon$.

Conclusion. The map is Lipschitz with constant $\|f - g\|_\infty$. The estimate works on the whole real line.

22.10 Exercise 22.10: Translation is continuous in the supremum metric

What we must show. For bounded uniformly continuous $f : \mathbb{R} \to \mathbb{R}$, prove uniform continuity of $F(x) = f_x$, where $f_x(y) = f(x + y)$.

Check each translated function.

Translation preserves continuity, and $|f_x(y)| \leq \|f\|_\infty$. Thus $f_x \in C(\mathbb{R})$ for every real x.

Keep the error bound uniform in the variable being translated.

Fix $\varepsilon > 0$. Uniform continuity of f supplies $\delta > 0$ such that $|u - v| < \delta$ implies $|f(u) - f(v)| < \varepsilon/2$ for all real u, v. If $|x - z| < \delta$, then for *every* y,

$$|f_x(y) - f_z(y)| = |f(x + y) - f(z + y)| < \varepsilon/2,$$

because $|(x+y)-(z+y)| = |x-z|$. Taking a supremum yields $d(F(x), F(z)) \leq \varepsilon/2 < \varepsilon$.

Conclusion. The same δ works for all x, z, proving uniform continuity. Using $\varepsilon/2$ avoids losing the strict inequality at a supremum.

22.11 Exercise 22.11: The closed unit ball and connectedness in $C(S)$

What we must show. Let $\mathcal{E} = \{f \in C(S) : \|f\|_\infty \leq 1\}$.

(a) Its complement is open.

Suppose $f \notin \mathcal{E}$. Then some $x_0 \in S$ has $|f(x_0)| > 1$; otherwise 1 would be an upper bound for all $|f(x)|$. Put $r = (|f(x_0)| - 1)/2 > 0$. If $d(f, g) < r$, then

$$|g(x_0)| \geq |f(x_0)| - |g(x_0) - f(x_0)| > |f(x_0)| - r > 1.$$

Thus every g in this metric ball is also outside $\mathcal{E}$. The complement is open, so $\mathcal{E}$ is closed.

(b) The entire space is path-connected.

For any $f, g \in C(S)$, Exercise 22.9 gives a continuous path $t \mapsto (1-t)f + tg$, $0 \leq t \leq 1$, from f to g. Thus $C(S)$ is path-connected, and Theorem 22.5 makes it connected.

(c) The same paths stay in the unit ball.

If $f, g \in \mathcal{E}$ and $0 \leq t \leq 1$, then at every x,

$$|(1-t)f(x) + tg(x)| \leq (1-t)|f(x)| + t|g(x)| \leq 1.$$

The path therefore lies in $\mathcal{E}$.

Conclusion. $\mathcal{E}$ is closed and path-connected, hence connected; $C(S)$ is connected as well.

22.12 Exercise 22.12: A topology determined by finitely many evaluations

What we must show. Verify the topology axioms for the specified pointwise neighborhoods. Write

$$N(f; F, \varepsilon) = \{g \in C(S) : |g(x) - f(x)| < \varepsilon \text{ for all } x \in F\},$$

where $F \subseteq S$ is finite and $\varepsilon > 0$.

(a) Recall what must be proved.

Discussion 13.7 describes the characteristic properties of open sets: the empty set and the entire space are open, arbitrary unions of open sets are open, and finite intersections of open sets are open. These properties define a topology even when no metric has been specified. Here a set U is declared open if each $f \in U$ has some $N(f; F, \varepsilon) \subseteq U$.

(b) Check all three properties.

The empty set is open because it has no points for which the requirement could fail. The whole space is open since every such neighborhood lies in it (one can take $F = \varnothing$). If $f \in \bigcup_\alpha U_\alpha$, then f belongs to some open U_{α_0}, which supplies a neighborhood contained in U_{α_0} and hence in the union.

For a finite intersection $U_1 \cap \cdots \cap U_m$ and a point f in it, choose $N(f; F_j, \varepsilon_j) \subseteq U_j$ for every j. The set $F = F_1 \cup \cdots \cup F_m$ is finite, and $\varepsilon = \min_j \varepsilon_j > 0$. A function in $N(f; F, \varepsilon)$ satisfies every constraint for every U_j, so this neighborhood lies in the intersection. The empty finite intersection is the whole space, already covered.

Why these are pointwise rather than uniform neighborhoods.

Each neighborhood restricts only finitely many values of a function. A small supremum-metric ball restricts values at *every* point. For example, on $S = \mathbb{R}$ any finite set of prescribed points misses some x_0; a continuous bounded bump supported near x_0 can be zero at every prescribed point yet have supremum 2. Thus no finite-point neighborhood of zero is contained in the supremum ball of radius 1.

Conclusion. The declared sets satisfy every topology axiom. The text's additional blanket remark about nonmetrizability needs qualification: for finite S this is the usual finite-dimensional topology, and for countable $S = \{s_1, s_2, \ldots\}$ it is metrized by $\sum_{j \geq 1} 2^{-j} \min(1, |f(s_j) - g(s_j)|)$. The requested topology verification is valid for every S.

22.13 Exercise 22.13: A connected graph with an oscillatory endpoint

What we must show. Analyze the graph with the oscillating branch at positive x and the added origin.

Clarify the printed notation.

The exercise prints $f(x) = (x, \sin(1/x))$ for $x > 0$ and $f(0) = 0$. Read literally, this is a map to $\mathbb{R}^2$ with $f(0) = (0, 0)$, and its graph lies in $\mathbb{R}^3$. The frequently intended scalar formulation is $f(x) = \sin(1/x)$, $f(0) = 0$. Both give the same connectedness answer: the literal graph is homeomorphic to

$$H = \{(x, \sin(1/x)) : x > 0\} \cup \{(0, 0)\}$$

by the map $(x, y) \mapsto (x, x, y)$, whose inverse on its image is $(a, b, c) \mapsto (a, c)$. Both maps are continuous coordinate maps. Thus we prove the assertions for H.

The added origin does not disconnect the branch.

The branch $A = \{(x, \sin(1/x)) : x > 0\}$ is a continuous image of the interval $(0, \infty)$, hence connected. Also $(0, 0) \in \overline{A}$, since $(1/(n\pi), 0) \to (0, 0)$. If H had a separation, connectedness would put all of A on one side. The other side would consist of the origin and be relatively open in H, which would provide a neighborhood of the origin missing A. This contradicts its being a closure point. Hence H is connected.

It has no path from the origin to the branch.

The proof of Exercise 22.4(c) applies without change. A proposed path has a last parameter t_0 at which its first coordinate is zero. Immediately after t_0 its first coordinate is positive and tends to zero. It must attain the values a_n and b_n from that exercise at parameters tending to t_0, forcing its second coordinate to take both 1 and -1 arbitrarily close to t_0. That contradicts continuity. The restriction $x \leq 1$ was not used in that argument.

Continuity of the original function fails at zero.

Along $x_n = 1/(\pi/2 + 2\pi n)$, the sine coordinate is 1, not tending to its value 0 at zero. The sequential criterion therefore rules out continuity at zero in either formulation; continuity holds for $x > 0$.

Conclusion. The graph is connected but not path-connected, and the function is not continuous on $[0, \infty)$.

22.14 Exercise 22.14: Small successive changes force an interval of cluster values

What we must show. For bounded (s_n) with $s_{n+1} - s_n \to 0$, show its set S of subsequential limits is a connected interval.

Plan. If the sequence repeatedly visits both sides of a level c, it must cross c repeatedly. Small jumps make some crossing terms arbitrarily close to c.

Fix any level between two subsequential limits.

Take $a, b \in S$ with $a < c < b$. For each integer $j \geq 1$, choose N_j large enough that $|s_{n+1} - s_n| < 1/j$ for all $n \geq N_j$. Since a and b are subsequential limits, there are arbitrarily late terms below c and arbitrarily late terms above c. Indeed, use errors smaller than $(c-a)/2$ and $(b-c)/2$ in their defining subsequences.

Locate crossings farther and farther out.

Choose $p_j > \max(N_j, q_{j-1})$ with $s_{p_j} < c$, and then choose $q_j > p_j$ with $s_{q_j} > c$ (start with $q_0 = 0$). Let k_j be the first index after p_j for which $s_{k_j} \geq c$; it exists because q_j is one such index. Then

$$s_{k_j-1} < c \leq s_{k_j}, \qquad 0 \leq s_{k_j} - c < s_{k_j} - s_{k_j-1} < 1/j.$$

The chosen indices strictly increase, so (s_{k_j}) is a subsequence and converges to c. Thus $c \in S$.

Conclude connectedness.

The preceding argument shows S contains every number strictly between any two of its points, which is exactly the interval property. Boundedness and Bolzano–Weierstrass ensure S is nonempty; the cluster-set theorem also makes it closed and bounded. Thus it is a closed bounded interval, possibly a singleton, and intervals are connected.

Conclusion. The cluster values cannot have a gap because vanishing step sizes prevent repeated crossings from jumping across such a gap.

23 Power series

For a power series $\sum_{n\geq 0} a_n x^n$, write $\beta = \limsup_{n\to\infty} |a_n|^{1/n}$. The radius of convergence is $R = 1/\beta$, with $1/0 = \infty$ and $1/\infty = 0$. The series converges absolutely for $|x| < R$ and diverges for $|x| > R$. *Neither endpoint is decided by the radius calculation*: substitute $x = R$ and $x = -R$ into the original series and test them separately. Finite initial terms do not affect convergence. Thus all series containing $1/n$ start at $n = 1$, and harmless choices of starting index do not change the answers. We use $n^{c/n} \to 1$ for fixed c, which follows by taking logarithms: $\log(n^{c/n}) = c \log n/n \to 0$.

23.1 Exercise 23.1: Eight radii and their endpoint tests

What we must show. Find the exact convergence interval in each case.

(a) The series $\sum n^2 x^n$.

Here $|a_n|^{1/n} = n^{2/n} \to 1$, so $R = 1$. At $x = 1$ the terms are n^2, and at $x = -1$ they are $(-1)^n n^2$. In both cases the terms fail to tend to zero, a necessary condition for series convergence. Thus the exact interval is $\boxed{(-1, 1)}$.

(b) The series $\sum (x/n)^n$.

The coefficient of x^n is $1/n^n$, whose nth root is $1/n \to 0$. Thus $R = \infty$. To see this directly, for fixed real x and sufficiently large n we have $|x|/n \leq 1/2$, and hence $|(x/n)^n| \leq 2^{-n}$. The geometric comparison proves absolute convergence. The exact interval is $\boxed{\mathbb{R}}$.

(c) The series $\sum (2^n/n^2)x^n$.

The coefficient root $2/n^{2/n}$ tends to 2, so $R = 1/2$. At $x = 1/2$ the series becomes $\sum 1/n^2$, which converges. At $x = -1/2$ it becomes $\sum (-1)^n/n^2$, which converges absolutely by the same p-series. The exact interval is $\boxed{[-1/2, 1/2]}$.

(d) The series $\sum (n^3/3^n)x^n$.

The coefficient root $n^{3/n}/3$ tends to $1/3$, giving $R = 3$. At $x = 3$ the terms are n^3; at $x = -3$ they are $(-1)^n n^3$. Neither tends to zero. The exact interval is $\boxed{(-3, 3)}$.

(e) The series $\sum (2^n/n!)x^n$.

For $x \neq 0$ the ratio of successive absolute terms is

$$\frac{2^{n+1}|x|^{n+1}/(n+1)!}{2^n|x|^n/n!} = \frac{2|x|}{n+1} \longrightarrow 0 < 1.$$

The ratio test proves absolute convergence for every such x. At $x = 0$ convergence is immediate. Thus $R = \infty$ and the interval is $\boxed{\mathbb{R}}$.

(f) The series $\sum x^n/((n+1)^2 2^n)$.

Its coefficient root is $1/(2(n+1)^{2/n}) \to 1/2$, so $R = 2$. At $x = 2$ it is $\sum 1/(n+1)^2$; at $x = -2$ it is $\sum (-1)^n/(n+1)^2$. Both converge absolutely. The interval is $\boxed{[-2, 2]}$.

(g) The series $\sum (3^n/(n4^n))x^n$.

The coefficient root $(3/4)n^{-1/n}$ tends to $3/4$, so $R = 4/3$. At $x = 4/3$ we obtain the divergent harmonic series $\sum 1/n$. At $x = -4/3$ we obtain $\sum(-1)^n/n$, which converges by the alternating-series test: $1/n$ decreases to zero. It is conditionally convergent because its absolute-value series is harmonic. The interval is $\boxed{[-4/3, 4/3)}$.

(h) The series $\sum((-1)^n/(n^2 4^n))x^n$.

Taking absolute values before the root removes the alternating sign: $|a_n|^{1/n} = 1/(4n^{2/n}) \to 1/4$. Hence $R = 4$. At $x = 4$ the series is $\sum(-1)^n/n^2$, and at $x = -4$ it is $\sum 1/n^2$, since the two factors $(-1)^n$ multiply to 1. Both converge absolutely, giving $\boxed{[-4, 4]}$.

Conclusion. The radii in order are $1, \infty, 1/2, 3, \infty, 2, 4/3, 4$. The separate endpoint tests determine the displayed intervals.

23.2 Exercise 23.2: Four more series, including missing powers

(a) The series $\sum \sqrt{n}\, x^n$.

The coefficient root is $n^{1/(2n)} \to 1$, giving $R = 1$. At $x = 1$ and $x = -1$, the term magnitudes are $\sqrt{n}$, which do not tend to zero. Thus the exact interval is $\boxed{(-1, 1)}$.

(b) The series $\sum x^n/(n\sqrt{n})$.

The denominator here is $n\sqrt{n} = n^{3/2}$, not an nth root. We have $|a_n|^{1/n} = n^{-3/(2n)} \to 1$, so $R = 1$. At either endpoint the absolute series is $\sum n^{-3/2}$, a convergent p-series because $3/2 > 1$. Thus the exact interval is $\boxed{[-1, 1]}$.

(c) The sparse series $\sum x^{n!}$.

For $|x| < 1$, since $n! \geq n$, $|x|^{n!} \leq |x|^n$. The right side has a convergent geometric sum, so absolute convergence follows. At $x = 1$ every term is 1. At $x = -1$, all terms with $n \geq 2$ are 1 because $n!$ is even. Both endpoint series diverge. For $|x| > 1$, the term magnitudes $|x|^{n!}$ do not approach zero. Therefore the radius is $R = 1$ and the exact interval is $\boxed{(-1, 1)}$.

(d) The series $\sum(3^n/\sqrt{n})x^{2n+1}$.

The exponent is $2n + 1$, so treating $3^n/\sqrt{n}$ as the coefficient of x^n would give the wrong radius. Instead write

$$\sum_{n\geq 1} \frac{3^n}{\sqrt{n}} x^{2n+1} = x \sum_{n\geq 1} \frac{(3x^2)^n}{\sqrt{n}}.$$

For $x \neq 0$, the absolute-term ratio tends to $3x^2$. Thus the series converges absolutely when $3x^2 < 1$ and diverges when $3x^2 > 1$. At $x = 0$ all terms vanish. At $x = 1/\sqrt{3}$, its terms equal $1/(\sqrt{3}\sqrt{n})$; at $x = -1/\sqrt{3}$, they equal $-1/(\sqrt{3}\sqrt{n})$. Each endpoint gives a constant multiple of the divergent p-series $\sum n^{-1/2}$. Thus

$$\boxed{R = 1/\sqrt{3}, \qquad \text{interval } (-1/\sqrt{3}, 1/\sqrt{3}).}$$

Conclusion. Missing powers must be respected. Testing the actual terms, rather than a misindexed coefficient sequence, avoids that error.

23.3 Exercise 23.3: The endpoint test for the sparse series in Example 6

What we must show. Determine the exact interval for $\sum_{n\geq 0} 2^{-n}x^{3n}$.

Recognize a geometric series in a new quantity.

Since $2^{-n}x^{3n} = (x^3/2)^n$, this is geometric with ratio $r = x^3/2$. It converges exactly when $|r| < 1$, equivalently $|x| < 2^{1/3}$. At $x = 2^{1/3}$ the terms are 1; at $x = -2^{1/3}$ they are $(-1)^n$. Neither term sequence tends to zero.

Conclusion. The exact interval is $(-2^{1/3}, 2^{1/3})$. The radius is $2^{1/3}$, not 2, because the powers are x^{3n}.

23.4 Exercise 23.4: Roots and ratios with oscillating coefficients

What we must show. Analyze $a_n = [(4 + 2(-1)^n)/5]^n$.

Separate even and odd indices before calculating.

For even n, $a_n = (6/5)^n$; for odd n, $a_n = (2/5)^n$. The term $a_0 = 1$ does not affect any limiting or convergence question.

(a) Compute the four requested extended limits.

The sequence $a_n^{1/n}$ alternates between $6/5$ and $2/5$, so

$$\limsup a_n^{1/n} = 6/5, \qquad \liminf a_n^{1/n} = 2/5.$$

For even n, the next index is odd and

$$\frac{a_{n+1}}{a_n} = \frac{(2/5)^{n+1}}{(6/5)^n} = \frac{2}{5}\,3^{-n} \longrightarrow 0.$$

For odd n, the next index is even and

$$\frac{a_{n+1}}{a_n} = \frac{(6/5)^{n+1}}{(2/5)^n} = \frac{6}{5}\,3^{n} \longrightarrow +\infty.$$

All coefficients are positive, so absolute values change nothing. Thus the ratio limsup is $+\infty$ and the ratio liminf is 0.

(b) Apply the necessary term test to the two numerical series.

Along even indices, $a_{2k} = (6/5)^{2k} \to +\infty$. Hence a_n does not tend to zero and $\sum a_n$ diverges. Also $(-1)^{2k}a_{2k} = a_{2k}$, so the signed series $\sum(-1)^n a_n$ diverges for the same reason. Alternating signs cannot make a series converge unless its terms tend to zero.

(c) Find the power-series interval.

The root limsup gives $R = 1/(6/5) = 5/6$. At either $x = 5/6$ or $x = -5/6$, the even terms satisfy

$$a_{2k}x^{2k} = (6/5)^{2k}(5/6)^{2k} = 1.$$

Neither endpoint series has terms tending to zero. Therefore

$$\boxed{R = 5/6, \qquad \text{interval } (-5/6, 5/6).}$$

Conclusion. The ordinary ratio limit does not exist. Its extreme subsequences illustrate why the root limsup is the general formula.

23.5 Exercise 23.5: Two coefficient conditions forcing radius at most one

(a) Infinitely many nonzero integer coefficients.

A nonzero integer has absolute value at least 1. There are therefore arbitrarily large indices n with $|a_n|^{1/n} \geq 1$. Consequently $\limsup |a_n|^{1/n} \geq 1$, and the reciprocal-radius formula yields $R \leq 1$. If that limsup is infinite, the radius is zero, which also satisfies the conclusion.

(b) A positive limsup of coefficient magnitudes.

If $\limsup |a_n| > 0$ (including $+\infty$), choose $c > 0$ below this limsup. Infinitely many indices $n_j \to \infty$ satisfy $|a_{n_j}| > c$. At those indices $|a_{n_j}|^{1/n_j} \geq c^{1/n_j}$, and $c^{1/n_j} \to 1$. Thus $\limsup_n |a_n|^{1/n} \geq 1$ and again $R \leq 1$.

Conclusion. The coefficients need not stay large at every index. A single infinite subsequence bounded away from zero already suffices.

23.6 Exercise 23.6: Endpoint asymmetry and nonnegative coefficients

(a) Convergence at the positive endpoint implies convergence at the negative one.

Suppose $a_n \geq 0$ and the finite radius is $R \geq 0$. If $R > 0$ and $\sum a_n R^n$ converges, then

$$\sum |a_n(-R)^n| = \sum a_n R^n < \infty.$$

Hence the series at $-R$ converges absolutely. If $R = 0$, the two endpoints are the same point and the assertion is immediate. The sign hypothesis is doing the work: the positive-endpoint sum is already a sum of absolute values.

(b) Construct the opposite endpoint pattern.

Take

$$\sum_{n\geq 1} \frac{(-1)^{n+1}}{n} x^n.$$

Its coefficient-root limsup is 1, so its radius is 1. At $x = 1$ the alternating harmonic series converges. At $x = -1$ the series is $-\sum 1/n$, which diverges. Thus the exact interval is $\boxed{(-1, 1]}$. These coefficients do not satisfy the nonnegative hypothesis of part (a), so there is no contradiction.

Conclusion. Endpoint behavior depends on signs as well as magnitudes.

23.7 Exercise 23.7: Powers of cosine and pointwise limits

What we must show. Analyze $f_n(x) = (\cos x)^n$ for each fixed real x.

(a) Points that are not multiples of π.

At such a point $-1 < \cos x < 1$. Therefore $|\cos x|^n \to 0$, and $|f_n(x)| \to 0$ implies $f_n(x) \to 0$.

(b) Even multiples of π.

If $x = 2k\pi$ for an integer k, then $\cos x = 1$, so $f_n(x) = 1$ for every n and the limit is 1.

(c) Odd multiples of π.

If $x = (2k+1)\pi$, then $\cos x = -1$, so $f_n(x) = (-1)^n$. The even subsequence is constantly 1 and the odd subsequence is constantly -1. Since they have different limits, the full limit does not exist.

Conclusion. The three cases exhaust the real line. Continuity of every f_n does not even guarantee pointwise convergence at every point.

23.8 Exercise 23.8: Convergence of functions does not imply convergence of derivatives

What we must show. Compare $f_n(x) = \sin(nx)/n$ with its derivative.

(a) Bound the function values independently of x.

For all real x,

$$|f_n(x)| \le 1/n \longrightarrow 0.$$

The squeeze theorem gives $f_n(x) \to 0$ for every x. In fact this estimate proves uniform convergence, because the upper bound is the same for all x.

(b) Differentiate and test a specific point.

The chain rule gives $f_n'(x) = \cos(nx)$. At $x = \pi$, $f_n'(\pi) = (-1)^n$, so the derivatives have no pointwise limit there. Even at $x = 0$ they are always 1, whereas the derivative of the limit function $f \equiv 0$ is 0.

Conclusion. The operation of taking a derivative cannot be passed through a function limit without additional hypotheses. The oscillations get small in height while their slopes do not.

23.9 Exercise 23.9: A vanishing pointwise limit with nonvanishing integrals

What we must show. Analyze $f_n(x) = nx^n$ on $[0, 1]$.

(a) Fix a point strictly below one.

At $x = 0$ all values are zero. If $0 < x < 1$, set $u_n = nx^n > 0$. Its successive ratio satisfies $u_{n+1}/u_n = (1 + 1/n)x \to x < 1$. Choose q with $x < q < 1$. For some N, $u_{n+1} \le qu_n$ for all $n \ge N$. Repeatedly applying this inequality gives $u_{N+j} \le u_N q^j \to 0$. Thus $f_n(x) \to 0$ for every $x \in [0, 1)$. At $x = 1$, however, $f_n(1) = n \to +\infty$, so no finite limit exists there.

(b) Evaluate each integral before taking the limit.

For fixed n, the ordinary power rule for integration yields

$$\int_0^1 f_n(x)\,dx = n\left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{n}{n+1} = 1 - \frac{1}{n+1} \longrightarrow 1.$$

Conclusion. The functions approach zero at every point short of the right endpoint, but their integrals approach 1. Mass concentrates closer and closer to 1; pointwise information alone does not justify interchanging limit and integral.

24 Uniform convergence

Before starting. Pointwise convergence means fixing one input x first and then letting n grow. Uniform convergence means finding one index that works for *all* inputs at once. We shall repeatedly estimate the largest error

$$E_n = \sup_{x \in S} |f_n(x) - f(x)|.$$

Uniform convergence is equivalent to $E_n \to 0$; Exercise 24.12 proves this. To disprove uniform convergence, it is enough to find a positive number ε_0 and inputs $x_n \in S$, for arbitrarily large n, with $|f_n(x_n) - f(x_n)| \geq \varepsilon_0$. The input is allowed to change with n. That is precisely the obstacle uniform convergence must overcome.

24.1 Exercise 24.1: One bound for every oscillation

What we must show. Prove uniform convergence to zero for $f_n(x) = (1 + 2\cos^2(nx))/\sqrt{n}$ on $\mathbb{R}$.

Plan. We need not follow the oscillations: the cosine is always between -1 and 1.

Bound the numerator independently of the input.

Since $0 \leq \cos^2(nx) \leq 1$, we have

$$|f_n(x) - 0| \leq \frac{3}{\sqrt{n}} \qquad (x \in \mathbb{R}).$$

Given $\varepsilon > 0$, choose an integer $N > (3/\varepsilon)^2$. For every $n > N$ and every real x, the last bound is smaller than ε. The choice of N depends only on ε.

Conclusion. $f_n \to 0$ uniformly on $\mathbb{R}$.

24.2 Exercise 24.2: A small slope on bounded and unbounded domains

What we must show. Analyze $f_n(x) = x/n$ on $[0, \infty)$.

(a) Fix the input before passing to the limit.

For fixed $x \geq 0$, the numerator is constant and $n \to \infty$, so $f_n(x) \to 0$. Thus the pointwise limit is $f(x) = 0$.

(b) Control all inputs in $[0, 1]$.

There $|f_n(x)| \leq 1/n$. Given $\varepsilon > 0$, any integer $N > 1/\varepsilon$ works for all $x \in [0, 1]$ and $n > N$.

(c) Let the input escape along with the index.

On $[0, \infty)$ take $x_n = n$. Then $f_n(x_n) = 1$, so the error never falls below $1/2$. This contradicts the requirement of uniform convergence.

Conclusion. The limit is zero everywhere; convergence is uniform on $[0, 1]$ but not on $[0, \infty)$.

24.3 Exercise 24.3: A transition concentrated near one

What we must show. Analyze $f_n(x) = 1/(1 + x^n)$ for $x \geq 0$.

(a) Separate the three possible behaviors of a power.

If $0 \leq x < 1$, then $x^n \to 0$, giving limit 1. At $x = 1$ every term is $1/2$. If $x > 1$, then $x^n \to \infty$, giving limit 0. Thus

$$f(x) = \begin{cases} 1 & 0 \leq x < 1, \\ 1/2 & x = 1, \\ 0 & x > 1. \end{cases}$$

(b) Test uniform convergence on $[0, 1]$.

Each f_n is continuous there because its denominator is positive. The limit is not continuous at 1: its values from the left are 1, whereas $f(1) = 1/2$. The uniform-limit theorem for continuous functions therefore excludes uniform convergence on $[0, 1]$.

(c) Use restriction to a smaller domain.

Uniform convergence on $[0, \infty)$ would imply uniform convergence on its subset $[0, 1]$, with exactly the same error bound. Part (b) rules that out.

Conclusion. The piecewise function above is the pointwise limit, and neither of the two requested uniform-convergence assertions holds.

24.4 Exercise 24.4: The complementary transition

What we must show. Analyze $f_n(x) = x^n/(1 + x^n)$ on $[0, \infty)$.

(a) Use an exact identity.

We have $f_n(x) = 1 - 1/(1 + x^n)$. The three cases of the preceding exercise therefore give

$$f(x) = \begin{cases} 0 & 0 \leq x < 1, \\ 1/2 & x = 1, \\ 1 & x > 1. \end{cases}$$

(b) Notice the endpoint jump.

The f_n are continuous on $[0, 1]$, but f is not continuous at 1. Consequently convergence cannot be uniform there, by Theorem 24.3.

(c) Restrict any proposed larger-domain convergence.

Uniform convergence on $[0, \infty)$ would restrict to uniform convergence on $[0, 1]$, contradicting part (b).

Conclusion. The displayed f is the limit; convergence is uniform on neither $[0, 1]$ nor $[0, \infty)$.

24.5 Exercise 24.5: A denominator containing both the index and a power

What we must show. Analyze $f_n(x) = x^n/(n + x^n)$ for $x \geq 0$.

(a) Find the pointwise limit.

For $0 \leq x \leq 1$, we have $0 \leq f_n(x) \leq 1/n$, so the limit is zero. For fixed $x > 1$, divide numerator and denominator by x^n:

$$f_n(x) = \frac{1}{1 + n/x^n} \longrightarrow 1.$$

Here $n/x^n \to 0$ because exponential growth dominates the linear factor; for example, the ratio of successive terms tends to $1/x < 1$. Thus $f(x) = 0$ on $[0, 1]$ and $f(x) = 1$ on $(1, \infty)$.

(b) Obtain a common bound on $[0, 1]$.

The estimate $|f_n(x) - f(x)| \leq 1/n$ works simultaneously for every input in that interval. Hence convergence is uniform there.

(c) Use the jump at an interior point of the larger domain.

Every f_n is continuous on $[0, \infty)$, but the pointwise limit jumps at 1. The uniform-limit theorem excludes uniform convergence on the larger domain.

Conclusion. Uniform convergence holds on $[0, 1]$, and fails on $[0, \infty)$.

24.6 Exercise 24.6: Shifted squares

What we must show. Find the limit of $f_n(x) = (x - 1/n)^2$ on $[0, 1]$ and decide uniform convergence.

(a) Expand to identify the limit.

For fixed x, $f_n(x) = x^2 - 2x/n + 1/n^2 \to x^2$. Thus $f(x) = x^2$.

(b) Bound the entire error, rather than just each separate input.

For $0 \leq x \leq 1$,

$$|f_n(x) - x^2| = \left| -\frac{2x}{n} + \frac{1}{n^2} \right| \leq \frac{2}{n} + \frac{1}{n^2} \leq \frac{3}{n}.$$

Given $\varepsilon > 0$, choose $N > 3/\varepsilon$. This bound is then less than ε for every $x \in [0, 1]$ and $n > N$.

Conclusion. $f_n \to (x \mapsto x^2)$ uniformly on $[0, 1]$.

24.7 Exercise 24.7: Subtracting a power

What we must show. Analyze $f_n(x) = x - x^n$ on $[0, 1]$.

(a) Check the endpoint separately.

For $0 \leq x < 1$, $x^n \to 0$, so the limit is x. At $x = 1$, $f_n(1) = 0$ for every n. Hence

$$f(x) = \begin{cases} x & 0 \leq x < 1, \\ 0 & x = 1. \end{cases}$$

(b) Apply the continuity obstruction.

Each f_n is a polynomial and therefore continuous on $[0,1]$. The limit is discontinuous at 1, because its left-hand limit is 1 and its value there is zero. By Theorem 24.3 the convergence is not uniform.

Conclusion. There is pointwise convergence to the displayed function, but no uniform convergence on $[0,1]$.

24.8 Exercise 24.8: Partial geometric sums at their divergent endpoint

What we must show. Analyze $f_n(x) = \sum_{k=0}^{n} x^k$ on $[0,1]$.

(a) Evaluate the finite sum where its formula is valid.

For $0 \le x < 1$,

$$f_n(x) = \frac{1 - x^{n+1}}{1 - x} \longrightarrow \frac{1}{1 - x}.$$

At $x = 1$ there are $n + 1$ summands equal to 1, so $f_n(1) = n + 1 \to +\infty$. Thus there is *no real-valued pointwise limit on all of* $[0,1]$. An infinite limit at a point is not a real-valued limit function in Definition 24.1.

(b) Uniform convergence would imply pointwise convergence.

Since real-valued pointwise convergence already fails at 1, uniform convergence on $[0,1]$ is impossible.

Conclusion. Convergence occurs pointwise only on $[0,1)$, to $1/(1-x)$; it does not occur pointwise, or uniformly, on the full requested interval.

24.9 Exercise 24.9: A moving peak whose integral vanishes

What we must show. Study $f_n(x) = nx^n(1-x)$ on $[0,1]$.

(a) Fix an input.

At $x = 0$ and $x = 1$ the function is zero. For $0 < x < 1$, the ratio of successive terms of nx^n is $x(1 + 1/n) \to x < 1$, so $nx^n \to 0$. Multiplication by the fixed number $1 - x$ gives $f_n(x) \to 0$. Thus $f = 0$ everywhere on $[0,1]$.

(b) Find a peak that does not shrink to zero.

Differentiating gives

$$f_n'(x) = nx^{n-1}(n - (n+1)x).$$

The sign changes from positive to negative at $x_n = n/(n+1)$, the maximum point. Its value is

$$f_n(x_n) = \frac{n}{n+1}\left(\frac{n}{n+1}\right)^n = \left(\frac{n}{n+1}\right)^{n+1} \longrightarrow e^{-1} > 0.$$

The last limit follows from $(1 + 1/n)^n \to e$. Because the maximum error does not approach zero, convergence is not uniform.

(c) Compute the integrals directly.

Despite the preceding failure,

$$\int_0^1 f_n(x)\,dx = n\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{n}{(n+1)(n+2)} \longrightarrow 0 = \int_0^1 f(x)\,dx.$$

Conclusion. The limit is zero, convergence is not uniform, and the integrals nevertheless converge to the integral of the limit. Uniform convergence is a sufficient condition for this interchange, not a necessary one.

24.10 Exercise 24.10: Adding uniform limits, and the danger with products

What we must show. Prove the sum rule and examine its product analogue.

(a) Split the desired error into two smaller errors.

Let $\varepsilon > 0$. Uniform convergence supplies N_1, N_2 such that $|f_n(x) - f(x)| < \varepsilon/2$ for $n > N_1$ and $|g_n(x) - g(x)| < \varepsilon/2$ for $n > N_2$, in each case for all $x \in S$. For $n > \max\{N_1, N_2\}$ the triangle inequality gives

$$|(f_n + g_n)(x) - (f + g)(x)| \le |f_n(x) - f(x)| + |g_n(x) - g(x)| < \varepsilon.$$

This is uniform convergence of the sums.

(b) Test the product claim before accepting it.

Products need not converge uniformly: on $S = \mathbb{R}$, let $f_n(x) = x$ and $g_n(x) = 1/n$. Their limits are $f(x) = x$ and $g(x) = 0$, both uniform. But $f_n(x)g_n(x) = x/n$ has value 1 at $x = n$, so it fails to converge uniformly to $fg = 0$. A small error multiplied by an unbounded function can become large; this is the missing issue in a naive product proof.

Conclusion. Sums preserve uniform convergence; unrestricted products do not.

24.11 Exercise 24.11: The product counterexample in full

What we must show. Verify every part of the example $f_n(x) = x$, $g_n(x) = 1/n$ on $\mathbb{R}$.

(a) Compute the two errors separately.

With $f(x) = x$ we have $|f_n(x) - f(x)| = 0$ for every n, x. With $g(x) = 0$ we have $|g_n(x) - g(x)| = 1/n$, independently of x. The first error is already zero, and the second tends to zero, so both convergences are uniform.

(b) Compute the product error at a suitable input.

For each n, use $x_n = n$:

$$|f_n(x_n)g_n(x_n) - f(x_n)g(x_n)| = |n/n - 0| = 1.$$

No common index can make all these errors smaller than $1/2$.

Conclusion. $f_n \to f$ and $g_n \to g$ uniformly, but $f_n g_n \not\to fg$ uniformly on $\mathbb{R}$.

24.12 Exercise 24.12: The supremum-error characterization

What we must show. Prove that $f_n \to f$ uniformly on S if and only if $E_n = \sup_{x \in S} |f_n(x) - f(x)| \to 0$. For an empty domain the assertion is vacuous; take its maximum error to be zero. For a nonempty domain, E_n is initially allowed to be $+\infty$.

Uniform convergence implies vanishing suprema.

Given $\varepsilon > 0$, use uniform convergence with $\varepsilon/2$. For sufficiently large n, all the errors are smaller than $\varepsilon/2$, so their supremum satisfies $0 \le E_n \le \varepsilon/2 < \varepsilon$. Notice the non-strict inequality at the supremum: a least upper bound need not be attained. The extra factor $1/2$ avoids any difficulty.

Vanishing suprema imply uniform convergence.

Given $\varepsilon > 0$, choose N with $E_n < \varepsilon$ for all $n > N$. Every individual error is at most its supremum, so $|f_n(x) - f(x)| \le E_n < \varepsilon$ for all $x \in S$ and $n > N$. This is exactly the definition of uniform convergence.

Conclusion. The two conditions are equivalent. The supremum is over the input variable first; the limit is then taken over the index.

24.13 Exercise 24.13: Uniform limits of uniformly continuous functions

What we must show. Show that a uniform limit f of uniformly continuous f_n on (a, b) is uniformly continuous there.

Plan. Approximate f by one fixed f_n, and let that function control nearby inputs. There will be three errors, each allocated $\varepsilon/3$.

Choose one approximating function before choosing the inputs.

Given $\varepsilon > 0$, uniform convergence supplies an index m with $|f_m(t) - f(t)| < \varepsilon/3$ for all $t \in (a, b)$. Uniform continuity of this particular f_m supplies $\delta > 0$ such that $|f_m(x) - f_m(y)| < \varepsilon/3$ whenever $x, y \in (a, b)$ and $|x - y| < \delta$.

Join the three estimates.

For such x, y,

$$|f(x) - f(y)| \le |f(x) - f_m(x)| + |f_m(x) - f_m(y)| + |f_m(y) - f(y)| < \varepsilon.$$

Neither m nor δ was chosen using x or y, so this is uniform continuity, rather than merely continuity at a single point.

Conclusion. The uniform limit is uniformly continuous. The same proof works on any subset of $\mathbb{R}$, not just an interval.

24.14 Exercise 24.14: A narrow peak near zero

What we must show. Analyze $f_n(x) = nx/(1 + n^2x^2)$ and the proposed limit $f = 0$.

(a) Fix x and distinguish zero from nonzero.

At $x = 0$ every term is zero. If $x \neq 0$, dividing by n^2x^2 gives

$$f_n(x) = \frac{1/(nx)}{1 + 1/(n^2x^2)} \longrightarrow 0.$$

Thus pointwise convergence holds on $\mathbb{R}$.

(b) Test the shrinking input $x_n = 1/n$ on $[0, 1]$.

At this input, $f_n(x_n) = 1/2$. Therefore uniform convergence to zero fails on $[0, 1]$. Each fixed positive input eventually lies past the peak, but the peak itself remains of height $1/2$.

(c) Keep a fixed positive distance from zero.

For $x \geq 1$,

$$0 \leq \frac{nx}{1 + n^2x^2} \leq \frac{nx}{n^2x^2} = \frac{1}{nx} \leq \frac{1}{n}.$$

The upper bound tends to zero independently of x, proving uniform convergence on $[1, \infty)$.

Conclusion. Pointwise convergence holds everywhere; uniform convergence fails on $[0, 1]$ and holds on $[1, \infty)$.

24.15 Exercise 24.15: A boundary layer with limiting height one

What we must show. Analyze $f_n(x) = nx/(1 + nx)$ on $[0, \infty)$.

(a) Identify the limit.

At zero the value is zero. If $x > 0$, then $f_n(x) = 1 - 1/(1 + nx) \to 1$. Thus $f(0) = 0$ and $f(x) = 1$ for $x > 0$.

(b) Inspect the interval containing the boundary layer.

The continuous f_n have a discontinuous pointwise limit at zero, so convergence cannot be uniform on $[0, 1]$. Explicitly, $x_n = 1/n$ gives $f_n(x_n) = 1/2$ and $f(x_n) = 1$, an error of $1/2$.

(c) Bound the error on $[1, \infty)$.

For $x \geq 1$ the limit is 1, and

$$|f_n(x) - 1| = \frac{1}{1 + nx} \leq \frac{1}{1 + n} \longrightarrow 0.$$

This is an input-independent bound and hence proves uniform convergence.

Conclusion. The limit jumps from 0 to 1 at zero; convergence is not uniform on $[0, 1]$ but is uniform on $[1, \infty)$.

24.16 Exercise 24.16: A boundary layer with an unbounded limiting function

What we must show. Analyze $f_n(x) = nx/(1 + nx^2)$ for $x \geq 0$.

(a) Compute the pointwise limit.

At zero the value is zero. For $x > 0$, division by n gives $f_n(x) = x/(1/n + x^2) \to 1/x$. Thus $f(0) = 0$ and $f(x) = 1/x$ for $x > 0$.

(b) Show failure of uniform convergence on $[0, 1]$.

One obstruction is the discontinuity, indeed unboundedness, of the limit near zero. More directly, at $x_n = 1/\sqrt{n}$,

$$f_n(x_n) = \frac{\sqrt{n}}{2}, \qquad f(x_n) = \sqrt{n}, \qquad |f_n(x_n) - f(x_n)| = \frac{\sqrt{n}}{2}.$$

These errors grow, so they certainly do not tend uniformly to zero.

(c) Estimate the error away from the singular endpoint.

For $x \geq 1$,

$$\left| \frac{nx}{1 + nx^2} - \frac{1}{x} \right| = \frac{1}{x(1 + nx^2)} \leq \frac{1}{1 + n} \longrightarrow 0.$$

Conclusion. The displayed piecewise f is the limit; convergence fails to be uniform on $[0, 1]$ and is uniform on $[1, \infty)$.

24.17 Exercise 24.17: Passing to the limit while the input also moves

What we must show. If $f_n \to f$ uniformly on $[a, b]$, the f_n are continuous, and $x_n \to x$ in that interval, prove $f_n(x_n) \to f(x)$.

First secure continuity of the limiting function.

By Theorem 24.3, the uniform limit f is continuous on $[a, b]$. Also $x \in [a, b]$, since this interval is closed. Thus continuity gives $f(x_n) \to f(x)$.

Separate the changing function from the changing input.

For every n,

$$|f_n(x_n) - f(x)| \leq |f_n(x_n) - f(x_n)| + |f(x_n) - f(x)|.$$

Given $\varepsilon > 0$, uniform convergence makes the first term smaller than $\varepsilon/2$ for all sufficiently large n, *even at the moving input* x_n. Continuity and $x_n \to x$ make the second term smaller than $\varepsilon/2$ for all sufficiently large n. Taking n beyond both thresholds proves the desired limit.

Conclusion. $f_n(x_n) \to f(x)$. Uniform convergence is what allows us to substitute a different input at every index.

25 More on uniform convergence

Three useful tools. A uniform error bound can be integrated on a finite interval: $|\int_a^b (f_n - f)| \le (b-a)\sup|f_n - f|$. A series of functions converges uniformly precisely when its partial sums do. Finally, the Weierstrass M-test says that $|g_k(x)| \le M_k$ for all inputs, together with $\sum M_k < \infty$, guarantees uniform convergence of $\sum g_k$. If all summands are continuous, its sum is continuous too. An M-test bound must not depend on the input x.

25.1 Exercise 25.1: Bounding an integral by the integral of an absolute value

What we must show. Deduce $|\int_a^b g| \le \int_a^b |g|$ from monotonicity of the integral. The integrals in the statement are assumed to exist.

Place the function between two integrable bounds.

For every x, $-|g(x)| \le g(x) \le |g(x)|$. Applying monotonicity first to the lower pair and then to the upper pair gives

$$-\int_a^b |g(x)|\,dx \le \int_a^b g(x)\,dx \le \int_a^b |g(x)|\,dx.$$

Write the middle integral as A and the nonnegative right-hand integral as B. The statement $-B \le A \le B$ is exactly the statement $|A| \le B$.

Conclusion. $\left|\int_a^b g(x)\,dx\right| \le \int_a^b |g(x)|\,dx.$

25.2 Exercise 25.2: Powers divided by their indices

What we must show. Show uniform convergence of $f_n(x) = x^n/n$ on $[-1,1]$.

Find a common error bound.

Since $|x| \le 1$, $|x^n/n| \le 1/n$ for every input in this interval. Therefore the pointwise limit is $f(x) = 0$. Given $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. Then $n > N$ implies $|f_n(x) - 0| \le 1/n < \varepsilon$ for *all* $x \in [-1,1]$ simultaneously.

Conclusion. The convergence is uniform and the limit function is identically zero.

25.3 Exercise 25.3: A rational trigonometric expression and its integral

What we must show. Study $f_n(x) = (n + \cos x)/(2n + \sin^2 x)$ on $\mathbb{R}$.

(a) Guess the limit, then calculate its exact error.

Dividing by n suggests the limit $1/2$. Subtract it with a common denominator:

$$f_n(x) - \frac{1}{2} = \frac{2\cos x - \sin^2 x}{2(2n + \sin^2 x)}.$$

Since $|2\cos x - \sin^2 x| \le 2 + 1 = 3$ and the denominator is at least $4n$,

$$\left|f_n(x) - \frac{1}{2}\right| \le \frac{3}{4n} \qquad (x \in \mathbb{R}).$$

The right-hand side tends to zero independently of x, proving uniform convergence.

(b) Apply the integral theorem after checking its hypotheses.

The denominator is positive, so every f_n is continuous on $[2, 7]$. The convergence is uniform on that interval by part (a). Theorem 25.2 therefore gives

$$\lim_{n\to\infty} \int_2^7 f_n(x)\,dx = \int_2^7 \frac{1}{2}\,dx = \frac{5}{2}.$$

Conclusion. The uniform limit on $\mathbb{R}$ is $1/2$, and the integral limit is $5/2$.

25.4 Exercise 25.4: Uniform convergence implies the uniform Cauchy property

What we must show. Show that $f_n \to f$ uniformly on S makes (f_n) uniformly Cauchy.

Use the same approximating limit for two indices.

Given $\varepsilon > 0$, choose N so that $|f_k(x) - f(x)| < \varepsilon/2$ whenever $k > N$ and $x \in S$. For *both* $m > N$ and $n > N$, the triangle inequality then gives

$$|f_n(x) - f_m(x)| \le |f_n(x) - f(x)| + |f(x) - f_m(x)| < \varepsilon \qquad (x \in S).$$

The same N works for every pair of sufficiently large indices and every input, which is exactly uniform Cauchyness.

Conclusion. Every uniformly convergent sequence is uniformly Cauchy.

25.5 Exercise 25.5: A uniform limit of bounded functions is bounded

What we must show. If all f_n are bounded on S and $f_n \to f$ uniformly, prove that f is bounded.

Plan. The bounds of all the f_n need not be the same. One suitably chosen approximating function is enough.

Choose one function within distance one of the limit.

Uniform convergence gives an index m such that $|f(x) - f_m(x)| < 1$ for all $x \in S$. Since f_m is bounded, some finite M satisfies $|f_m(x)| \le M$ for every $x \in S$.

Add the approximation error to that fixed bound.

For every input,

$$|f(x)| \le |f(x) - f_m(x)| + |f_m(x)| < 1 + M.$$

This single finite bound establishes boundedness of f.

Conclusion. f is bounded; a bound is $M + 1$ for the chosen approximant.

25.6 Exercise 25.6: Absolute summability gives continuity at the endpoints too

What we must show. Relate $\sum |a_k| < \infty$ to the power series on $[-1, 1]$.

(a) Apply the M-test on the full closed interval.

For every $x \in [-1,1]$, $|a_k x^k| \le |a_k|$. The numerical majorant series $\sum |a_k|$ converges by assumption. The M-test therefore gives uniform convergence of $\sum a_k x^k$ on $[-1,1]$. Every partial sum is a polynomial and hence continuous. Its uniform limit is continuous on $[-1,1]$, including the endpoints with the relative notion of continuity.

(b) Check the coefficient condition in the given example.

For $a_n = 1/n^2$, $n \ge 1$, the numerical series $\sum 1/n^2$ converges by the p-series test, since $2 > 1$. Part (a) therefore applies directly to $\sum_{n=1}^{\infty} x^n/n^2$.

Conclusion. Both the general series and the specified example have continuous sums on the whole closed interval $[-1,1]$.

25.7 Exercise 25.7: A uniformly convergent trigonometric series

What we must show. Prove that $\sum_{n=1}^{\infty} \cos(nx)/n^2$ defines a continuous function on $\mathbb{R}$.

Bound every summand on the unbounded domain.

The elementary estimate $|\cos(nx)| \le 1$ gives $|\cos(nx)/n^2| \le 1/n^2$ for all real x. Since $\sum 1/n^2$ converges, the M-test proves uniform convergence on $\mathbb{R}$.

Transfer continuity from finite sums to the limit.

Each summand is continuous, so every partial sum is continuous. Theorem 24.3 applied to this uniformly convergent sequence of partial sums makes the sum continuous on $\mathbb{R}$.

Conclusion. The series converges uniformly on $\mathbb{R}$ to a continuous function.

25.8 Exercise 25.8: A power series uniform on its entire convergence interval

What we must show. Find the radius and endpoint behavior of $\sum_{n=1}^{\infty} x^n/(n^2 2^n)$.

Compute the radius using coefficient roots.

With $a_n = 1/(n^2 2^n)$,

$$|a_n|^{1/n} = \frac{1}{2n^{2/n}} \longrightarrow \frac{1}{2},$$

since $n^{1/n} \to 1$. Thus the radius is $R = 1/(1/2) = 2$.

Bound the terms at all inputs, including the two endpoints.

For $|x| \le 2$,

$$\left|\frac{x^n}{n^2 2^n}\right| \le \frac{1}{n^2}.$$

The M-test proves uniform convergence on $[-2,2]$, not merely on smaller intervals. The summands are polynomials, so the sum is continuous there.

Conclusion. The radius is 2, and the series converges uniformly to a continuous function on its full interval of convergence $[-2,2]$.

25.9 Exercise 25.9: Geometric series on a smaller and a maximal open interval

What we must show. Compare uniform convergence of $\sum_{n=0}^{\infty} x^n$ on $[-a, a]$ and $(-1, 1)$, where $0 < a < 1$.

(a) Estimate the geometric remainder on the smaller interval.

For $|x| < 1$, the sum is $1/(1-x)$ and the partial sum through N has remainder

$$\frac{1}{1-x} - \sum_{n=0}^{N} x^n = \frac{x^{N+1}}{1-x}.$$

For $|x| \le a$, its absolute value is at most $a^{N+1}/(1-a)$, which tends to zero because $0 < a < 1$. The estimate is uniform in x.

(b) Show that terms do not even tend uniformly to zero on $(-1, 1)$.

If the partial sums converged uniformly, their successive differences x^N would converge uniformly to zero: indeed each difference is bounded by the two uniform errors of the corresponding partial sums. But at $x_N = 2^{-1/N} \in (0, 1)$, we have $x_N^N = 1/2$ for every $N \ge 1$. Thus the necessary uniform vanishing of the terms fails.

Conclusion. Uniform convergence holds on every $[-a, a]$ with $a < 1$, and fails on $(-1, 1)$. Pointwise convergence on the latter interval still holds.

25.10 Exercise 25.10: A non-geometric series compared with a geometric one

What we must show. Analyze $\sum x^n/(1+x^n)$ on $[0, 1)$. The starting index may be taken as $n = 0$; its term is just the constant $1/2$ and does not affect any convergence decision below.

(a) Prove convergence for a fixed input.

For $0 \le x < 1$ and $n \ge 1$,

$$0 \le \frac{x^n}{1+x^n} \le x^n.$$

The majorant $\sum_{n=1}^{\infty} x^n$ converges, so comparison proves convergence of the given nonnegative series at every such fixed x.

(b) Obtain a majorant independent of the input in $[0, a]$.

If $0 \le x \le a < 1$, then $x^n/(1+x^n) \le a^n$. Since $\sum a^n$ converges, the M-test proves uniform convergence on $[0, a]$.

(c) Test the necessary uniform vanishing of the summands.

Take $x_n = 2^{-1/n} \in (0, 1)$. Then $x_n^n = 1/2$ and

$$\frac{x_n^n}{1+x_n^n} = \frac{1/2}{3/2} = \frac{1}{3}.$$

Thus the terms do not tend uniformly to zero on $[0, 1)$. Uniform convergence of the series would force that property by subtracting successive partial sums, as in Exercise 25.9. Hence uniform convergence fails.

Conclusion. There is pointwise convergence on $[0,1)$ and uniform convergence on every $[0,a]$ with $a<1$, but no uniform convergence on $[0,1)$.

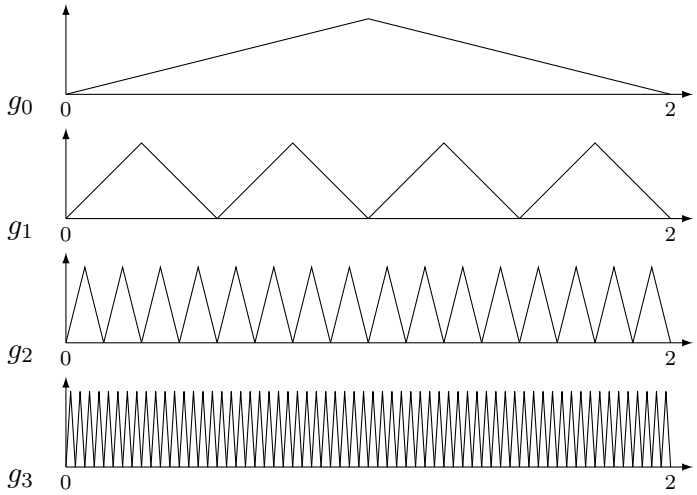
25.11 Exercise 25.11: Four increasingly compressed triangular waves

What we must show. Sketch the $g_n(x) = g(4^n x)$ from Example 3 and prove continuity of $f(x) = \sum_{n=0}^{\infty}(3/4)^n g_n(x)$.

(a) Describe the graphs by exact vertices.

The base function g is the triangular wave of period 2: for every integer j, $g(2j)=0$, $g(2j+1)=1$, and its graph is straight between consecutive integer inputs. Thus g_n has zeros at $2j/4^n$, peaks of height 1 at $(2j+1)/4^n$, and period $2/4^n$. Multiplication of the input compresses the horizontal scale; it does not change the heights.

The sketches below show $0 \le x \le 2$, with the same horizontal and vertical scales in all four rows; repeat each graph periodically in both directions. For g_3, the closely spaced peaks are intentional.



(b) Bound the vertically weighted waves.

The triangular wave is continuous, with $0 \le g \le 1$. Thus every g_n is continuous as the composition of g with the continuous map $x \mapsto 4^n x$, and

$$\left|\left(\frac{3}{4}\right)^n g_n(x)\right| \le \left(\frac{3}{4}\right)^n \qquad (x \in \mathbb{R}).$$

The numerical geometric series $\sum_{n=0}^{\infty}(3/4)^n$ converges. The M-test gives uniform convergence on $\mathbb{R}$, and the uniform-limit theorem gives continuity of f on $\mathbb{R}$.

Conclusion. The graphs have periods 2, 1/2, 1/8, 1/32 and height 1. The weighted infinite sum is continuous everywhere. No differentiability claim is needed for this exercise.

25.12 Exercise 25.12: Integrating a uniformly convergent series

What we must show. If continuous g_k have a uniformly convergent sum g on $[a,b]$, prove $\int_a^b g = \sum_{k=1}^{\infty}\int_a^b g_k$.

Apply the known theorem to partial sums, not directly to infinite sums.

Set $s_n(x) = \sum_{k=1}^{n} g_k(x)$. Each s_n is continuous, and the hypothesis means $s_n \to g$ uniformly. Theorem 25.2 therefore gives $\int_a^b s_n \to \int_a^b g$.

Use linearity only for the finite sum, then take its limit.

For every n,

$$\int_a^b s_n(x)\,dx = \sum_{k=1}^{n} \int_a^b g_k(x)\,dx.$$

Consequently the numerical series on the right converges, and its sum is $\int_a^b g(x)\,dx$ by the definition of an infinite series as a limit of finite partial sums.

Conclusion. Term-by-term integration is valid under the stated uniform convergence and continuity assumptions.

25.13 Exercise 25.13: Adding two uniformly convergent series

What we must show. Show that $\sum(g_k + h_k)$ converges uniformly if $\sum g_k$ and $\sum h_k$ do.

Name the two limits and their partial sums.

Write $G_n = \sum_{k=1}^{n} g_k \to G$ uniformly and $H_n = \sum_{k=1}^{n} h_k \to H$ uniformly. The partial sum of the proposed series is exactly $G_n + H_n$.

Estimate the error of that partial sum.

For every input x,

$$|(G_n + H_n)(x) - (G + H)(x)| \le |G_n(x) - G(x)| + |H_n(x) - H(x)|.$$

Given $\varepsilon > 0$, make each error smaller than $\varepsilon/2$ uniformly, and take the larger of the two index thresholds. This gives uniform convergence to $G + H$.

Conclusion. The sum series is uniformly convergent, with sum equal to the sum of the two original sum functions.

25.14 Exercise 25.14: Multiplying a series by one bounded function

What we must show. If $\sum g_k$ converges uniformly on S and h is bounded there, prove uniform convergence of $\sum hg_k$.

Write the new partial sums as a product with the old ones.

Let $G_n = \sum_{k=1}^{n} g_k \to G$ uniformly and choose a finite $M \ge 0$ with $|h(x)| \le M$ for all $x \in S$. Then

$$\sum_{k=1}^{n} h(x)g_k(x) = h(x)G_n(x), \qquad |h(x)G_n(x) - h(x)G(x)| \le M|G_n(x) - G(x)|.$$

Choose the uniform error tolerance after accounting for the bound.

If $M = 0$, every new term is zero and there is nothing to prove. If $M > 0$, choose N so that $|G_n(x) - G(x)| < \varepsilon/M$ for every $x \in S$ and $n > N$. The displayed inequality then bounds the new error by ε, uniformly in x.

Conclusion. The new series converges uniformly to hG. No continuity of h is required, only its boundedness.

25.15 Exercise 25.15: Monotone convergence on a compact interval: Dini's theorem

What we must show. Prove both monotone versions of the uniform-convergence assertion.

(a) First note that decreasing terms converging to zero are nonnegative.

For each fixed x, $f_n(x) \geq \lim_m f_m(x) = 0$. We must show that the nonnegative errors $f_n(x)$ become small at one common index.

Use continuity to turn each strict error bound into an open set.

Fix $\varepsilon > 0$, and let

$$U_n = \{x \in [a, b] : f_n(x) < \varepsilon\}.$$

Continuity makes U_n relatively open in $[a, b]$. Since $f_{n+1} \leq f_n$, these sets increase: $U_n \subseteq U_{n+1}$. Pointwise convergence to zero implies every x lies in some U_n, so $\{U_n : n \geq 1\}$ is an open cover of $[a, b]$.

Compactness turns the varying pointwise indices into one common index.

The closed bounded interval is compact, so finitely many of the U_n cover it. Let N be the largest of their indices. Because the sets are nested, U_N alone covers the whole interval. Thus $0 \leq f_n(x) \leq f_N(x) < \varepsilon$ for all $x \in [a, b]$ and $n \geq N$. This is uniform convergence to zero.

(b) Reduce increasing convergence to the decreasing case.

Suppose now $f_n(x)$ increases to a continuous function $f(x)$. Set $h_n = f - f_n$. Each h_n is continuous because both f and f_n are. At every input h_n decreases to zero, and $h_n \geq 0$. Part (a) gives $h_n \to 0$ uniformly, which is exactly $f_n \to f$ uniformly.

Conclusion. Both claims hold. Compactness and continuity of the limiting function are essential ingredients in turning the pointwise monotonicity into a uniform bound.

26 Differentiation and integration of power series

The permission comes before the calculation. Inside the radius of convergence of a power series, we may differentiate term by term, and we may integrate term by term from zero to any interior point. Uniform convergence on each smaller closed interval is the reason integration works. Endpoints require a separate argument: convergence inside an interval does not by itself settle either endpoint. We use Abel's theorem only after checking that the numerical endpoint series converges.

26.1 Exercise 26.1: Term-by-term integration at a positive input

What we must show. If $f(t) = \sum_{k=0}^{\infty} a_k t^k$ has radius $R > 0$, prove $\int_0^x f(t)\,dt = \sum_{k=0}^{\infty} a_k x^{k+1}/(k+1)$ when $0 < x < R$.

Work on an interval lying strictly inside the radius.

Fix such an x, and put $s_n(t) = \sum_{k=0}^{n} a_k t^k$. The power series converges uniformly on $[-x, x]$ by Theorem 26.1, and therefore also on $[0, x]$. Each s_n is a continuous polynomial.

Pass the limit through the integral using the established theorem.

Theorem 25.2 yields

$$\int_0^x f(t)\,dt = \lim_{n\to\infty} \int_0^x s_n(t)\,dt.$$

For a finite sum, linearity and the power rule give

$$\int_0^x s_n(t)\,dt = \sum_{k=0}^{n} a_k \int_0^x t^k\,dt = \sum_{k=0}^{n} \frac{a_k}{k+1} x^{k+1}.$$

Taking the limit of these finite sums gives precisely the claimed infinite series.

Conclusion. The term-by-term integration formula holds for every positive $x < R$. The strict inequality $x < R$ is what secured the uniform convergence used in the proof.

26.2 Exercise 26.2: A differentiated geometric series and three numerical sums

What we must show. Derive $\sum_{n=1}^{\infty} nx^n$ and use it to evaluate the requested sums.

(a) Differentiate first, then restore the missing power of x.

For $|x| < 1$, the geometric series is $\sum_{n=0}^{\infty} x^n = 1/(1-x)$. Its radius is 1, so Theorem 26.5 allows differentiation at every such x:

$$\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}.$$

Multiplying by x gives

$$\boxed{\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}} \qquad (|x| < 1).$$

(b) Substitute $x = 1/2$.

The value lies strictly inside the radius, so the formula gives

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = \frac{1/2}{(1-1/2)^2} = \frac{1/2}{1/4} = 2.$$

(c) Substitute the two signed inputs separately.

At $x = 1/3$,

$$\sum_{n=1}^{\infty} \frac{n}{3^n} = \frac{1/3}{(2/3)^2} = \frac{3}{4}.$$

At $x = -1/3$,

$$\sum_{n=1}^{\infty} \frac{(-1)^n n}{3^n} = \frac{-1/3}{(1+1/3)^2} = \frac{-1/3}{16/9} = -\frac{3}{16}.$$

Conclusion. The three numerical sums are 2, 3/4, and $-3/16$.

26.3 Exercise 26.3: One more derivative gives squared indices

What we must show. Find $\sum_{n=1}^{\infty} n^2 x^n$ and evaluate it at 1/2 and 1/3.

(a) Differentiate the identity from Exercise 26.2.

For $|x| < 1$, term-by-term differentiation is legitimate, because differentiation preserves the radius of a power series. Hence

$$\sum_{n=1}^{\infty} n^2 x^{n-1} = \frac{d}{dx}\left[\frac{x}{(1-x)^2}\right] = \frac{1}{(1-x)^2} + \frac{2x}{(1-x)^3} = \frac{1+x}{(1-x)^3}.$$

Multiplication by x gives

$$\boxed{\sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3}} \qquad (|x| < 1).$$

(b) Evaluate the two substitutions.

Both inputs are strictly inside the radius. At $x = 1/2$ the numerator is $(1/2)(3/2) = 3/4$ and the denominator is $(1/2)^3 = 1/8$, so

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = \frac{3/4}{1/8} = 6.$$

At $x = 1/3$ the numerator is $4/9$ and the denominator is $8/27$, so

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} = \frac{4/9}{8/27} = \frac{3}{2}.$$

Conclusion. The general formula is $x(1+x)/(1-x)^3$, and the requested values are 6 and 3/2.

26.4 Exercise 26.4: Integrating the Gaussian power series

What we must show. Express e^{-x^2} and $F(x) = \int_0^x e^{-t^2}\,dt$ as power series.

(a) Substitute into the exponential series.

The exponential identity, with its constant term included, is

$$e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!} \qquad (u \in \mathbb{R}).$$

Taking $u = -x^2$ gives

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^{2n} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots .$$

This substitution is valid for every real x because the original exponential series converges for every real input.

(b) Verify the radius before integrating.

For fixed x, the ratio of successive absolute terms is $x^2/(n+1) \to 0$, so this power series has infinite radius. Theorem 26.4 therefore applies between zero and every real x, whether positive or negative. It gives

$$\boxed{F(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)n!}} = x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \cdots .$$

The lower limit contributes zero for each monomial, so no integration constant has been omitted.

Conclusion. The two displayed expansions hold for all $x \in \mathbb{R}$.

26.5 Exercise 26.5: A function equal to its own derivative, without naming it

What we must show. For $f(x) = \sum_{n=0}^{\infty} x^n/n!$, prove $f' = f$ directly from the series.

Check that term-by-term differentiation is allowed everywhere.

For each fixed x, the ratio of successive absolute terms is $|x|/(n+1) \to 0$. Thus the radius is infinite, and Theorem 26.5 allows term-by-term differentiation at every real input.

Differentiate and carefully change the index.

The constant term differentiates to zero, and for $n \geq 1$, $n/n! = 1/(n-1)!$. Therefore

$$f'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = \sum_{j=0}^{\infty} \frac{x^j}{j!} = f(x).$$

In the last step we set $j = n-1$, so the first new index is zero.

Conclusion. $f' = f$ on $\mathbb{R}$. No identification of f with the exponential function was used.

26.6 Exercise 26.6: A trigonometric identity derived without trigonometry

What we must show. For

$$s(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \qquad c(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

prove the requested derivative relations and $s^2 + c^2 = 1$.

(a) Differentiate each series with its radius justified.

For fixed x, ratios of successive absolute terms tend to zero in both series, since their denominators acquire two factorial factors while their numerators acquire only a factor x^2. Thus both radii are infinite. Theorem 26.5 gives

$$s'(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = c(x).$$

For c, the $n = 0$ term differentiates to zero, so

$$\begin{aligned} c'(x) &= \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{(2n-1)!} \\ &= -\sum_{j=0}^{\infty} \frac{(-1)^j x^{2j+1}}{(2j+1)!} = -s(x). \end{aligned}$$

Here $j = n - 1$, and $(-1)^{j+1} = -(-1)^j$ explains the minus sign.

(b) Differentiate the sum of squares.

The product rule gives

$$(s^2 + c^2)' = 2ss' + 2cc' = 2sc + 2c(-s) = 0.$$

(c) Determine the constant from one input.

A differentiable function with derivative zero on an interval is constant there, by the Mean Value theorem. Apply this on the interval joining zero to any real x. Thus $s(x)^2 + c(x)^2 = s(0)^2 + c(0)^2$. The defining series give $s(0) = 0$ and $c(0) = 1$, so that constant is 1.

Conclusion. $s' = c$, $c' = -s$, and $s(x)^2 + c(x)^2 = 1$ for every real x, proved using the series alone.

26.7 Exercise 26.7: Why absolute value has no power-series representation at zero

What we must show. Decide whether $|x| = \sum_{n=0}^{\infty} a_n x^n$ can hold for all real x.

A power series with positive radius is differentiable at its center.

If such a representation held for every real input, its radius would be infinite, in particular positive. Theorem 26.5 would make its sum differentiable at zero.

Calculate the two difference quotients of absolute value.

Since $|0| = 0$,

$$\frac{|h| - |0|}{h} = \begin{cases} 1 & h > 0, \\ -1 & h < 0. \end{cases}$$

The right-hand and left-hand limits differ, so $|x|$ is not differentiable at zero. This contradicts the consequence of a power-series representation.

Conclusion. No such power series exists, even on any two-sided neighborhood of zero. On the separate half-lines the formulas are $|x| = x$ and $|x| = -x$, but these do not combine into one power series about zero. Polynomial approximation, discussed next, is a different and weaker requirement.

26.8 Exercise 26.8: The arctangent series and its endpoints

What we must show. Derive the arctangent series inside $(-1, 1)$ and settle both endpoints.

(a) Substitute into the geometric series.

For $|x| < 1$, put $y = -x^2$, so $|y| < 1$. Then

$$\sum_{n=0}^{\infty} (-1)^n x^{2n} = \sum_{n=0}^{\infty} y^n = \frac{1}{1-y} = \frac{1}{1+x^2}.$$

(b) Integrate on a strictly interior interval.

The power series has radius 1, so Theorem 26.4 gives, for $|x| < 1$,

$$\int_0^x \frac{dt}{1+t^2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}.$$

The integral on the left is $\arctan x - \arctan 0 = \arctan x$. Therefore

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots \qquad (-1 < x < 1).$$

(c) Establish convergence at 1 before using Abel's theorem.

At $x = 1$ the numerical series is $\sum_{n=0}^{\infty} (-1)^n/(2n+1)$. Its positive term sizes decrease to zero, so the alternating-series test proves convergence. Abel's theorem then identifies its sum with the limit of the interior sum as $x \uparrow 1$. Since arctangent is continuous,

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \lim_{x \uparrow 1} \arctan x = \frac{\pi}{4}.$$

Multiplication by 4 gives the familiar formula

$$\boxed{\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)}.$$

(d) Evaluate at the negative endpoint with its sign intact.

Every power in the series is odd. Substitution $x = -1$ therefore makes the series the negative of the convergent series in part (c). Its sum is $-\pi/4 = \arctan(-1)$. It converges conditionally at both endpoints: the series of absolute values $\sum 1/(2n+1)$ diverges by comparison with the harmonic series, since $1/(2n+1) \geq 1/(2n+2)$.

Conclusion. The arctangent expansion holds on $[-1, 1]$, with endpoint values $\pi/4$ and $-\pi/4$ justified separately, rather than by interior convergence alone.

27 Weierstrass's approximation theorem

Approximation is different from equality with a power series. A continuous function on a closed bounded interval can be approximated uniformly by polynomials, even when it has corners. The approximating polynomials need not be successive partial sums of a single power series. The Bernstein polynomials on $[0,1]$ are

$$B_n f(x) = \sum_{k=0}^{n} f(k/n)\binom{n}{k}x^k(1-x)^{n-k} \qquad (n \geq 1).$$

For $x \in [0,1]$, the weights $\binom{n}{k}x^k(1-x)^{n-k}$ are nonnegative and sum to 1. Thus $B_n f(x)$ is a weighted average of sampled values of f. Products with exponent zero are interpreted as 1, including at endpoints.

27.1 Exercise 27.1: Transferring approximation from $[0,1]$ to $[a,b]$

What we must show. Deduce the general Weierstrass approximation theorem from the Bernstein theorem on $[0,1]$.

Move the input interval to the standard one.

Assume first $a < b$. The affine function $\phi(t) = a + (b-a)t$ maps $[0,1]$ onto $[a,b]$, with inverse $\phi^{-1}(x) = (x-a)/(b-a)$. If f is continuous on $[a,b]$, then $g = f \circ \phi$ is continuous on $[0,1]$. Theorem 27.4 supplies polynomials $q_n = B_n g$ converging uniformly to g there.

Move the approximating polynomials back.

Define

$$p_n(x) = q_n\left(\frac{x-a}{b-a}\right).$$

A polynomial composed with an affine function is again a polynomial. For every $x \in [a,b]$, let $t = (x-a)/(b-a) \in [0,1]$. Then

$$|p_n(x) - f(x)| = |q_n(t) - g(t)|.$$

Because the substitution is a bijection between the intervals, the suprema of these errors on the two intervals are equal. The supremum on $[0,1]$ tends to zero, so the one on $[a,b]$ does too.

Include the degenerate interval if needed.

If $a = b$, the constant polynomial $p_n(x) = f(a)$ is already exact on the one-point interval.

Conclusion. Every continuous function on $[a,b]$ is a uniform limit of polynomials.

27.2 Exercise 27.2: One sequence that works on every bounded subset of $\mathbb{R}$

What we must show. For continuous f on $\mathbb{R}$, find polynomials p_n converging uniformly to f on each bounded subset.

Use a larger interval and a smaller tolerance at each stage.

For each integer $n \geq 1$, Weierstrass approximation on $[-n, n]$ gives one polynomial p_n with

$$|p_n(x) - f(x)| < \frac{1}{n} \qquad (|x| \leq n).$$

We choose one such polynomial for each n.

Verify convergence on an arbitrary bounded set.

Let $E \subseteq \mathbb{R}$ be bounded. Choose M so that $|x| \leq M$ for every $x \in E$. Given $\varepsilon > 0$, choose an integer N larger than both M and $1/\varepsilon$. For $n > N$, every $x \in E$ belongs to $[-n, n]$, so $|p_n(x) - f(x)| < 1/n < \varepsilon$.

Conclusion. The constructed sequence converges uniformly on every bounded subset of $\mathbb{R}$. The index threshold can depend on that bounded set; this does not assert uniform convergence on all of $\mathbb{R}$.

27.3 Exercise 27.3: Why the entire real line is too large

What we must show. Show that neither $\sin x$ nor e^x is a uniform limit of polynomials on $\mathbb{R}$.

(a) A polynomial uniformly close to sine would have to be constant.

Suppose $p_n \to \sin$ uniformly. Then for some index m, $|p_m(x) - \sin x| < 1/2$ for every real x. Consequently $|p_m(x)| \leq |\sin x| + 1/2 \leq 3/2$ everywhere. A nonconstant polynomial is unbounded on $\mathbb{R}$: its leading term dominates all terms of smaller degree as $|x| \to \infty$. Thus p_m must be a constant c. At $x = \pi/2$ and $x = 3\pi/2$, the proposed approximation requires $|c - 1| < 1/2$ and $|c + 1| < 1/2$. But the triangle inequality would then give $2 \leq |1 - c| + |c + 1| < 1$, an impossibility.

(b) No polynomial can stay within a fixed error of the exponential.

Let p be any polynomial. Choose an integer $d \geq 0$ at least its degree, and $C > 0$ such that $|p(x)| \leq Cx^d$ for $x \geq 1$; the sum of the absolute values of its coefficients, enlarged if necessary, supplies such a C. The exponential series has nonnegative terms at nonnegative inputs, so

$$e^x \geq \frac{x^{d+1}}{(d+1)!} \qquad (x \geq 0).$$

Therefore, for $x \geq 1$,

$$e^x - p(x) \geq \frac{x^{d+1}}{(d+1)!} - Cx^d = x^d \left(\frac{x}{(d+1)!} - C \right) \longrightarrow +\infty.$$

In particular, no polynomial has $|p(x) - e^x| < 1$ for all real x. Uniform convergence would eventually require exactly such a polynomial, so it is impossible.

Conclusion. Uniform polynomial approximation on each bounded interval does not extend to uniform approximation on the whole real line in either example.

27.4 Exercise 27.4: Correcting the two endpoint values

What we must show. Construct approximating polynomials with $p_n(a) = f(a)$ and $p_n(b) = f(b)$.

Start with any uniformly approximating sequence.

For $a < b$, choose polynomials $q_n \to f$ uniformly on $[a, b]$. Write $E_n = \sup_{x\in[a,b]} |q_n(x) - f(x)|$, so $E_n \to 0$. The desired corrections at the two endpoints are $A_n = f(a) - q_n(a)$ and $B_n = f(b) - q_n(b)$.

Interpolate those corrections with a straight line.

Define

$$s_n(x) = \frac{b-x}{b-a}A_n + \frac{x-a}{b-a}B_n, \qquad p_n(x) = q_n(x) + s_n(x).$$

The function s_n is affine, so p_n is a polynomial. At $x = a$ its correction is A_n, and at $x = b$ its correction is B_n. Thus $p_n(a) = f(a)$ and $p_n(b) = f(b)$ exactly.

Show that the correction stays small throughout the interval.

On $[a, b]$ the two coefficients in s_n are nonnegative and sum to 1. Also $|A_n|, |B_n| \le E_n$. Hence $|s_n(x)| \le E_n$ for all such x, and

$$|p_n(x) - f(x)| \le |q_n(x) - f(x)| + |s_n(x)| \le 2E_n \longrightarrow 0.$$

The bound is uniform in x. If $a = b$, the constant polynomial $f(a)$ settles the one-point case directly.

Conclusion. The endpoint-corrected polynomials retain uniform convergence.

27.5 Exercise 27.5: Bernstein polynomials for the first two powers

What we must show. Compute $B_n f$ when $f(x) = x$ and when $f(x) = x^2$.

(a) Use the first binomial moment.

For $f(t) = t$,

$$B_n f(x) = \frac{1}{n}\sum_{k=0}^{n} k\binom{n}{k}x^k(1-x)^{n-k}.$$

For $k \ge 1$, $k\binom{n}{k} = n\binom{n-1}{k-1}$. Substitution and the binomial theorem give

$$\sum_{k=0}^{n} k\binom{n}{k}x^k(1-x)^{n-k} = nx\sum_{j=0}^{n-1}\binom{n-1}{j}x^j(1-x)^{n-1-j} = nx.$$

Therefore $B_n f(x) = x$ for every $n \ge 1$: linear functions of this form are reproduced exactly.

(b) Decompose the square into two binomial moments.

For $f(t) = t^2$, use $k^2 = k(k-1) + k$. For $n \ge 2$, the factorial identity $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$ gives

$$\sum_{k=0}^{n} k(k-1)\binom{n}{k}x^k(1-x)^{n-k} = n(n-1)x^2.$$

The same identity is zero on both sides for $n = 1$. Adding the first moment nx and dividing by n^2 yields

$$B_n f(x) = \frac{n(n-1)x^2 + nx}{n^2} = x^2 + \frac{x(1-x)}{n}.$$

As a check, $0 \le x(1-x) \le 1/4$ on $[0, 1]$, so the error is at most $1/(4n)$, consistent with uniform convergence.

Conclusion. $B_n(t \mapsto t)(x) = x$ and $B_n(t \mapsto t^2)(x) = x^2 + x(1-x)/n$.

27.6 Exercise 27.6: A necessary condition for uniform Bernstein approximation

What we must show. If $B_n f \to f$ uniformly on $[0,1]$, prove that f is continuous, without initially assuming continuity of f.

Notice what kind of function each approximant is.

For each fixed n, the finitely many sampled values $f(k/n)$ are real constants. Thus the formula defining $B_n f$ is a finite linear combination of polynomials, so $B_n f$ is a polynomial and is continuous on $[0,1]$.

Use the uniform-limit theorem.

The assumption is uniform convergence of this sequence of continuous functions to f. Theorem 24.3 therefore makes f continuous on $[0,1]$.

Conclusion. Continuity is necessary for uniform convergence of the Bernstein polynomials to the original function. Together with Theorem 27.4, it is also sufficient.

27.7 Exercise 27.7: A Bernstein bound from nonnegative weights

What we must show. If $|f(t)| \le M$ for every $t \in [0,1]$, prove $|B_n f(x)| \le M$ for all $n \ge 1$ and $x \in [0,1]$.

Use the triangle inequality with signs of the weights checked.

For $0 \le x \le 1$, every weight $w_k = \binom{n}{k} x^k (1-x)^{n-k}$ is nonnegative. Also each sample point k/n belongs to $[0,1]$, so $|f(k/n)| \le M$. Hence

$$|B_n f(x)| \le \sum_{k=0}^{n} |f(k/n)| w_k \le M \sum_{k=0}^{n} w_k.$$

Evaluate the total weight.

By the binomial theorem, $\sum_{k=0}^{n} w_k = (x + (1-x))^n = 1$. The preceding bound is therefore M.

Conclusion. Every Bernstein polynomial obeys the same bound M as f on $[0,1]$. The interval matters: outside it the weights need not be nonnegative.

28 Basic properties of the derivative

A derivative measures a limiting slope, not merely a formal manipulation of a formula. At an interior point a,

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

The limit must be a finite real number and must agree from both sides. Differentiability implies continuity; the converse is false. The sum, product, quotient, and chain rules apply only after their hypotheses have been checked. At a point where a formula changes, return to this definition.

28.1 Exercise 28.1: Where absolute values create corners

What we must show. Find all points of nondifferentiability for the six functions.

Plan. Away from zeros of the expression inside an absolute value, its sign stays fixed nearby. Replace the absolute value by either that expression or its negative, and use ordinary differentiation. At a zero, compare the two one-sided slopes.

(a) The function $e^{|x|}$.

For $x > 0$ the function is e^x, and for $x < 0$ it is e^{-x}. Both formulas are differentiable on their respective open half-lines. At zero, the right derivative is 1, whereas the left derivative is -1. The only exceptional point is 0.

(b) The function $\sin|x|$.

On $x > 0$ this is $\sin x$, and on $x < 0$ it is $-\sin x$. The one-sided derivatives at zero are 1 and -1. Again the exceptional set is $\{0\}$.

(c) The function $|\sin x|$.

The possible corners are the zeros $k\pi$, $k \in \mathbb{Z}$. Near $a = k\pi$, write $x = a + h$. Then

$$\frac{|\sin(a+h)| - |\sin a|}{h} = \frac{|\sin h|}{h}.$$

As $h \to 0^+$ this tends to 1; as $h \to 0^-$ it tends to -1. All and only the points $k\pi$ are exceptional.

(d) The function $|x| + |x - 1|$.

Splitting at 0 and 1 gives

$$|x| + |x-1| = \begin{cases} 1 - 2x, & x < 0, \\ 1, & 0 \le x \le 1, \\ 2x - 1, & x > 1. \end{cases}$$

The slopes change from -2 to 0 at 0, and from 0 to 2 at 1. The exceptional set is $\{0, 1\}$.

(e) The function $|x^2 - 1|$.

It equals $x^2 - 1$ for $|x| \ge 1$ and $1 - x^2$ for $|x| < 1$. At -1 the slopes are -2 from the left and 2 from the right; at 1 they are -2 from the left and 2 from the right. Thus the exceptional set is $\{-1, 1\}$.

(f) The function $|x^3 - 8|$.

The inner expression changes sign only at 2. The function is $8 - x^3$ to the left and $x^3 - 8$ to the right, so its slopes at 2 are -12 and 12. Only 2 is exceptional.

Conclusion. In order: $\{0\}$, $\{0\}$, $\{k\pi : k \in \mathbb{Z}\}$, $\{0, 1\}$, $\{-1, 1\}$, and $\{2\}$. The piecewise descriptions also specify the corresponding graph shapes.

28.2 Exercise 28.2: Four derivatives directly from the definition

What we must show. Evaluate the requested limiting difference quotients without using differentiation rules.

(a) The cube at 2.

For $x \neq 2$, factor the difference of cubes:

$$\frac{x^3 - 8}{x - 2} = x^2 + 2x + 4 \longrightarrow 4 + 4 + 4 = 12.$$

Thus $f'(2) = 12$.

(b) The affine function at a.

For $h \neq 0$,

$$\frac{g(a + h) - g(a)}{h} = \frac{a + h + 2 - (a + 2)}{h} = 1.$$

The quotient is already constant, so $g'(a) = 1$ for every real a.

(c) The function $x^2 \cos x$ at zero.

Here $f(0) = 0$, and for $x \neq 0$ the quotient is $x \cos x$. Since $|x \cos x| \leq |x| \to 0$, the squeeze theorem gives $f'(0) = 0$.

(d) The rational function at 1.

For $r(x) = (3x + 4)/(2x - 1)$, we have $r(1) = 7$. In a sufficiently small neighborhood of 1, the denominator is nonzero, and

$$\frac{r(x) - r(1)}{x - 1} = \frac{3x + 4 - 7(2x - 1)}{(2x - 1)(x - 1)} = \frac{-11}{2x - 1} \longrightarrow -11.$$

Conclusion. The derivatives are 12, 1, 0, and -11, respectively.

28.3 Exercise 28.3: Square roots and cube roots

What we must show. Derive the root differentiation formulas from limiting quotients and examine the cube root at zero.

(a) Rationalize the square-root quotient.

Fix $a > 0$. For $x \geq 0$ with $x \neq a$,

$$\frac{\sqrt{x} - \sqrt{a}}{x - a} = \frac{1}{\sqrt{x} + \sqrt{a}} \longrightarrow \frac{1}{2\sqrt{a}}.$$

Continuity of the square root justifies the last limit; its denominator tends to the positive number $2\sqrt{a}$. Thus $h'(a) = \frac{1}{2}a^{-1/2}$.

(b) Factor a difference of cubes.

Fix $a \neq 0$, and put $u = x^{1/3}$, $v = a^{1/3}$. Since $x - a = u^3 - v^3 = (u - v)(u^2 + uv + v^2)$,

$$\frac{x^{1/3} - a^{1/3}}{x - a} = \frac{1}{x^{2/3} + x^{1/3}a^{1/3} + a^{2/3}} \longrightarrow \frac{1}{3a^{2/3}}.$$

The real cube-root function is continuous, and $3a^{2/3} > 0$, even when $a < 0$. This proves $f'(a) = \frac{1}{3}a^{-2/3}$.

(c) The origin is different.

At zero the quotient is $x^{1/3}/x = 1/x^{2/3}$. It tends to $+\infty$ from both sides, not to a finite real number. A vertical tangent is not a derivative in the definition being used.

Conclusion. The formulas hold on the stated nonzero domains; the cube-root function is not differentiable at 0.

28.4 Exercise 28.4: A differentiable function with a discontinuous derivative

What we must show. Analyze $f(x) = x^2 \sin(1/x)$ off zero, with $f(0) = 0$.

(a) Differentiate only where the displayed formula is valid.

For $a \neq 0$, the product and chain rules give

$$f'(a) = 2a \sin(1/a) + a^2 \cos(1/a)(-a^{-2}) = 2a \sin(1/a) - \cos(1/a).$$

The reciprocal, sine, and polynomial factors are differentiable near such an a, so the rules apply.

(b) Return to the definition at zero.

For $x \neq 0$,

$$\frac{f(x) - f(0)}{x} = x \sin(1/x), \qquad |x \sin(1/x)| \leq |x| \longrightarrow 0.$$

Thus $f'(0) = 0$.

(c) Test continuity using a particular sequence.

Take $x_n = 1/(2\pi n) \to 0$. Then $\sin(1/x_n) = 0$ and $\cos(1/x_n) = 1$, so $f'(x_n) = -1$ for every n. These values do not tend to $f'(0) = 0$. Therefore f' is not continuous at zero.

Conclusion. The function is differentiable everywhere, but its derivative fails to be continuous at zero.

28.5 Exercise 28.5: Why an informal chain-rule cancellation can fail

What we must show. Explain the problem with a quotient involving $f(x) - f(0)$ for the function in Exercise 28.4 and $g(x) = x$.

(a) Verify the separate derivatives exist.

Exercise 28.4 proves that f is differentiable everywhere, including zero. The identity function g has derivative 1 everywhere.

(b) Locate infinitely many zeros.

For any nonzero integer n, let $x = 1/(\pi n)$. Then

$$f(x) = \frac{1}{\pi^2 n^2} \sin(\pi n) = 0 = f(0).$$

Such nonzero x occur arbitrarily close to zero.

(c) Inspect the proposed quotient before taking a limit.

At every one of these points,

$$\frac{g(f(x)) - g(f(0))}{f(x) - f(0)} = \frac{0}{0},$$

which is undefined. Thus it is not a function on a full deleted neighborhood of zero, as a naive factorization of the chain-rule difference quotient would require. This is the sense in which the displayed unrestricted limit is meaningless in the exercise. If one explicitly restricts its domain to points with $f(x) \neq f(0)$, the quotient equals 1 and has relative limit 1; that different statement does not make the missing values defined. A rigorous chain-rule proof fills in the outer difference quotient at the problematic value using $g'(f(0))$, or uses a differentiability remainder.

Conclusion. Differentiability of f does not guarantee $f(x) \neq f(0)$ for nearby $x \neq 0$. Division by this difference needs separate justification.

28.6 Exercise 28.6: Continuity without differentiability at an oscillating point

What we must show. Examine $f(x) = x \sin(1/x)$ for $x \neq 0$, with $f(0) = 0$.

(a) Continuity.

The estimate $|f(x) - f(0)| \leq |x|$ shows that $f(x) \to f(0)$ as $x \to 0$. For an explicit epsilon–delta proof, choose $\delta = \varepsilon$.

(b) Differentiability.

The difference quotient at zero is $\sin(1/x)$. Along

$$u_n = \frac{1}{\pi/2 + 2\pi n}, \qquad v_n = \frac{1}{3\pi/2 + 2\pi n},$$

both sequences tend to zero, but their quotient values are 1 and -1, respectively. A function with a limit must have the same limit along every approaching sequence. Hence the derivative does not exist.

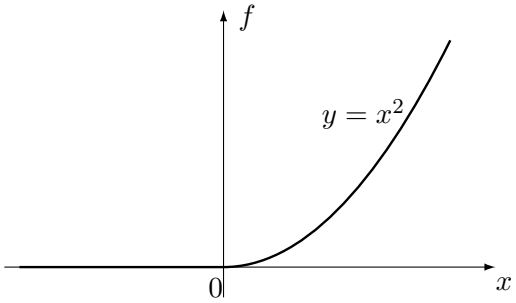
Conclusion. The factor x controls the function values, but dividing by x in the derivative removes precisely that control.

28.7 Exercise 28.7: Joining a horizontal half-line to a parabola

What we must show. For $f(x) = 0$ on $x < 0$ and $f(x) = x^2$ on $x \geq 0$, draw and analyze f and f'.

(a) Sketch the function.

The left branch is the negative half of the x-axis. The right branch is the right half of the parabola $y = x^2$, joined at $(0, 0)$.



(b) Differentiate at the join from the definition.

For $h \neq 0$,

$$\frac{f(h) - f(0)}{h} = \begin{cases} 0, & h < 0, \\ h, & h > 0. \end{cases}$$

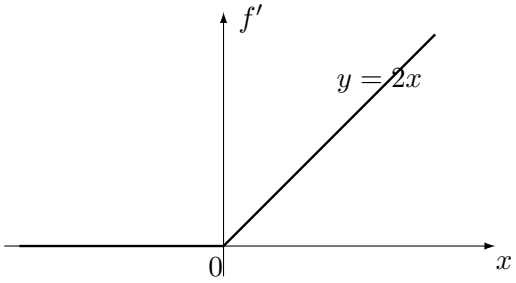
Both sides tend to zero, so $f'(0) = 0$.

(c) Calculate and sketch the derivative.

On the open half-lines, ordinary differentiation gives

$$f'(x) = \begin{cases} 0, & x \leq 0, \\ 2x, & x > 0. \end{cases}$$

Its graph is a horizontal left half-line joined to a straight right half-line of slope 2:



(d) Separate continuity from another derivative.

Both branches of f' approach zero at the origin, its actual value there. Thus f' is continuous on all of $\mathbb{R}$. Its difference quotient at zero, however, is 0 for $h < 0$ and 2 for $h > 0$. These do not agree. Consequently f' is differentiable on $\mathbb{R} \setminus \{0\}$ but not on all of $\mathbb{R}$.

Conclusion. f is continuously differentiable, but its second derivative does not exist at zero.

28.8 Exercise 28.8: A rational–irrational example

What we must show. Study $f(x) = x^2$ at rational x and $f(x) = 0$ at irrational x.

(a) Continuity at zero uses a common bound.

In both cases $0 \le f(x) \le x^2$, and $f(0) = 0$. Given $\varepsilon > 0$, choosing $\delta = \sqrt{\varepsilon}$ makes $|x| < \delta$ imply $|f(x) - f(0)| < \varepsilon$.

(b) At a nonzero point, compare dense kinds of inputs.

Fix $a \neq 0$. Density supplies rational $r_n \to a$ and irrational $t_n \to a$. Then

$$f(r_n) = r_n^2 \longrightarrow a^2, \qquad f(t_n) = 0 \longrightarrow 0.$$

Since $a^2 \neq 0$, these two limits disagree. Thus f has no limit at a and cannot be continuous there.

(c) Differentiate at zero without assuming a neighborhood formula.

For $h \neq 0$ the quotient is h if h is rational and 0 otherwise. In either case its absolute value is at most $|h|$, so its limit is zero. Hence $f'(0) = 0$. At every other point differentiability fails because continuity already fails.

Conclusion. The derivative exists exactly at 0, where it equals 0. It would be invalid to claim $f'(x) = 2x$ just because one branch has that formula.

28.9 Exercise 28.9: Separating an inner polynomial and an outer power

What we must show. Differentiate $h(x) = (x^4 + 13x)^7$ and identify the composition.

(a) Compute the derivative.

The answer is

$$h'(x) = 7(x^4 + 13x)^6(4x^3 + 13).$$

The factor $4x^3 + 13$ is essential: it accounts for the rate of change of the inner quantity.

(b) Justify each factor.

Set $f(x) = x^4 + 13x$ and $g(u) = u^7$. These are polynomials, so they are differentiable everywhere. Since $h = g \circ f$, the chain rule gives $h'(x) = g'(f(x))f'(x)$, with $g'(u) = 7u^6$ and $f'(x) = 4x^3 + 13$. Substitution gives part (a).

28.10 Exercise 28.10: A power of a trigonometric–exponential sum

What we must show. Repeat the preceding decomposition for $h(x) = (\cos x + e^x)^{12}$.

(a) Differentiate.

$$h'(x) = 12(\cos x + e^x)^{11}(-\sin x + e^x).$$

(b) Verify the chain rule applies.

Let $f(x) = \cos x + e^x$ and $g(u) = u^{12}$. Both are differentiable everywhere, $f'(x) = -\sin x + e^x$, and $g'(u) = 12u^{11}$. Thus $h = g \circ f$ and the displayed formula is exactly $g'(f(x))f'(x)$.

28.11 Exercise 28.11: A chain of three functions

What we must show. State and prove the derivative rule for $h \circ g \circ f$ at a under the given differentiability assumptions.

First combine the two inner functions.

Since f is differentiable at a and g at $f(a)$, the chain rule makes $g \circ f$ differentiable at a and gives

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Differentiability also gives the continuity needed to ensure the compositions are defined near the relevant points.

Then apply the rule to the remaining outer function.

The assumption about h is differentiability at $(g \circ f)(a)$. A second application yields

$$(h \circ g \circ f)'(a) = h'(g(f(a)))(g \circ f)'(a) = h'(g(f(a)))g'(f(a))f'(a).$$

Conclusion. Each derivative is evaluated at the input that its particular function actually receives.

28.12 Exercise 28.12: Three nested functions with an exponent

What we must show. Differentiate $H(x) = \cos(e^{x^5-3x})$; the entire expression $x^5 - 3x$ is in the exponent.

(a) Work from the outside inward.

Differentiating cosine introduces a minus sine; differentiating the exponential reproduces it; differentiating the polynomial gives $5x^4 - 3$. Therefore

$$H'(x) = -\sin(e^{x^5-3x})\, e^{x^5-3x}(5x^4 - 3).$$

(b) Make the three-function justification explicit.

Take $f(x) = x^5 - 3x$, $g(u) = e^u$, and $h(v) = \cos v$. They are differentiable on $\mathbb{R}$, so the hypotheses of Exercise 28.11 hold at every x. Substituting f', g', and h' into its formula proves the calculation.

28.13 Exercise 28.13: Ensuring a composition is defined nearby

What we must show. Find an open interval around a on which $g \circ f$ is defined.

Choose a target neighborhood that stays in the domain of g.

Because that domain contains an open interval around $f(a)$, there is $\eta > 0$ such that $(f(a)-\eta, f(a)+\eta)$ lies in the domain of g.

Use continuity of f to stay inside that neighborhood.

Continuity at a supplies $\delta_1 > 0$ such that $|x-a| < \delta_1$ implies $|f(x)-f(a)| < \eta$, for x in the domain of f. Its domain also contains $(a-\delta_2, a+\delta_2)$ for some $\delta_2 > 0$. Let $\delta = \min(\delta_1, \delta_2)$. Every $x \in (a-\delta, a+\delta)$ belongs to the domain of f and has $f(x)$ in the domain of g. Thus $g(f(x))$ is defined throughout that open interval.

Conclusion. Continuity of g is unnecessary for this domain assertion; only its available domain matters.

28.14 Exercise 28.14: Increment quotients and symmetric quotients

What we must show. Derive both requested limits from differentiability at a.

(a) Replace the approaching point by an increment.

Set $x = a + h$. Then $x \to a$ if and only if $h \to 0$, and $x \neq a$ if and only if $h \neq 0$. Consequently the definition immediately gives

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = f'(a).$$

(b) Split the symmetric difference into two ordinary ones.

For nonzero h small enough that both arguments are in the domain,

$$\frac{f(a+h)-f(a-h)}{2h} = \frac{1}{2}\left(\frac{f(a+h)-f(a)}{h} + \frac{f(a-h)-f(a)}{-h}\right).$$

Each quotient in parentheses tends to $f'(a)$ by part (a), the second with increment $-h$. The limit is therefore $\frac{1}{2}(f'(a)+f'(a)) = f'(a)$.

Conclusion. Differentiability implies the symmetric limit formula. The converse need not hold: $|x|$ at zero has symmetric quotient 0 but no derivative.

28.15 Exercise 28.15: Leibniz's higher product rule

What we must show. Prove the binomial-coefficient formula for the nth derivative of a product.

Plan. Induction extends the usual product rule. Each differentiation can land on either factor; Pascal's identity combines the two contributions to an interior term.

Start the induction.

For $n = 1$ the formula is $(fg)' = fg' + f'g$, precisely the product rule. We use $f^{(0)} = f$ and $g^{(0)} = g$.

Differentiate the induction formula.

Assume the identity for order n, and suppose the derivatives needed for order $n+1$ exist. On the neighborhood where the lower derivatives are defined, differentiate each term:

$$(fg)^{(n+1)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k+1)} g^{(n-k)} + \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n+1-k)}.$$

Reindex the first sum with $j = k + 1$. Its end term at $j = n + 1$ is $f^{(n+1)}g$, and the second sum's end term at $j = 0$ is $fg^{(n+1)}$. For $1 \le j \le n$, the coefficient of $f^{(j)}g^{(n+1-j)}$ is

$$\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j}.$$

The two end coefficients are also the corresponding binomial coefficients, both 1. Thus

$$(fg)^{(n+1)}(a) = \sum_{j=0}^{n+1} \binom{n+1}{j} f^{(j)}(a)g^{(n+1-j)}(a).$$

This closes the induction. If only the final derivative is assumed at a, the same argument is applied to the lower-order identity near a and differentiated at a; it does not assume the highest derivatives exist throughout a larger interval.

Conclusion. For every permitted order, $(fg)^{(n)}(a) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(a)g^{(n-k)}(a)$.

28.16 Exercise 28.16: Differentiability as a linear approximation with vanishing error

What we must show. Prove the equivalence in the exercise, interpreting the proposed coefficient as a fixed real number A until differentiability has been established.

From a derivative to an error function.

If $f'(a)$ exists, let $A = f'(a)$ and define

$$\epsilon(x) = A - \frac{f(x) - f(a)}{x - a} \quad (x \neq a), \qquad \epsilon(a) = 0.$$

By the derivative definition, $\epsilon(x) \to 0$ as $x \to a$. Multiplication by $x - a$ gives

$$f(x) - f(a) = (x - a)[A - \epsilon(x)]$$

for $x \neq a$; at a both sides are zero, so the identity holds on all of I.

From the error representation to a derivative.

Conversely, suppose such a real A and such an error function are given. For $x \neq a$, divide the identity by $x - a$. The quotient becomes $A - \epsilon(x)$ and tends to A. Thus $f'(a)$ exists and equals A, making the notation $f'(a)$ in the exercise legitimate.

Conclusion. The change in f equals its linear part plus an error whose ratio to $x - a$ tends to zero. No continuity of the error away from a is required.

29 The Mean Value Theorem

The Mean Value Theorem (MVT) connects an average slope to an actual derivative. If f is continuous on $[a, b]$ and differentiable on (a, b), with $a < b$, there exists $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

The theorem asserts existence, not a prescribed choice of c. Both hypotheses matter. Ross uses "increasing" to mean nondecreasing; "strictly increasing" requires strict inequalities. A derivative sign on an entire interval controls monotonicity there; a sign at one point does not have the same force.

29.1 Exercise 29.1: Checking the MVT conclusion in six examples

What we must show. Compute the secant slope and determine whether an interior derivative equals it.

(a) The square on $[-1, 2]$.

The slope is $(4 - 1)/3 = 1$. Since $f'(x) = 2x$, take $c = 1/2 \in (-1, 2)$. Polynomial continuity and differentiability verify all MVT hypotheses.

(b) Sine on $[0, \pi]$.

The secant slope is 0, and $f'(\pi/2) = \cos(\pi/2) = 0$. Sine satisfies all hypotheses.

(c) Absolute value on $[-1, 2]$.

The secant slope is $(2 - 1)/3 = 1/3$. At nonzero interior points the derivative is either -1 or 1; at 0 it is undefined. Thus no derivative equals $1/3$. The failed hypothesis is differentiability throughout the open interval, although continuity holds.

(d) The reciprocal on $[-1, 1]$.

The endpoint expression gives $(1 - (-1))/2 = 1$. Wherever it exists, the derivative is $-1/x^2 < 0$, so it cannot equal 1. The function is not even defined at 0; in particular it is neither continuous on the whole closed interval nor differentiable on the whole open interval.

(e) The reciprocal on $[1, 3]$.

Now both hypotheses hold. The slope is $(1/3 - 1)/2 = -1/3$, and $-1/c^2 = -1/3$ is satisfied by $c = \sqrt{3} \in (1, 3)$.

(f) The sign function on $[-2, 2]$.

The secant slope is $(1 - (-1))/4 = 1/2$. At every nonzero point its derivative is 0; at 0 it has a jump and no derivative. Thus the conclusion fails. Continuity on the closed interval and differentiability throughout the open interval both fail.

Conclusion. The conclusion holds in (a), (b), and (e), at the exhibited points; it fails in (c), (d), and (f).

29.2 Exercise 29.2: Cosine is Lipschitz with constant one

What we must show. Prove $|\cos x - \cos y| \le |x - y|$ for all real x, y.

Handle equal points and order unequal ones.

If $x = y$, both sides vanish. Otherwise interchange their names if needed so $x < y$. Cosine is continuous on $[x, y]$ and differentiable on (x, y). The MVT gives $c \in (x, y)$ with

$$\cos y - \cos x = -\sin c\,(y - x).$$

Taking absolute values and using $|\sin c| \le 1$ proves the inequality. The absolute-value form is unchanged when the original points are interchanged.

29.3 Exercise 29.3: Two guaranteed derivative values

What we must show. Use $f(0) = 0$, $f(1) = f(2) = 1$ to locate derivative values $1/2$ and $1/7$.

(a) Use the endpoints 0 and 2.

Differentiability on $\mathbb{R}$ implies continuity there, so the MVT applies on $[0, 2]$. It supplies $c \in (0, 2)$ with

$$f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{1}{2}.$$

(b) First find slopes above and below $1/7$.

The MVT on $[0, 1]$ gives $u \in (0, 1)$ with $f'(u) = 1$. On $[1, 2]$ it gives $v \in (1, 2)$ with $f'(v) = 0$. Derivatives have the intermediate value property (Darboux's theorem, Theorem 29.8), even when they are not continuous. Therefore some $w \in (u, v) \subset (0, 2)$ satisfies $f'(w) = 1/7$.

An elementary way to implement the same argument.

Let $H(x) = f(x) - x/7$ on $[u, v]$. Then $H'(u) = 6/7 > 0$ and $H'(v) = -1/7 < 0$. By the derivative definition, H takes values larger than $H(u)$ just to the right of u, and values larger than $H(v)$ just to the left of v. Its maximum on the compact interval therefore cannot occur at either endpoint. At an interior maximum w, Fermat's theorem gives $H'(w) = 0$, or $f'(w) = 1/7$. This explains why continuity of f' is not needed.

29.4 Exercise 29.4: Introducing an exponential integrating factor

What we must show. Find $c \in (a, b)$ with $f'(c) + f(c)g'(c) = 0$ when $f(a) = f(b) = 0$.

Plan. The derivative of fe^g contains exactly the desired expression, multiplied by a positive factor.

Verify Rolle's theorem for the auxiliary function.

Define $H(x) = f(x)e^{g(x)}$. The differentiability assumptions on the open interval I ensure that H is continuous on $[a, b] \subset I$ and differentiable on (a, b). Also $H(a) = H(b) = 0$. Rolle's theorem gives $c \in (a, b)$ with $H'(c) = 0$.

Differentiate and divide by a nonzero factor.

By the product and chain rules,

$$0 = H'(c) = e^{g(c)}[f'(c) + f(c)g'(c)].$$

Because $e^{g(c)} > 0$, division yields the required equation.

29.5 Exercise 29.5: A quadratic bound forces constancy

What we must show. Show a function satisfying $|f(x) - f(y)| \le (x-y)^2$ for all real x, y is constant.

Prove differentiability rather than assume it.

Fix $a \in \mathbb{R}$. For $h \neq 0$, the assumed estimate gives

$$\left|\frac{f(a+h) - f(a)}{h}\right| \le \frac{h^2}{|h|} = |h| \longrightarrow 0.$$

Thus $f'(a)$ exists and equals 0. Since a was arbitrary, $f' = 0$ everywhere.

Use the MVT between any two points.

For $x < y$, the now-established differentiability gives the MVT hypotheses on $[x, y]$. Hence $f(y) - f(x) = f'(c)(y - x) = 0$ for some c between them. Every pair has equal values, so f is constant.

29.6 Exercise 29.6: The straight line in the MVT proof

What we must show. Write the line through $(a, f(a))$ and $(b, f(b))$ explicitly.

Use point–slope form.

The required function is

$$L(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

Indeed $L(a) = f(a)$ and $L(b) = f(a) + f(b) - f(a) = f(b)$. Its derivative is the constant secant slope $(f(b) - f(a))/(b - a)$. Subtracting this line from f gives a function with equal endpoint values, which is precisely why Rolle's theorem can be used in the MVT proof.

29.7 Exercise 29.7: Vanishing higher derivatives imply polynomial form

What we must show. Determine f when its second or third derivative vanishes on an interval.

(a) If $f'' = 0$, first make f' constant.

The zero-derivative corollary to the MVT, applied to f', gives a constant A with $f'(x) = A$ on I. Now $[f(x) - Ax]' = 0$ on I, so the same corollary gives another constant B with $f(x) - Ax = B$. Thus $f(x) = Ax + B$.

(b) If $f''' = 0$, repeat one level higher.

Apply the corollary to f'' to obtain $f''(x) = C$. Then

$$[f'(x) - Cx]' = 0,$$

so $f'(x) = Cx + D$. Finally,

$$\left[f(x) - \frac{C}{2}x^2 - Dx\right]' = 0,$$

which gives $f(x) = \frac{C}{2}x^2 + Dx + E$. Conversely, every polynomial of this form has third derivative zero, so these are exactly the possibilities.

Conclusion. The interval assumption allows the MVT to connect every pair of points. On disconnected domains the constants could differ between components.

29.8 Exercise 29.8: Derivative signs and the three remaining monotonicity rules

What we must show. Prove Corollary 29.7(ii)–(iv).

Use one MVT equation for all three cases.

Choose arbitrary $x_1 < x_2$ in the interval. The MVT gives $c \in (x_1, x_2)$ with

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

The factor $x_2 - x_1$ is strictly positive, so it preserves the sign of $f'(c)$.

(ii) Strictly negative derivative.

If $f' < 0$ everywhere, the right side is negative. Thus $f(x_2) < f(x_1)$ for every ordered pair, which is strict decrease.

(iii) Nonnegative derivative.

If $f' \geq 0$ everywhere, then $f(x_2) - f(x_1) \geq 0$, so $f(x_1) \leq f(x_2)$. This is increase in Ross's nonstrict sense.

(iv) Nonpositive derivative.

If $f' \leq 0$ everywhere, the equation gives $f(x_2) \leq f(x_1)$, which is decrease. Equality is permitted in the last two cases.

29.9 Exercise 29.9: The inequality $ex \leq e^x$

What we must show. Show the product $e \cdot x$ never exceeds e^x.

Subtract and locate the minimum.

Let $F(x) = e^x - ex$. Then $F'(x) = e^x - e$. Because the exponential is strictly increasing, this derivative is negative for $x < 1$, zero at 1, and positive for $x > 1$. The monotonicity corollaries imply F decreases toward 1 and increases afterward. Therefore its global minimum is

$$F(1) = e - e = 0.$$

Hence $e^x - ex \geq 0$ for every real x.

Conclusion. The inequality holds for every real x, with equality exactly when $x = 1$. Here ex denotes multiplication, not a second copy of e^x.

29.10 Exercise 29.10: A positive derivative at one point does not imply local increase

What we must show. Study $f(x) = x^2 \sin(1/x) + x/2$ for $x \neq 0$, with $f(0) = 0$.

(a) Differentiate at zero.

The difference quotient is $x \sin(1/x) + 1/2$, whose first term tends to zero by $|x \sin(1/x)| \leq |x|$. Thus $f'(0) = 1/2 > 0$.

(b) Find negative derivatives arbitrarily close to zero.

For $x \neq 0$, Exercise 28.4 and the sum rule give

$$f'(x) = 2x \sin(1/x) - \cos(1/x) + \frac{1}{2}.$$

At $x_n = 1/(2\pi n)$ this equals $-1/2$. Every open interval containing zero contains such an x_n as an interior point. An increasing function that is differentiable at an interior point has nonnegative derivative there: its difference quotients are nonnegative for increments of either sign. Thus the negative derivative at x_n rules out increase on that interval.

(c) Compare the hypotheses carefully.

Corollary 29.7(i) requires $f'(x) > 0$ at *every* point of the interval. This example gives positivity only at zero and negative values arbitrarily nearby, so there is no contradiction.

29.11 Exercise 29.11: Sine lies below its argument on the positive half-line

What we must show. Prove $\sin x \leq x$ for $x \geq 0$.

Apply a derivative sign to the difference.

Let $F(x) = x - \sin x$. Then $F'(x) = 1 - \cos x \geq 0$ on $\mathbb{R}$, so F is increasing. For $x \geq 0$ this implies $F(x) \geq F(0) = 0$, which rearranges to the required inequality. At $x = 0$ equality is immediate.

29.12 Exercise 29.12: Three related trigonometric inequalities

What we must show. Establish the tangent inequality, monotonicity of $x/\sin x$, and the sine lower bound on a quarter period.

(a) Compare tangent with the identity.

Let $F(x) = \tan x - x$ on $[0, \pi/2)$. It is continuous at zero, $F(0) = 0$, and for $0 < x < \pi/2$,

$$F'(x) = \sec^2 x - 1 = \tan^2 x > 0.$$

Applying the MVT on $[0, x]$ gives $F(x) > F(0) = 0$. Thus $x < \tan x$.

(b) Differentiate the quotient with a positive denominator.

On $(0, \pi/2)$,

$$\left(\frac{x}{\sin x}\right)' = \frac{\sin x - x \cos x}{\sin^2 x}.$$

Both $\cos x$ and $\sin^2 x$ are positive. Part (a), multiplied by $\cos x > 0$, gives $x \cos x < \sin x$. Therefore the numerator is positive, and the quotient is strictly increasing.

(c) Compare with the endpoint value.

The quotient extends continuously to $x = \pi/2$ with value $\pi/2$. Its increase on the interior yields $x/\sin x \leq \pi/2$ for $0 < x \leq \pi/2$, for instance by taking an increasing sequence of comparison points tending to the endpoint. Multiplying by the positive $\sin x$ gives $x \leq (\pi/2)\sin x$. At $x = 0$ both sides vanish.

Conclusion. The last inequality is equivalently $\sin x \geq 2x/\pi$ on $[0, \pi/2]$.

29.13 Exercise 29.13: Comparing functions by comparing derivatives

What we must show. Use equal initial values and $f' \leq g'$ to show $f(x) \leq g(x)$ for $x \geq 0$.

Build a single function with a known sign of derivative.

Set $H = g - f$. Then $H(0) = 0$ and $H' = g' - f' \geq 0$. The monotonicity corollary makes H increasing. Hence for $x \geq 0$, $H(x) \geq H(0) = 0$, or $f(x) \leq g(x)$.

Conclusion. Equal initial values anchor the comparison. Without them, derivative inequalities alone do not fix which graph starts higher.

29.14 Exercise 29.14: Bounding a function between two straight lines

What we must show. Given $f(0) = 0$ and $1 \leq f' \leq 2$, prove $x \leq f(x) \leq 2x$ for $x \geq 0$.

Use the MVT on the interval from zero to the chosen point.

At $x = 0$ the assertion is immediate. For $x > 0$, differentiability ensures the hypotheses on $[0, x]$, so for some $c \in (0, x)$,

$$f(x) - f(0) = f'(c)x.$$

Since $x > 0$, multiplying $1 \leq f'(c) \leq 2$ by x preserves order. Substituting $f(0) = 0$ gives $x \leq f(x) \leq 2x$.

29.15 Exercise 29.15: The power rule for rational exponents

What we must show. Prove the derivative formula for $h(x) = x^{m/n}$ at every permitted nonzero real input, including negative inputs when n is odd.

Differentiate the real root by the inverse-function theorem.

Let $u(x) = x^{1/n}$ on its real domain. Fix a permitted $a \neq 0$ and write $t = u(a)$, so $t^n = a$ and $t \neq 0$. The function $v(t) = t^n$, restricted to the positive half-line if n is even and considered on $\mathbb{R}$ if n is odd, is one-to-one and continuous. Near this nonzero t it is differentiable with $v'(t) = nt^{n-1} \neq 0$. The inverse derivative theorem gives

$$u'(a) = \frac{1}{nt^{n-1}}.$$

Compose with an integer power.

The function $h = u^m$ is differentiable near a; negative m is allowed because u is nonzero there. The integer power rule and chain rule give

$$h'(a) = mt^{m-1}\frac{1}{nt^{n-1}} = \frac{m}{n}t^{m-n} = \frac{m}{n}a^{m/n-1}.$$

The final equality uses the real-root definition of the rational powers. It remains valid for negative a when n is odd, without invoking an undefined logarithm of a negative number.

Conclusion. $h'(a) = ra^{r-1}$ everywhere requested. The exclusion of zero avoids both inverse-derivative failure and negative-power domain issues.

29.16 Exercise 29.16: The derivative of arctangent

What we must show. Apply the inverse derivative theorem to tangent on $(-\pi/2, \pi/2)$.

Check invertibility and derivative hypotheses.

On this interval, $f(x) = \tan x$ is continuous and strictly increasing, has range $\mathbb{R}$, and has derivative $\sec^2 x > 0$. It therefore has the inverse $g = \arctan : \mathbb{R} \to (-\pi/2, \pi/2)$, and Theorem 29.9 applies at every inverse pair $y = \tan x$.

Express the derivative using the new input.

The theorem gives $g'(y) = 1/\sec^2 x$. Since $\sec^2 x = 1 + \tan^2 x = 1 + y^2$, this becomes

$$\frac{d}{dy}\arctan y = \frac{1}{1+y^2} \qquad (y \in \mathbb{R}).$$

29.17 Exercise 29.17: Matching values and slopes at a join

What we must show. Prove the criterion for differentiability of h, formed from f to the left of a and g at and to its right.

Necessity of matching values.

Suppose h is differentiable at a. It is then continuous there. By continuity of f,

$$f(a) = \lim_{x\to a^-} f(x) = \lim_{x\to a^-} h(x) = h(a) = g(a).$$

Thus the two branches meet at the same height.

Necessity of matching slopes.

Using $h(a) = f(a) = g(a)$, the left difference quotient for h has limit $f'(a)$, while the right one has limit $g'(a)$. Both must equal $h'(a)$, so $f'(a) = g'(a)$.

Sufficiency.

Conversely, suppose both equalities hold. For $x < a$ the difference quotient of h at a is $(f(x) - f(a))/(x - a)$; for $x > a$ it is $(g(x) - g(a))/(x - a)$. Their limits are the equal numbers $f'(a)$ and $g'(a)$. The two-sided limit therefore exists, and $h'(a)$ is their common value.

Conclusion. Matching heights prevents a jump; matching slopes prevents a corner. Both conditions are necessary and together are sufficient.

29.18 Exercise 29.18: Iterating a contraction on the real line

What we must show. Prove convergence of $s_{n+1} = f(s_n)$ and obtain a fixed point when $a = \sup_{x\in\mathbb{R}} |f'(x)| < 1$.

(a), first stage: convert the derivative bound to a distance bound.

Since $|f'| \geq 0$, we have $0 \leq a < 1$. The MVT gives, for any $u, v \in \mathbb{R}$,

$$|f(u) - f(v)| \leq a|u - v|.$$

For equal inputs the inequality is automatic. Applying it to successive iterates gives

$$|s_{n+1} - s_n| \leq a|s_n - s_{n-1}| \leq \cdots \leq a^n|s_1 - s_0|.$$

(a), second stage: sum the small successive gaps.

Assume first $0 < a < 1$. For integers $m > n$, the triangle inequality and a geometric series give

$$|s_m - s_n| \leq \sum_{k=n}^{m-1} |s_{k+1} - s_k| \leq |s_1 - s_0| \sum_{k=n}^{m-1} a^k \leq \frac{|s_1 - s_0|a^n}{1 - a}.$$

This bound tends to zero independently of $m > n$. For any $\varepsilon > 0$, choose N to make the last expression less than ε for $n \geq N$; this is exactly the Cauchy criterion. Completeness of $\mathbb{R}$ gives a limit s. If $a = 0$, the distance bound says f is constant, so all iterates from s_1 onward are equal and convergence is immediate.

(b) Pass to the limit in the iteration.

Differentiability implies continuity, and therefore

$$f(s) = f\left(\lim_n s_n\right) = \lim_n f(s_n) = \lim_n s_{n+1} = s.$$

Thus s is a fixed point. In fact it is unique: if $t = f(t)$, then $|s - t| = |f(s) - f(t)| \leq a|s - t|$; since $1 - a > 0$, this forces $s = t$.

Conclusion. Every starting value produces a convergent sequence with the same fixed-point limit. The geometric bound explicitly explains why the iteration settles down.

30 L'Hospital's rule

L'Hospital's rule compares a quotient with a quotient of derivatives; it does not say that every limit may be evaluated by differentiating. In the 0/0 case, both numerator and denominator must tend to zero. Ross also permits the case in which the absolute value of the denominator tends to infinity, without requiring a numerator limit. In either case the functions must be differentiable on the relevant deleted neighborhood, the derivative of the denominator must be nonzero there, and the derivative quotient must have a limit. That limit may be finite or infinite.

For positive quantities with variable exponents, take a logarithm first. If $A(x) > 0$, then $A(x)^{B(x)} = \exp(B(x) \log A(x))$; after finding the exponent's limit, use continuity of the exponential.

30.1 Exercise 30.1: Four introductory limits

What we must show. Evaluate each limit and justify the method, using the actual numerator and denominator in the printed exercise.

(a) The quotient $(e^{2x} - \cos x)/x$ at zero.

Both numerator and denominator tend to zero, they are differentiable, and the denominator derivative is $1 \neq 0$. L'Hospital's rule therefore gives

$$\lim_{x \to 0} \frac{e^{2x} - \cos x}{x} = \lim_{x \to 0} (2e^{2x} + \sin x) = 2.$$

(b) The quotient $(1 - \cos x)/x^2$ at zero.

The original form is 0/0, and the derivative of x^2 is nonzero for $x \neq 0$. The derivative quotient has the known limit

$$\lim_{x \to 0} \frac{\sin x}{2x} = \frac{1}{2}.$$

L'Hospital's rule thus gives 1/2 for the original limit. The last step uses the basic sine limit, not another unjustified cancellation at zero.

(c) The quotient x^3/e^{2x} at positive infinity.

At every differentiation stage the exponential denominator tends to infinity and has nonzero derivative. Successive derivative quotients reduce the polynomial degree:

$$\frac{x^3}{e^{2x}}, \quad \frac{3x^2}{2e^{2x}}, \quad \frac{6x}{4e^{2x}}, \quad \frac{6}{8e^{2x}}.$$

The final quotient tends to zero. Applying the rule backward through these three stages proves that the original limit is 0.

(d) A difference of square roots at zero.

For $0 < |x| < 1$, multiply by the conjugate:

$$\frac{\sqrt{1+x} - \sqrt{1-x}}{x} = \frac{(1+x) - (1-x)}{x(\sqrt{1+x} + \sqrt{1-x})} = \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \longrightarrow 1.$$

This direct algebraic method is shorter than L'Hospital's rule and also keeps the real domain explicit.

Conclusion. The four limits are 2, 1/2, 0, and 1.

30.2 Exercise 30.2: Cancellation, repeated differentiation, and a variable power

What we must show. Evaluate the four limits, checking why each reduction is valid.

(a) The quotient $x^3/(\sin x - x)$.

The numerator and denominator and their first two derivative pairs tend to zero. Differentiating each pair produces

$$\frac{x^3}{\sin x - x}, \quad \frac{3x^2}{\cos x - 1}, \quad \frac{6x}{-\sin x}, \quad \frac{6}{-\cos x}.$$

On a sufficiently small deleted neighborhood of zero, each denominator derivative used in passing to the next pair is nonzero: these are $\cos x - 1$, $-\sin x$, and $-\cos x$. The final quotient tends to -6, so three valid applications of L'Hospital's rule yield -6.

(b) The quotient $(\tan x - x)/x^3$.

This is $0/0$ near zero, where tangent is defined. The denominator derivative $3x^2$ is nonzero for $x \neq 0$. The derivative quotient is

$$\frac{\sec^2 x - 1}{3x^2} = \frac{1}{3}\left(\frac{\tan x}{x}\right)^2 \longrightarrow \frac{1}{3},$$

because $\tan x/x = (\sin x/x)/\cos x \to 1$. Thus the requested limit is $1/3$.

(c) The difference $1/\sin x - 1/x$.

Combine the fractions before applying any limit rule:

$$\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x \sin x}.$$

The first derivative quotient is $(1 - \cos x)/(\sin x + x\cos x)$, again of form $0/0$. Its derivative quotient is

$$\frac{\sin x}{2\cos x - x \sin x} \longrightarrow 0.$$

The final denominator tends to 2 and hence is nonzero near zero. The intermediate denominator $\sin x + x\cos x$ is nonzero for sufficiently small nonzero x, because its ratio to x tends to 2. Thus both uses of L'Hospital's rule are legitimate and the original limit is 0.

(d) The expression $(\cos x)^{1/x^2}$.

For $|x| < \pi/2$, the base is positive, so its logarithm is $\log(\cos x)/x^2$. This is $0/0$, and its derivative quotient is

$$\frac{-\tan x}{2x} \longrightarrow -\frac{1}{2}.$$

L'Hospital's rule gives the logarithmic limit $-1/2$. Continuity of the exponential now gives the original limit $e^{-1/2}$.

Conclusion. The answers are -6, $1/3$, 0, and $e^{-1/2}$.

30.3 Exercise 30.3: Recognizing when differentiation is unnecessary

What we must show. Evaluate the four limits, including a correction to a printed selected answer.

(a) The quotient $(x - \sin x)/x$ at positive infinity.

For the expression printed in the exercise,

$$\frac{x - \sin x}{x} = 1 - \frac{\sin x}{x}, \qquad \left|\frac{\sin x}{x}\right| \le \frac{1}{x} \longrightarrow 0.$$

The limit is therefore 1. The selected-answer entry 0 in the linked edition is inconsistent with this printed expression. Differentiating would give $1 - \cos x$, which has no limit; this simply means L'Hospital's rule is not useful here. Its failure to supply a derivative-quotient limit says nothing against the elementary estimate above.

(b) The power $x^{\sin(1/x)}$ at positive infinity.

Its logarithm is

$$\sin(1/x)\log x = \frac{\sin(1/x)}{1/x}\,\frac{\log x}{x}.$$

The first factor tends to 1. The second tends to 0 by L'Hospital's rule, since $(\log x)'/x' = 1/x \to 0$ and $x \to \infty$. Thus the logarithm tends to zero and the power tends to $e^0 = 1$.

(c) The quotient $(1 + \cos x)/(e^x - 1)$ as $x \to 0^+$.

The numerator tends to 2, while the denominator tends to zero through positive values. Consequently the limit is $+\infty$. Explicitly, near zero the numerator is greater than 1. For any prescribed $M > 0$, continuity of e^x makes $0 < e^x - 1 < 1/M$ for sufficiently small $x > 0$, giving the quotient greater than M. This is not a 0/0 situation.

(d) The quotient $(1 - \cos 2x - 2x^2)/x^4$ at zero.

For transparency, list all derivatives through the first nonzero limiting values:

order	numerator derivative	denominator derivative
0	$1 - \cos 2x - 2x^2$	x^4
1	$2\sin 2x - 4x$	$4x^3$
2	$4\cos 2x - 4$	$12x^2$
3	$-8\sin 2x$	$24x$
4	$-16\cos 2x$	24

At orders 0 through 3 both entries tend to zero. Every denominator derivative required by the rule is nonzero for nonzero x near zero. The last ratio tends to $-16/24 = -2/3$. Four applications, read backward, prove that this is the requested limit.

Conclusion. For the printed questions, the limits are 1, 1, $+\infty$, and $-2/3$.

30.4 Exercise 30.4: Changing a right-hand zero limit into an infinite limit

What we must show. For $g(y) = f(1/y)$, prove equivalence of the two limits, including infinite limits.

Plan. The substitution is reversible: $y \to +\infty$ corresponds exactly to $x = 1/y \to 0^+$.

From a right-hand limit of f to a limit of g.

Suppose first $f(x) \to L \in \mathbb{R}$ as $x \to 0^+$. Given $\varepsilon > 0$, choose $\delta > 0$, with $\delta < a$ if a is finite, such that $0 < x < \delta$ implies $|f(x) - L| < \varepsilon$. If $y > 1/\delta$, then $0 < 1/y < \delta$, so $|g(y) - L| < \varepsilon$. This proves $g(y) \to L$ as $y \to \infty$.

Reverse the argument.

Suppose $g(y) \to L \in \mathbb{R}$. Choose $Y > 0$ inside its domain such that $y > Y$ implies $|g(y) - L| < \varepsilon$. Let $\delta > 0$ be at most $1/Y$ and small enough that $(0, \delta) \subset (0, a)$. Then $0 < x < \delta$ implies $1/x > Y$, and

$$|f(x) - L| = |g(1/x) - L| < \varepsilon.$$

Thus $f(x) \to L$ as $x \to 0^+$.

Use the same substitution for infinite limits.

For a limit $+\infty$, replace $|f(x) - L| < \varepsilon$ and $|g(y) - L| < \varepsilon$ by $f(x) > M$ and $g(y) > M$, respectively, for arbitrary real thresholds M. The same reciprocal choices of Y and δ give both implications. For $-\infty$, use the inequalities $f(x) < M$ and $g(y) < M$. No other part of the proof changes.

Conclusion. Either limit exists exactly when the other does, and their finite or infinite values are equal. The convention $a^{-1} = 0$ when $a = \infty$ gives precisely the correct transformed domain.

30.5 Exercise 30.5: Three exponential limits

What we must show. Evaluate the limits by taking logarithms on domains where the bases are positive.

(a) The expression $(1 + 2x)^{1/x}$ as $x \to 0$.

For $|x| < 1/2$, the base is positive. Its logarithm is $\log(1 + 2x)/x$, of form $0/0$. The denominator derivative is 1, and the numerator derivative is $2/(1 + 2x) \to 2$. Thus the logarithm tends to 2, and the original expression tends to e^2.

(b) The expression $(1 + 2/y)^y$ as $y \to \infty$.

Set $x = 1/y \to 0^+$. The expression becomes $(1 + 2x)^{1/x}$, whose right-hand limit is e^2 by part (a). Exercise 30.4 rigorously justifies the reciprocal change of variable.

(c) The expression $(e^x + x)^{1/x}$ as $x \to \infty$.

The base is positive for $x > 0$. Factor out e^x inside its logarithm:

$$\frac{\log(e^x + x)}{x} = \frac{x + \log(1 + xe^{-x})}{x} = 1 + \frac{\log(1 + xe^{-x})}{x}.$$

L'Hospital's rule gives $x/e^x \to 0$, so $\log(1 + xe^{-x}) \to 0$. Dividing this term by $x \to \infty$ still gives zero; for example its absolute value is eventually at most $1/x$. The logarithmic limit is 1, and the original limit is e.

Conclusion. The three limits are e^2, e^2, and e.

30.6 Exercise 30.6: Recovering limits of a function and its derivative from their sum

What we must show. Given $f(x) + f'(x) \to L \in \mathbb{R}$ at positive infinity, prove $f(x) \to L$ and $f'(x) \to 0$.

Plan. Multiplication by e^x makes the sum appear as a product derivative.

Apply the denominator-infinity version of the rule.

Write

$$f(x) = \frac{f(x)e^x}{e^x}.$$

Both numerator and denominator are differentiable on (c, ∞), the denominator tends to $+\infty$, and its derivative e^x never vanishes. The derivative quotient is

$$\frac{(f(x)e^x)'}{(e^x)'} = \frac{e^x(f'(x) + f(x))}{e^x} = f'(x) + f(x) \longrightarrow L.$$

Theorem 30.2 therefore gives $f(x) \to L$. It is unnecessary to know whether the numerator itself has a limit: Ross's denominator-infinity formulation does not require that.

Subtract two now-known finite limits.

Since $f'(x) = [f(x) + f'(x)] - f(x)$, ordinary limit arithmetic gives $f'(x) \to L - L = 0$.

Conclusion. Both conclusions follow without assuming a limit of f or f' at the outset.

30.7 Exercise 30.7: Why zeros of the denominator derivative cannot be canceled away

What we must show. Analyze $f(x) = x + \cos x \sin x$ and $g(x) = e^{\sin x} f(x)$ and explain the apparent L'Hospital paradox.

(a) Both functions tend to positive infinity.

Since $|\cos x \sin x| \le 1$, we have $f(x) \ge x - 1 \to \infty$. Also $e^{\sin x} \ge e^{-1} > 0$, so for $x > 1$,

$$g(x) \ge e^{-1}(x - 1) \longrightarrow \infty.$$

(b) Differentiate and factor.

Using $1 = \sin^2 x + \cos^2 x$,

$$f'(x) = 1 - \sin^2 x + \cos^2 x = 2\cos^2 x.$$

The product and chain rules give

$$\begin{aligned} g'(x) &= e^{\sin x} \cos x \, f(x) + e^{\sin x} \, 2\cos^2 x \\ &= e^{\sin x} \cos x [2\cos x + f(x)]. \end{aligned}$$

(c) Cancel only at points where cancellation is permitted.

If $x > 3$, then $2\cos x + f(x) \geq -2 + x - 1 = x - 3 > 0$. When also $\cos x \neq 0$, the denominator derivative is nonzero and

$$\frac{f'(x)}{g'(x)} = \frac{2e^{-\sin x}\cos x}{2\cos x + f(x)}.$$

If $\cos x = 0$, however, both derivatives vanish and the left side is undefined. The algebra does not define this original derivative quotient at those points.

(d) Distinguish a continuous extension from the actual quotient.

The expression on the right in part (c) is defined for every $x > 3$, and satisfies

$$\left|\frac{2e^{-\sin x}\cos x}{2\cos x + f(x)}\right| \leq \frac{2e}{x-3} \longrightarrow 0.$$

Nevertheless, $g'(\pi/2 + k\pi) = 0$ for arbitrarily large integers k. Thus there is no tail on which g' is everywhere nonzero, as required by L'Hospital's rule. For the original functions, $f(x) > 0$ on $x > 1$, and cancellation of the nonzero $f(x)$ gives

$$\frac{f(x)}{g(x)} = e^{-\sin x}.$$

Along $u_n = \pi/2 + 2\pi n$ this is e^{-1}; along $v_n = 3\pi/2 + 2\pi n$ it is e. Both sequences tend to infinity, and the values have different limits. Therefore f/g has no limit.

Conclusion. The extended derivative expression tends to zero, but the required nonvanishing hypothesis fails. There is no contradiction to L'Hospital's rule.

31 Taylor's theorem

A Taylor polynomial matches a function's derivatives at its center. Matching those derivatives does not by itself show that the infinite series represents the function. The missing step is to show that the remainder tends to zero. With Ross's convention the polynomial in R_n stops at degree $n-1$:

$$R_n(x) = f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!}(x-c)^k.$$

Taylor's theorem bounds this error using the nth derivative at a point between c and x. Later exercises use Newton's method, which replaces a curve by its tangent line and uses that line's zero as the next approximation.

31.1 Exercise 31.1: The cosine series and its error

What we must show. Find the Taylor series at 0 and prove that it equals $\cos x$ everywhere.

Compute the coefficients.

Successive derivatives of cosine are $\cos x, -\sin x, -\cos x, \sin x$, then repeat. At zero, the odd derivatives vanish and the $2k$th derivative is $(-1)^k$. Thus the Taylor series is

$$\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots.$$

Control the error, rather than assuming the series is correct.

Every derivative of cosine has absolute value at most 1. Taylor's theorem therefore gives $|R_n(x)| \leq |x|^n/n!$ for each fixed real x. This bound tends to zero: if $a_n = |x|^n/n!$, then $a_{n+1}/a_n = |x|/(n+1) \leq 1/2$ eventually (and $x = 0$ is immediate). The tail is consequently bounded by a geometric sequence tending to zero.

Conclusion. The displayed series equals $\cos x$ for every $x \in \mathbb{R}$.

31.2 Exercise 31.2: The hyperbolic sine and cosine series

What we must show. Obtain series at zero for $\sinh x = (e^x - e^{-x})/2$ and $\cosh x = (e^x + e^{-x})/2$, valid on all of $\mathbb{R}$.

Add and subtract the exponential series.

For each real x, the exponential series is absolutely convergent: $e^x = \sum_{k\geq 0} x^k/k!$ and $e^{-x} = \sum_{k\geq 0}(-1)^k x^k/k!$. Subtracting cancels the even terms; adding cancels the odd terms. Hence

$$\sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}, \qquad \cosh x = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}.$$

Termwise addition here is legitimate because it is addition of convergent series at a fixed x.

Check that these really are Taylor coefficients.

Differentiation interchanges sinh and cosh. Their values at zero are 0 and 1, respectively. This gives exactly the displayed coefficients. Alternatively, on $[-M, M]$ all derivatives of either function are bounded by e^M, so Taylor's bound is $e^M |x|^n / n! \to 0$. For any given x, choose $M > |x|$; this proves validity at that x.

Conclusion. Both series represent their functions at every real number.

31.3 Exercise 31.3: Why the common derivative bound is unavailable

What we must show. Explain the theorem choice for $f(x) = \log(1 + x)$ in Example 2.

Inspect the hypothesis of Corollary 31.4.

That corollary requires a single constant bounding *all* derivatives on the interval in question. But

$$f^{(n)}(x) = (-1)^{n+1}(n-1)!(1+x)^{-n}, \qquad \left|f^{(n)}(0)\right| = (n-1)!.$$

Even at the center 0 these numbers have no common finite bound. No interval containing zero can satisfy the corollary's hypothesis.

Use the more precise remainder instead.

For evaluating at $x = 1$, Taylor's theorem gives a $y_n \in (0, 1)$ with

$$|R_n(1)| = \frac{1}{n(1+y_n)^n} \le \frac{1}{n} \longrightarrow 0.$$

The factorial growth of the derivative is canceled by the $n!$ in the remainder. The corollary's stronger hypothesis would miss that cancellation.

Conclusion. Theorem 31.3 applies directly; the common-bound hypothesis of the corollary fails.

31.4 Exercise 31.4: Smooth cutoffs and a smooth step

What we must show. Construct the four requested infinitely differentiable functions.

Plan. Start from the smooth function in Example 3,

$$\phi(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \le 0. \end{cases}$$

It is important that all its derivatives at zero vanish; this is what makes the joining point smooth, despite the piecewise formula.

(a) Move the start of the positive part to a.

Set $f_a(x) = \phi(x - a)$. A translation preserves smoothness. If $x \le a$ its argument is nonpositive, so $f_a(x) = 0$; if $x > a$, its argument is positive and $f_a(x) > 0$.

(b) Reverse the direction and move the endpoint to b.

Set $g_b(x) = \phi(b - x)$. The affine change of variable preserves smoothness. This is zero for $x \ge b$ and strictly positive for $x < b$.

(c) Require both positivity conditions.

Set $h_{a,b}(x) = f_a(x)g_b(x)$. Products of smooth functions are smooth. Both factors are positive exactly when $a < x < b$. Outside this interval at least one factor is zero, so the product is zero.

(d) Turn the two cutoffs into a step.

Set

$$h^*_{a,b}(x) = \frac{f_a(x)}{f_a(x) + g_b(x)}.$$

The denominator never vanishes. Indeed, if $x \le a$ then $x < b$ and $g_b(x) > 0$; if $x \ge b$ then $x > a$ and $f_a(x) > 0$; between a and b both are positive. The permitted quotient rule for smooth functions therefore applies. For $x \le a$ the quotient is 0; for $x \ge b$ its numerator equals its denominator, so it is 1.

Conclusion. These four explicit formulas have all the required properties, including smoothness at a and b.

31.5 Exercise 31.5: A nonzero smooth function with zero Taylor series

What we must show. For $g(x) = e^{-1/x^2}$ when $x \neq 0$, $g(0) = 0$, find all derivatives at zero and compare g with its Taylor series.

(a) Recognize a smooth composition.

Using ϕ from Example 3, we have $g(x) = \phi(x^2)$ for every x. Since both ϕ and $x \mapsto x^2$ are smooth, so is g. Repeated chain and product rules show that each positive-order derivative of this composition is a finite sum of polynomial factors times $\phi^{(j)}(x^2)$, for integers $j \ge 1$. At zero every such term vanishes, because $\phi^{(j)}(0) = 0$. Also $g(0) = 0$. Thus $g^{(n)}(0) = 0$ for all $n = 0, 1, 2, \ldots$.

(b) Compare the function and the series point by point.

Every coefficient $g^{(n)}(0)/n!$ is zero, so the Taylor series is the identically zero series. It equals g at 0. For $x \neq 0$, $e^{-1/x^2} > 0$, so it cannot equal $g(x)$ there.

Conclusion. Smoothness alone does not imply agreement with a Taylor series, even arbitrarily close to its center.

31.6 Exercise 31.6: A telescoping proof of Taylor's theorem

What we must show. Verify the older proof using its auxiliary function F. Fix $x > 0$. As in the theorem's proof, define M by

$$f(x) = \sum_{k=0}^{n-1} \frac{x^k}{k!} f^{(k)}(0) + \frac{Mx^n}{n!}.$$

(a) Differentiate and cancel adjacent terms.

Write the auxiliary function as

$$F(t) = \sum_{k=0}^{n-1} \frac{(x-t)^k}{k!} f^{(k)}(t) + M\frac{(x-t)^n}{n!}, \qquad 0 \le t \le x.$$

For $k \geq 1$, the product rule gives

$$\frac{d}{dt}\left(\frac{(x-t)^k}{k!}f^{(k)}(t)\right) = -\frac{(x-t)^{k-1}}{(k-1)!}f^{(k)}(t) + \frac{(x-t)^k}{k!}f^{(k+1)}(t).$$

The negative term cancels the positive term from index $k-1$. Only the last positive term and the derivative of the M term remain:

$$F'(t) = \frac{(x-t)^{n-1}}{(n-1)!}(f^{(n)}(t) - M).$$

All needed derivatives exist on an open interval containing $[0, x]$; in particular F is continuous on the closed interval and differentiable in its interior. For $n = 1$ the formula follows directly from $F(t) = f(t) + M(x-t)$.

(b) Evaluate the endpoints.

The definition of M says $F(0) = f(x)$. At $t = x$ all positive powers of $x-t$ vanish, leaving $F(x) = f(x)$. Thus $F(0) = F(x)$.

(c) Apply Rolle's theorem.

There is $y \in (0, x)$ with $F'(y) = 0$. Since $x - y > 0$, the prefactor in the derivative formula is nonzero; hence $f^{(n)}(y) = M$. Substituting into the defining equation for M yields Taylor's remainder.

Conclusion. The auxiliary function proves the required mean-value expression for the remainder.

31.7 Exercise 31.7: Newton's method for the square root of two

What we must show. Recover the recursion from Exercise 9.5. For $f(x) = x^2 - 2$, we have $f'(x) = 2x$. Newton's update at a nonzero t_n is

$$t_{n+1} = t_n - \frac{t_n^2 - 2}{2t_n} = \frac{2t_n^2 - t_n^2 + 2}{2t_n} = \frac{t_n^2 + 2}{2t_n}.$$

With initial point $t_1 = 1$ this is exactly the earlier sequence. Every iterate is positive, because a positive t_n gives a positive numerator and denominator. Thus no zero denominator is encountered.

Conclusion. The recursion is Newton's method for $x^2 - 2 = 0$, starting at 1.

31.8 Exercise 31.8: Two real roots of a quartic

What we must show. Count the real roots of $p(x) = x^4 + x^3 - 1$ and approximate each to six decimals.

(a) Establish the shape and count crossings.

The derivative is $p'(x) = x^2(4x + 3)$. It is negative for $x < -3/4$ and positive for $x > -3/4$, except for the isolated zero at $x = 0$. Thus p is strictly decreasing on $(-\infty, -3/4]$ and strictly increasing on $[-3/4, \infty)$. For strict increase across zero, apply the Mean Value theorem separately on either side and combine the inequalities. Now $p(-2) = 7 > 0$, $p(-1) = -1 < 0$, $p(0) = -1 < 0$, and $p(1) = 1 > 0$. The Intermediate Value theorem supplies one root in $(-2, -1)$ and one in $(0, 1)$. Strict monotonicity on the two branches rules out any others.

(b) Run the explicit update.

Newton's formula is

$$x_{n+1} = x_n - \frac{x_n^4 + x_n^3 - 1}{4x_n^3 + 3x_n^2}.$$

Two runs, retaining guard digits, give

n	$x_0 = -1.5$	$x_0 = 0.8$
0	−1.500000000000	0.800000000000
1	−1.398148148148	0.819758064516
2	−1.380747193802	0.819173040661
3	−1.380277903561	0.819172513397
4	−1.380277569098	0.819172513396

As a check independent of apparent convergence, direct substitution gives opposite signs at the endpoints of each interval $(-1.380277570, -1.380277568)$ and $(0.819172512, 0.819172515)$. Each interval contains the unique root on its branch and lies wholly within the rounding range for the answer below.

Conclusion. The two roots, rounded to six decimal places, are −1.380278 and 0.819173.

31.9 Exercise 31.9: Solving the cosine fixed-point equation

What we must show. Approximate the solution of $x = \cos x$ to six decimals. Set $f(x) = x - \cos x$. Since $f'(x) = 1 + \sin x > 0$ on $(0, \pi/2)$, the root there is unique. Newton's iteration is

$$x_{n+1} = x_n - \frac{x_n - \cos x_n}{1 + \sin x_n}.$$

Starting at $x_0 = 1$ gives successively

$$0.750363867840, \quad 0.739112890911, \quad 0.739085133385, \quad 0.739085133215.$$

The signs of f at 0.739085132 and 0.739085135 are negative and positive, respectively. Continuity and strict increase bracket the root, so the displayed digits are not justified solely by two similar iterates.

Conclusion. The root is $x = 0.739085$ to six decimal places.

31.10 Exercise 31.10: Solving an exponential equation

What we must show. Approximate the root of $xe^x = 2$ in $(0, 1)$ to six decimals. Use $f(x) = xe^x - 2$, with $f'(x) = (1 + x)e^x > 0$ on $(0, 1)$. Thus there is at most one root there; the opposite endpoint signs give existence by continuity. Newton's update is

$$x_{n+1} = x_n - \frac{x_n e^{x_n} - 2}{(1 + x_n)e^{x_n}}.$$

Starting with $x_0 = 1$ gives

$$0.867879441171, \quad 0.852783373416, \quad 0.852605526369, \quad 0.852605502014.$$

Direct substitution gives $f(0.852605501) < 0 < f(0.852605504)$, so the unique root is in that small interval.

Conclusion. The six-decimal value is 0.852606.

31.11 Exercise 31.11: A convergent Newton sequence must approach a zero

What we must show. Under the stated bounded-derivative hypotheses, prove $f(x) = 0$ when the Newton iterates converge to the interior point x.

Rearrange the recursion before taking limits.

Since $f'(x_{n-1}) \neq 0$, the update is defined and implies

$$f(x_{n-1}) = (x_{n-1} - x_n)f'(x_{n-1}).$$

Choose $B < \infty$ with $|f'(t)| \leq B$ throughout (a, b). Convergence of (x_n) means both x_n and x_{n-1} tend to x, so their difference tends to zero. Consequently

$$|f(x_{n-1})| \leq B\,|x_{n-1} - x_n| \longrightarrow 0.$$

Identify this limit using continuity.

Differentiability at the interior point x implies continuity there, so also $f(x_{n-1}) \to f(x)$. Uniqueness of a real limit gives $f(x) = 0$. We did not assume f' was continuous or bounded away from zero; the boundedness from above was the estimate actually used.

Conclusion. Every convergent sequence of these iterates with an interior limit converges to a root.

31.12 Exercise 31.12: An interior point is no farther than an endpoint

What we must show. If $a < c < b$, prove $|c - x| \leq \max\{|a - x|\,, |b - x|\}$ for every real x.

Case 1: $x \leq c$**.**

Then $c - x \geq 0$ and $b - x > 0$, so $|c - x| = c - x \leq b - x = |b - x|$.

Case 2: $x > c$**.**

Then $x - c > 0$ and $x - a > 0$, so $|c - x| = x - c \leq x - a = |a - x|$. These cases exhaust the real line. In either case the relevant endpoint distance is at most the maximum of the two endpoint distances.

Conclusion. The claimed inequality follows for all $x \in \mathbb{R}$.

32 The Riemann integral

For a bounded function f and a partition $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$, let $I_j = [t_{j-1}, t_j]$. The lower and upper sums are

$$L(f,P) = \sum_j (\inf_{I_j} f)\,|I_j|\,, \qquad U(f,P) = \sum_j (\sup_{I_j} f)\,|I_j|\,,$$

where $|I_j| = t_j - t_{j-1}$ denotes interval length. The lower integral is the supremum of all lower sums; the upper integral is the infimum of all upper sums. Write these as $\underline{I}(f)$ and $\overline{I}(f)$. Integrability means they are equal. A practical equivalent test is that $U(f,P) - L(f,P)$ can be made arbitrarily small. No antiderivative formula is assumed in the Darboux calculations below.

32.1 Exercise 32.1: The integral of a cube from sums

What we must show. Compute both Darboux integrals of x^3 on $[0,b]$, where $b > 0$.

Choose equal subintervals and identify maxima and minima.

The cube is increasing on $[0,b]$. For $t_k = kb/n$, its maximum on $[t_{k-1}, t_k]$ is t_k^3 and its minimum is t_{k-1}^3. Therefore

$$U_n = \frac{b^4}{n^4}\sum_{k=1}^{n} k^3 = \frac{b^4(n+1)^2}{4n^2}, \qquad L_n = \frac{b^4}{n^4}\sum_{k=0}^{n-1} k^3 = \frac{b^4(n-1)^2}{4n^2}.$$

Here we used $\sum_{k=1}^{n} k^3 = [n(n+1)/2]^2$.

Squeeze the integrals.

For every n, $L_n \le \underline{I}(f) \le \overline{I}(f) \le U_n$. Both outer sequences tend to $b^4/4$, because $(n \pm 1)^2/n^2 \to 1$. Thus the two integrals coincide at that value.

Conclusion. The function is integrable and $\int_0^b x^3\,dx = b^4/4$.

32.2 Exercise 32.2: A rational-only linear function

What we must show. For $f(x) = x$ on rational points and $f(x) = 0$ on irrational points of $[0,b]$, compute its lower and upper integrals.

Use density on each interval.

Every nondegenerate interval $[u,v] \subseteq [0,b]$ contains irrational points, so its infimum is zero. Its rational points can approach v from below; their values approach v, while no value exceeds v. Thus its supremum is v, whether or not v itself is rational. For any partition, consequently,

$$L(f,P) = 0, \qquad U(f,P) = \sum_{j=1}^{n} t_j(t_j - t_{j-1}).$$

In particular, the lower integral is 0.

Find the infimum of the upper sums.

Since $t_j \geq t_{j-1} \geq 0$,

$$t_j(t_j - t_{j-1}) \geq \frac{t_j + t_{j-1}}{2}(t_j - t_{j-1}) = \frac{t_j^2 - t_{j-1}^2}{2}.$$

Summing telescopes to $U(f, P) \geq b^2/2$. For the equal partition,

$$U(f, P_n) = \frac{b^2}{n^2} \sum_{j=1}^{n} j = \frac{b^2(n+1)}{2n} \longrightarrow \frac{b^2}{2}.$$

An infimum bounded below by $b^2/2$ and approached by such sums equals $b^2/2$.

Conclusion. (a) The lower integral is 0 and the upper integral is $b^2/2$. (b) They differ for $b > 0$, so f is not integrable.

32.3 Exercise 32.3: A rational-only quadratic function

What we must show. Repeat the calculation for $g(x) = x^2$ on rational points and $g(x) = 0$ on irrational points of $[0, b]$.

Identify each interval's extrema.

Density and nonnegativity give $\inf_{[u,v]} g = 0$ and $\sup_{[u,v]} g = v^2$. Thus all lower sums vanish and $U(g, P) = \sum_j t_j^2(t_j - t_{j-1})$.

Compute the infimum of these upper sums.

For $0 \leq u \leq v$ we have $v^2 \geq (v^2 + vu + u^2)/3$, whence

$$v^2(v - u) \geq \frac{v^3 - u^3}{3}.$$

This telescopes to $U(g, P) \geq b^3/3$ for every P. For equal subintervals,

$$U(g, P_n) = \frac{b^3}{n^3} \sum_{j=1}^{n} j^2 = \frac{b^3 n(n+1)(2n+1)}{6n^3} \longrightarrow \frac{b^3}{3}.$$

Conclusion. (a) The lower and upper integrals are 0 and $b^3/3$. (b) The function is not integrable on $[0, b]$ for $b > 0$.

32.4 Exercise 32.4: Refinement by induction

What we must show. Extend the one-point refinement argument in Lemma 32.2 to any finite refinement $P \subseteq Q$.

State precisely what is being inducted on.

Let r be the number of points of Q not already in P. We claim

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P).$$

For $r = 0$ the partitions are equal; the middle inequality follows because each interval's infimum is at most its supremum.

Insert one more point.

Suppose the result is established for r extra points. If Q has $r+1$ extra points, choose one of them, q, and set $Q' = Q \setminus \{q\}$. By the induction hypothesis, $L(f,P) \le L(f,Q')$ and $U(f,Q') \le U(f,P)$. The proved one-point case, applied to Q' and Q, gives $L(f,Q') \le L(f,Q)$ and $U(f,Q) \le U(f,Q')$. Combining the inequalities proves the two outer comparisons for Q. The middle comparison always holds interval by interval.

Conclusion. Adding finitely many partition points can only increase a lower sum and decrease an upper sum.

32.5 Exercise 32.5: Why the lower integral cannot exceed the upper

What we must show. Apply the order property of suprema and infima to Theorem 32.4.

Name the two sets.

Let

$$S = \{L(f,P) : P \text{ is a partition of } [a,b]\}, \qquad T = \{U(f,Q) : Q \text{ is a partition of } [a,b]\}.$$

Both are nonempty and bounded because f is bounded and the total interval length is $b-a$.

Compare sums from different partitions.

For any P, Q, their union $R = P \cup Q$ refines both. Lemma 32.2 gives

$$L(f,P) \le L(f,R) \le U(f,R) \le U(f,Q).$$

Thus every $s \in S$ is at most every $t \in T$. By Exercise 4.8, $\sup S \le \inf T$. To see the order reasoning directly, each $t \in T$ is an upper bound for S, so $\sup S \le t$ for all $t \in T$; therefore $\sup S$ is a lower bound for T, and hence is at most $\inf T$.

Conclusion. $\underline{I}(f) = \sup S \le \inf T = \overline{I}(f)$.

32.6 Exercise 32.6: Upper and lower sums whose gap tends to zero

What we must show. Prove integrability and identify the limits of the given sums. The sums U_n and L_n need not come from the same partition. Nevertheless, the cross-partition comparison just established gives

$$L_n \le \underline{I}(f) \le \overline{I}(f) \le U_n.$$

Subtracting shows $0 \le \overline{I}(f) - \underline{I}(f) \le U_n - L_n \to 0$. The fixed middle difference must therefore be zero. Write the common integral as I. Then

$$0 \le U_n - I \le U_n - L_n, \qquad 0 \le I - L_n \le U_n - L_n.$$

Each right-hand side tends to zero, so both $U_n \to I$ and $L_n \to I$.

Conclusion. f is integrable and $\int_a^b f = \lim_n U_n = \lim_n L_n$.

32.7 Exercise 32.7: Changing finitely many values

What we must show. Show that changing an integrable function at finitely many points preserves its integral.

First understand the finite-support error.

Put $h = g - f$. It is zero except at finitely many points $c_1, \dots, c_m$, so it is bounded; choose $H > 0$ with $|h| \le H$. Given any $\eta > 0$, make a partition that contains the exceptional points and nearby points on each available side, so the total length of cells meeting these points is less than η. This is possible because there are only finitely many points. If a point is a or b, only its one adjacent side is needed. On every other cell $h = 0$; on the exceptional cells $-H \le h \le H$. Thus

$$-H\eta \le L(h, P) \le U(h, P) \le H\eta.$$

This proves that a finite-support function has lower and upper integrals both zero, by letting $\eta \downarrow 0$.

Combine the estimate with the original integrable function.

Here is a direct argument that does not assume a later linearity theorem. Given $\varepsilon > 0$, choose a partition for f with upper-minus-lower gap less than $\varepsilon/2$. Refine it to the partition just described, with exceptional-cell total length less than η and $2H\eta < \varepsilon/2$. Refinement preserves the bound for f. On any cell J, $g = f + h$ implies

$$\sup_J g \le \sup_J f + \sup_J h, \qquad \inf_J g \ge \inf_J f + \inf_J h.$$

Consequently $U(g, P) - L(g, P) < \varepsilon/2 + 2H\eta < \varepsilon$. This proves g is integrable.

Compare the two integral values.

For the same partition, $L(g, P) \ge L(f, P) - H\eta$ and $U(g, P) \le U(f, P) + H\eta$. Both integrals lie between the respective sums. Hence their difference in absolute value is at most $U(f, P) - L(f, P) + H\eta$. The original gap and η can both be made arbitrarily small.

Conclusion. $\int_a^b g = \int_a^b f$. Isolated values do not change area.

32.8 Exercise 32.8: Restriction to a smaller interval

What we must show. Prove integrability of f on $[c, d] \subseteq [a, b]$. Assume $c < d$; the zero-length case has integral zero by convention. Given $\varepsilon > 0$, choose a partition P of $[a, b]$ with $U(f, P) - L(f, P) < \varepsilon$. Add c and d to form the refinement Q. Its gap is still less than ε. The points of Q lying in $[c, d]$ form a partition Q_0 of that interval. Its gap is a sub-sum of the gap for Q:

$$U_{[c,d]}(f, Q_0) - L_{[c,d]}(f, Q_0) = \sum_{J \subseteq [c,d]} (\sup_J f - \inf_J f)\,|J| \le U_{[a,b]}(f, Q) - L_{[a,b]}(f, Q).$$

Every term omitted on the right is nonnegative. Thus the smaller interval has partitions with gap below every positive ε.

Conclusion. The Darboux criterion proves that $f|_{[c,d]}$ is integrable.

33 Properties of the Riemann integral

The Darboux criterion converts integrability into an estimate of the total oscillation of a function over short intervals. Once integrability has been established, linearity, comparison, and interval additivity make calculations much easier. Keep two issues separate: first justify that an integral exists; then use its algebraic properties. A pointwise limit alone does not usually allow us to move a limit through an integral.

33.1 Exercise 33.1: Decreasing functions are integrable

What we must show. Complete the monotone-function integrability proof. Let f be decreasing on $[a, b]$, meaning nonincreasing. It is bounded between $f(b)$ and $f(a)$. If these values agree, f is constant and every upper and lower sum agrees. Otherwise $D = f(a) - f(b) > 0$. For a partition $P = \{t_0, \dots, t_n\}$, decreasingness gives

$$U(f, P) - L(f, P) = \sum_{j=1}^{n} [f(t_{j-1}) - f(t_j)](t_j - t_{j-1}) \le \operatorname{mesh}(P) \sum_{j=1}^{n} [f(t_{j-1}) - f(t_j)].$$

The last sum telescopes to D. Given $\varepsilon > 0$, choose an equal partition with mesh less than ε/D. Then the gap is below ε.

Conclusion. The Darboux criterion proves integrability, including the constant case.

33.2 Exercise 33.2: Scaling suprema and infima

What we must show. For a nonempty bounded $S \subseteq \mathbb{R}$ and $c > 0$, prove $\sup(cS) = c \sup S$ and $\inf(cS) = c \inf S$.

Prove the supremum statement using both defining properties.

Write $M = \sup S$. Every $s \in S$ satisfies $s \le M$; multiplication by the positive number c gives $cs \le cM$. Thus cM is an upper bound for cS. To show it is the least one, let $\varepsilon > 0$. Since $M - \varepsilon/c < M$, it is not an upper bound for S, so there is $s \in S$ with $M - \varepsilon/c < s \le M$. Multiplying gives $cM - \varepsilon < cs \le cM$. Hence no number below cM can be an upper bound for cS, proving the desired equality.

Repeat with lower bounds.

If $m = \inf S$, then $m \le s$ implies $cm \le cs$, so cm is a lower bound for cS. Given $\varepsilon > 0$, the number $m + \varepsilon/c$ is not a lower bound for S. Thus some $s \in S$ satisfies $m \le s < m + \varepsilon/c$, and $cm \le cs < cm + \varepsilon$. No number greater than cm is a lower bound for cS.

Conclusion. Both scaling identities follow; positivity of c was needed to preserve the inequality directions.

33.3 Exercise 33.3: Step functions and postage

What we must show. Integrate a finite step function and the postage-stamp example.

(a) Work one subinterval at a time.

Suppose $f = c_j$ on (u_{j-1}, u_j) for the finite partition in the exercise. On the closed interval $[u_{j-1}, u_j]$, the function differs from the constant c_j at most at its two endpoints. Exercise 32.7 therefore proves integrability there and gives integral $c_j(u_j - u_{j-1})$. Interval additivity, applied successively to the finitely many intervals, proves

$$\int_a^b f(x)\,dx = \sum_{j=1}^{m} c_j(u_j - u_{j-1}).$$

There is no condition on the finitely many partition-point values, except that they are real numbers.

(b) Apply the formula to the postage function.

On the four open unit intervals between 0 and 4, the values of P are $A, A+B, A+2B, A+3B$. Each interval has length 1; the value at 4 does not change the integral. Consequently

$$\int_0^4 P(x)\,dx = A + (A+B) + (A+2B) + (A+3B) = 4A + 6B.$$

Conclusion. Step functions are integrable, with integral equal to the sum of height times width; the postage integral is $4A + 6B$.

33.4 Exercise 33.4: An integrable absolute value does not suffice

What we must show. Find a nonintegrable f on $[0, 1]$ with integrable $|f|$. Set $f(x) = 1$ at rational points and $f(x) = -1$ at irrational points. Every nondegenerate interval contains both sorts of points. Thus every upper sum is 1 and every lower sum is -1; the two Darboux integrals differ, so f is not integrable. However, $|f(x)| = 1$ for every x, and a constant function is integrable, with integral 1 on $[0, 1]$.

Conclusion. This example shows that the converse of integrability of f implying integrability of $|f|$ is false.

33.5 Exercise 33.5: Bounding an integral without evaluating it

What we must show. Bound the absolute integral of $x^2 \sin^8(e^x)$ on $[-2\pi, 2\pi]$. The integrand is continuous, hence integrable. Since $0 \le \sin^8(e^x) \le 1$, we have $|x^2 \sin^8(e^x)| \le x^2$. The absolute-integral inequality and comparison give

$$\left|\int_{-2\pi}^{2\pi} x^2 \sin^8(e^x)\,dx\right| \le \int_{-2\pi}^{2\pi} x^2\,dx = 2\frac{(2\pi)^3}{3} = \frac{16\pi^3}{3}.$$

The integral of the square follows from the Darboux calculation in Section 32 (or, once established, the Fundamental Theorem).

Conclusion. The requested bound is $16\pi^3/3$.

33.6 Exercise 33.6: Absolute value does not increase oscillation

What we must show. Show, for a nonempty set S on which f is bounded,

$$\sup_S |f| - \inf_S |f| \le \sup_S f - \inf_S f.$$

Compare two values before taking extrema.

For arbitrary $x, y \in S$, the reverse triangle inequality gives

$$|f(x)| - |f(y)| \le |f(x) - f(y)| \le M - m, \quad M = \sup_S f, \quad m = \inf_S f.$$

The last inequality holds because both $f(x)$ and $f(y)$ lie in $[m, M]$. Fix y and take the supremum over x. This yields $\sup_S |f| \le |f(y)| + M - m$ for every $y \in S$. Taking the infimum over y gives $\sup_S |f| \le \inf_S |f| + M - m$. No maximum or minimum needs to be attained.

Conclusion. Rearrangement gives exactly the oscillation inequality needed in the proof of Theorem 33.5.

33.7 Exercise 33.7: Squaring preserves integrability

What we must show. Prove the Darboux-gap estimate and deduce integrability of f^2. Assume $|f(x)| \le B$ on $[a, b]$, with $B > 0$.

(a) Estimate the oscillation on one cell.

For any x, y in a partition cell J,

$$\left|f(x)^2 - f(y)^2\right| = |f(x) + f(y)|\,|f(x) - f(y)| \le 2B(\sup_J f - \inf_J f).$$

Taking the supremum over pairs $x, y \in J$ gives

$$\sup_J f^2 - \inf_J f^2 \le 2B(\sup_J f - \inf_J f).$$

Indeed, the difference of a supremum and an infimum equals the supremum of all differences; the values need not be attained. Multiply by the length of J and sum over the partition to obtain

$$U(f^2, P) - L(f^2, P) \le 2B[U(f, P) - L(f, P)].$$

(b) Use the integrability criterion.

If f is integrable, choose P with $U(f, P) - L(f, P) < \varepsilon/(2B)$. The estimate makes the gap for f^2 smaller than ε. Also f^2 is bounded by B^2, as required by the criterion.

Conclusion. f^2 is integrable whenever f is integrable.

33.8 Exercise 33.8: Products, maxima, and minima

What we must show. Prove integrability of three combinations of integrable f, g.

(a) Turn a product into a linear combination of squares.

Linearity already proves that $f + g$ and $f - g$ are integrable. By Exercise 33.7 their squares are integrable. Since

$$fg = \frac{(f + g)^2 - (f - g)^2}{4},$$

linearity proves integrability of fg. This argument establishes the existence of the integral, without claiming that an integral of a product is a product of integrals.

(b) Express maxima and minima using absolute value.

For any two real numbers u, v, $\max(u,v) = (u+v+|u-v|)/2$ and $\min(u,v) = (u+v-|u-v|)/2$. Check these identities separately when $u \geq v$ and $u < v$. Substituting $f(x), g(x)$ gives pointwise formulas. The absolute value of the integrable function $f-g$ is integrable by Theorem 33.5, and linearity now proves both conclusions.

Conclusion. The functions fg, $\max(f,g)$, and $\min(f,g)$ are integrable.

33.9 Exercise 33.9: Uniform limits of integrable functions

What we must show. Show that a uniform limit f is integrable and its integral is the limit of the integrals.

First check boundedness.

Choose one index m such that $|f - f_m| < 1$ everywhere. Since f_m is bounded, this also bounds f.

Make a Darboux gap small.

Given $\varepsilon > 0$, choose m with $\sup_{[a,b]} |f - f_m| < \varepsilon/[4(b-a)]$. Call this upper bound η. For every cell J,

$$\sup_J f \leq \sup_J f_m + \eta, \qquad \inf_J f \geq \inf_J f_m - \eta.$$

Since f_m is integrable, choose P with its gap below $\varepsilon/2$. Then

$$U(f,P) - L(f,P) \leq U(f_m,P) - L(f_m,P) + 2\eta(b-a) < \varepsilon.$$

The Darboux criterion proves integrability of f.

Estimate the difference of integrals.

Now $f - f_n$ is integrable, and

$$\left| \int_a^b f - \int_a^b f_n \right| \leq \int_a^b |f - f_n| \leq (b-a) \sup_{[a,b]} |f - f_n| \longrightarrow 0.$$

Uniform convergence is what makes the final supremum tend to zero. Theorem 25.2 treated continuous functions; this proof needs only integrability. The dominated-convergence result has weaker convergence hypotheses but additional boundedness and limit-integrability assumptions.

Conclusion. Uniform limits preserve Riemann integrability and allow the limit to pass through the integral.

33.10 Exercise 33.10: An oscillatory function that is still integrable

What we must show. Prove integrability on $[-1,1]$ of $f(x) = \sin(1/x)$ for $x \neq 0$, with $f(0) = 0$.

Plan. Isolate the only troublesome point in a short interval. Given $\varepsilon > 0$, choose $0 < \delta < 1$ with $4\delta < \varepsilon/2$. The function is continuous on $[-1,-\delta]$ and $[\delta,1]$, so those intervals have partitions whose

combined Darboux gaps are less than $\varepsilon/2$. Combine these partitions with the middle cell $[-\delta, \delta]$. On that cell $-1 \le f \le 1$, so its gap contribution is at most $2(2\delta) = 4\delta < \varepsilon/2$. The complete partition therefore has gap below ε.

Conclusion. The function is Riemann integrable. Infinitely many oscillations near one point can be confined to intervals of arbitrarily small length.

33.11 Exercise 33.11: An integrable function beyond the piecewise criteria

What we must show. Study $f(x) = x\,\mathrm{sgn}(\sin(1/x))$ for $x \neq 0$ and $f(0) = 0$, with $\mathrm{sgn}(0) = 0$.

(a) Rule out piecewise continuity.

At $x_k = 1/(k\pi) > 0$, the sine changes sign and the two one-sided limits of f are x_k and $-x_k$, in one order or the other. They differ because $x_k > 0$, so each x_k is a discontinuity. Every interval $(0, \delta)$ contains infinitely many of these points. A finite partition always leaves an interval $(0, \delta)$ without any interior partition point, and continuity on its interior is impossible. Thus f is not piecewise continuous on $[-1, 1]$.

(b) Rule out piecewise monotonicity.

The sign of $\sin(1/x)$ alternates on the infinitely many successive intervals between the x_k. In any $(0, \delta)$ choose ordered points $u < v < w$ in three consecutive sign intervals with signs $+, -, +$. Then $f(u) > 0$, $f(v) < 0$, and $f(w) > 0$. These values are neither nondecreasing nor nonincreasing. Hence f is not monotonic on any $(0, \delta)$, which again rules out any finite monotonicity partition.

(c) Estimate the Darboux gap directly.

Given $\varepsilon > 0$, choose $\delta \in (0, 1)$ with $4\delta^2 < \varepsilon/2$. On each of $[-1, -\delta]$ and $[\delta, 1]$ there are only finitely many zeros of $\sin(1/x)$. Between them the function equals x or $-x$; it is thus a finite piecewise continuous function, hence integrable there. Choose outer partitions with combined gaps below $\varepsilon/2$. On $[-\delta, \delta]$, $|f| \le |x| \le \delta$, so the gap contribution is at most $(2\delta)(2\delta) = 4\delta^2$. The combined gap is below ε.

Conclusion. f is neither piecewise continuous nor piecewise monotonic, but it is Riemann integrable. The two piecewise conditions are sufficient, not necessary.

33.12 Exercise 33.12: The rational-denominator function has zero integral

What we must show. For the function of Exercise 17.14, prove the two non-piecewise claims and compute the integral on every $[a, b]$. Recall that $f = 0$ at irrationals and $f(p/q) = 1/q$ for rationals in lowest terms with $q > 0$, including $f(0) = 1$.

(a) Examine every nondegenerate subinterval.

The function is discontinuous at every rational point: its value there is positive, whereas nearby irrational values are zero. Every nondegenerate interval contains a rational, so it cannot be an interval of continuity. Thus no finite continuity partition exists. It cannot be monotonic on any such interval either: choose a rational r strictly inside it and irrational points $u < r < v$ in it. Then $f(u) = f(v) = 0 < f(r)$, violating both possible monotonicities. Thus a finite monotonicity partition is also impossible.

(b) Isolate the finitely many large values.

Every cell contains an irrational, so every lower sum is zero. To make an upper sum small, let $\varepsilon > 0$ and choose N with $(b-a)/N < \varepsilon/2$. There are only finitely many rationals in $[a, b]$ with denominator $q \le N$: for each fixed q, only finitely many integers p have $a \le p/q \le b$. Make a partition whose cells meeting these finitely many points have total length less than $\varepsilon/2$. On these cells, use the bound $f \le 1$. On all other cells, all nonzero values have denominator $q > N$, and hence $f \le 1/N$. Therefore

$$U(f, P) \le 1 \cdot \frac{\varepsilon}{2} + \frac{1}{N}(b-a) < \varepsilon.$$

This proves the upper integral is also zero.

Conclusion. f is integrable on every bounded closed interval and $\int_a^b f = 0$, although it is neither of the two piecewise types.

33.13 Exercise 33.13: Equal integrals force an interior intersection

What we must show. For continuous f, g with equal integrals on $[a, b]$, find $x \in (a, b)$ with $f(x) = g(x)$. Let $h = f - g$. It is continuous and $\int_a^b h = 0$. Suppose h never vanishes in (a, b). By the Intermediate Value theorem it cannot take both a positive and a negative value there, since that would force a zero between them. Thus it is either positive everywhere in (a, b) or negative everywhere there. In the positive case, continuity gives $h \ge 0$ at the endpoints as well. At any chosen interior point x_0, continuity gives a small interval of positive length on which $h \ge h(x_0)/2 > 0$. Its integral is strictly positive, and the rest is nonnegative, contradicting the zero integral. Apply the same reasoning to $-h$ in the negative case.

Conclusion. The assumption was impossible, so $f(x) = g(x)$ for some interior x.

33.14 Exercise 33.14: A weighted mean-value theorem

What we must show. Prove the weighted formula, identify the ordinary case, and test the sign hypothesis.

(a) Separate the zero-weight case.

Let $G = \int_a^b g \ge 0$. If $G = 0$, nonnegative continuous g must be identically zero: a positive value would, by continuity, contribute a positive integral on a short interval. Thus $\int_a^b fg = 0$ too, and every interior x satisfies the formula. If $G > 0$, put $c = G^{-1} \int_a^b fg$. We need $f(x) = c$ at an interior point. Suppose no such point exists. As in Exercise 33.13, continuity forces $f - c$ to have one strict sign throughout (a, b). Because $G > 0$, g is positive at some interior point t_0; otherwise it would vanish on (a, b) and hence also at both endpoints. Thus $(f - c)g$ has one weak sign throughout $[a, b]$ and the corresponding strict sign on a neighborhood of t_0. Its integral cannot be zero. But linearity gives

$$\int_a^b (f - c)g = \int_a^b fg - cG = 0,$$

a contradiction. Therefore an interior x has $f(x) = c$, and $\int_a^b fg = f(x) \int_a^b g$.

(b) Recover the ordinary integral mean-value theorem.

Take $g \equiv 1$, so $G = b - a$. The formula becomes $f(x) = (b-a)^{-1} \int_a^b f$, exactly Theorem 33.9.

(c) Test a weight that changes sign.

On $[-1,1]$, set $f(t)=g(t)=t$. Then $\int_{-1}^{1} fg=\int_{-1}^{1} t^2\,dt=2/3$, but $f(x)\int_{-1}^{1} g=x\cdot 0=0$ for every x.

Conclusion. The weighted theorem holds for nonnegative continuous weights; part (c) shows that dropping the sign condition can destroy it.

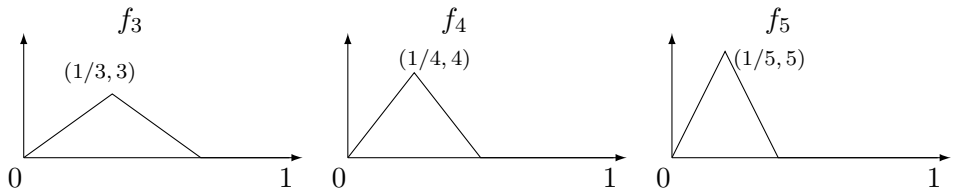
33.15 Exercise 33.15: Narrow triangular peaks

What we must show. Sketch the first three peaks, find their pointwise limit, and compare limits of integrals. The line-segment description gives the convenient formula

$$f_n(x)=\begin{cases} n^2x, & 0\le x\le 1/n,\\ 2n-n^2x, & 1/n\le x\le 2/n,\\ 0, & 2/n\le x\le 1. \end{cases}$$

(a) Draw the vertices and join them by straight segments.

The following sketches use the same horizontal and vertical scales; the vertical unit is 8 picture units and the horizontal unit is 100. The vertices are $(0,0),(1/n,n),(2/n,0),(1,0)$.



(b) Fix the point before choosing how large n must be.

At $x=0$, every value is zero. For a fixed $x>0$, choose $N>2/x$. If $n\ge N$, then $2/n<x$ and the last branch of the formula gives $f_n(x)=0$. Thus $f_n(x)\to 0$ for every $x\in[0,1]$.

(c) Compute areas and check the theorem's hypothesis.

Each graph is a triangle of base $2/n$ and height n, followed by a zero segment. Its integral is

$$\int_0^1 f_n(x)\,dx=\frac{1}{2}\frac{2}{n}\,n=1.$$

Hence the limit of the integrals is 1, whereas the integral of the pointwise limit is 0. There is no contradiction with Ross's Dominated Convergence theorem: it requires a single constant M bounding $|f_n(x)|$ for all n,x, but $f_n(1/n)=n$ is unbounded as n varies.

Conclusion. Pointwise convergence to zero does not force the integrals to tend to zero; the increasingly tall peaks retain area 1.

34 Fundamental Theorem of Calculus

There are two complementary facts to keep straight. If f is continuous, then $H(x) = \int_a^x f(t)\,dt$ is differentiable and $H'(x) = f(x)$. If $G' = f$ and the relevant integrability hypotheses hold, then $\int_a^b f = G(b) - G(a)$. An accumulation integral can remain continuous even at a jump of its integrand, although it may then have a corner. For reversed endpoints we use $\int_b^a f = -\int_a^b f$.

34.1 Exercise 34.1: Recovering the antiderivative evaluation formula

What we must show. Deduce $\int_a^b g' = g(b) - g(a)$ when g' is continuous. Define $F(x) = \int_a^x g'(t)\,dt$ for $x \in [a, b]$. Continuity of g' guarantees that it is integrable and, by Theorem 34.3, $F'(x) = g'(x)$ for interior x. Thus $(F - g)' = 0$ on (a, b). Both F and g are continuous on the closed interval, so Corollary 29.5 (the zero-derivative consequence of the Mean Value theorem) says $F - g$ is constant there. At a, $F(a) - g(a) = -g(a)$; hence $F(b) - g(b) = -g(a)$. Rearranging gives

$$\int_a^b g'(t)\,dt = F(b) = g(b) - g(a).$$

Conclusion. The continuous-integrand version of the Fundamental Theorem implies the antiderivative evaluation formula in this case.

34.2 Exercise 34.2: Two local averages of an exponential

What we must show. Calculate the two limits using derivatives of accumulation integrals.

(a) Center the integral at zero.

Let $H(x) = \int_0^x e^{t^2}\,dt$. The integrand is continuous, so $H'(0) = e^0 = 1$ and $H(0) = 0$. By the definition of derivative,

$$\lim_{x\to 0} \frac{1}{x} \int_0^x e^{t^2}\,dt = \lim_{x\to 0} \frac{H(x) - H(0)}{x - 0} = H'(0) = 1.$$

The signed-integral convention makes this valid from both sides.

(b) Center the integral at three.

Let $K(y) = \int_3^y e^{t^2}\,dt$. Then $K(3) = 0$ and $K'(3) = e^9$. Therefore

$$\lim_{h\to 0} \frac{1}{h} \int_3^{3+h} e^{t^2}\,dt = \lim_{h\to 0} \frac{K(3 + h) - K(3)}{h} = e^9.$$

Conclusion. The limits are 1 and e^9, respectively.

34.3 Exercise 34.3: A continuous accumulation function with one corner

What we must show. Integrate the piecewise function, sketch its accumulation function, and locate all derivatives. The integrand is 0 for $t < 0$, t for $0 \le t \le 1$, and 4 for $t > 1$.

(a) Integrate over the pieces actually traversed.

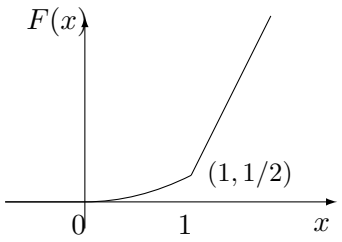
For $x \le 0$, $F(x) = \int_0^x 0\,dt = 0$. For $0 \le x \le 1$, $F(x) = \int_0^x t\,dt = x^2/2$. For $x \ge 1$, split at 1:

$$F(x) = \int_0^1 t\,dt + \int_1^x 4\,dt = \tfrac{1}{2} + 4(x-1) = 4x - \tfrac{7}{2}.$$

The isolated value of the integrand at 1 has no effect on the latter integral.

(b) Join the graph pieces at their actual heights.

The graph is horizontal at height zero to the left of 0, then follows a parabola to $(1, 1/2)$, then a straight line of slope 4. At 0 both neighboring formulas give 0; at 1 both give $1/2$. The formulas are continuous on their separate pieces, so F is continuous at every real number.



The drawing scales the horizontal coordinate by 40 and the vertical coordinate by 20, so the right line has its correct slope in data coordinates.

(c) Check the joining points separately.

Away from $0, 1$, differentiation gives $0, x, 4$ on the three open pieces. At zero, the difference quotient is 0 for negative increments and $h/2$ for positive increments, so $F'(0) = 0$. At 1 the left-hand difference quotient tends to 1 and the right-hand one tends to 4. Thus no derivative exists there.

Conclusion. F is continuous on $\mathbb{R}$ and differentiable precisely on $\mathbb{R} \setminus \{1\}$, with $F' = 0$ for $x \le 0$, $F' = x$ for $0 < x < 1$, and $F' = 4$ for $x > 1$.

34.4 Exercise 34.4: Two corners in another accumulation function

What we must show. Repeat the analysis when $f(t) = t$ for $t < 0$, $f(t) = t^2 + 1$ for $0 \le t \le 2$, and $f(t) = 0$ for $t > 2$.

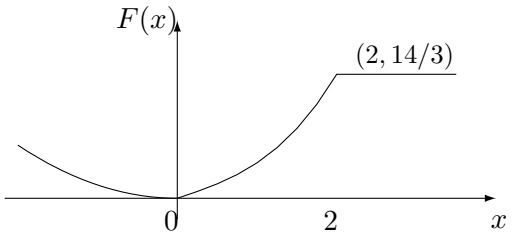
(a) Keep the orientation of the integral for negative x.

For $x < 0$, $F(x) = -\int_x^0 t\,dt = x^2/2$. For $0 \le x \le 2$, integration gives $F(x) = x^3/3 + x$. For $x > 2$, the integrand vanishes after 2, so $F(x) = F(2) = 8/3 + 2 = 14/3$. Thus

$$F(x) = \begin{cases} x^2/2, & x \le 0, \\ x^3/3 + x, & 0 \le x \le 2, \\ 14/3, & x \ge 2. \end{cases}$$

(b) Sketch and test continuity.

The left parabola decreases to $(0, 0)$; the middle cubic expression increases to $(2, 14/3)$; the remaining graph is horizontal. The two formulas at zero both give zero, and those at 2 both give $14/3$. Thus the function is continuous everywhere.



This is a schematic plot of the exact formulas; the middle curve is represented by closely spaced points.

(c) Compare slopes at the joins.

On the three open pieces the derivative is respectively x, x^2+1, and 0. At 0, the one-sided derivatives are 0 and 1. At 2, they are 5 and 0. The unequal one-sided derivatives exclude differentiability at each joining point.

Conclusion. F is continuous on all of $\mathbb{R}$, differentiable on $\mathbb{R}\setminus\{0,2\}$, and has the piecewise derivative just computed.

34.5 Exercise 34.5: Moving both endpoints

What we must show. Differentiate $F(x)=\int_{x-1}^{x+1} f(t)\,dt$ for continuous f on $\mathbb{R}$. Choose the fixed-base primitive $H(u)=\int_0^u f(t)\,dt$. The Fundamental Theorem gives $H'(u)=f(u)$ for every u. Interval additivity, with signed integrals when necessary, gives

$$F(x)=H(x+1)-H(x-1).$$

Apply the chain rule to each term. The derivatives of $x+1$ and $x-1$ are both 1, so

$$F'(x)=H'(x+1)-H'(x-1)=f(x+1)-f(x-1).$$

Conclusion. F is differentiable everywhere, with the displayed derivative.

34.6 Exercise 34.6: A trigonometric upper endpoint

What we must show. Differentiate $G(x)=\int_0^{\sin x} f(t)\,dt$ for continuous f. As above, let $H(u)=\int_0^u f(t)\,dt$, so $H'=f$ everywhere. Now $G=H\circ\sin$, and the chain rule yields

$$G'(x)=H'(\sin x)\cos x=f(\sin x)\cos x.$$

This works even when $\sin x<0$ because H uses oriented integrals, and at zeros of $\cos x$ because the chain rule includes zero inner derivatives.

Conclusion. G is differentiable on all of $\mathbb{R}$ and $G'(x)=f(\sin x)\cos x$.

34.7 Exercise 34.7: A substitution with a reversed direction

What we must show. Evaluate $\int_0^1 x\sqrt{1-x^2}\,dx$ by change of variables. Choose $u=1-x^2$, so $du=-2x\,dx$ and $x\,dx=-du/2$. As x goes from 0 to 1, u goes from 1 to 0. Keeping these transformed

endpoints gives

$$\int_0^1 x\sqrt{1-x^2}\,dx = -\frac{1}{2}\int_1^0 u^{1/2}\,du = \frac{1}{2}\int_0^1 u^{1/2}\,du = \frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_0^1 = \frac{1}{3}.$$

The substitution is justified by the change-of-variables theorem: its inner map is continuously differentiable and $\sqrt{u}$ is continuous on the transformed interval $[0,1]$. No derivative of $\sqrt{u}$ at zero is needed.

Conclusion. The value of the integral is $1/3$.

34.8 Exercise 34.8: Two choices of antiderivative in integration by parts

What we must show. Evaluate $I = \int_0^1 x\arctan x\,dx$ using two versions of v.

(a) Choose $u = \arctan x$ and $v = x^2/2$.

Then $u' = 1/(1+x^2)$ and $v' = x$. Integration by parts gives

$$I = \left[\frac{x^2}{2}\arctan x\right]_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{1+x^2}\,dx.$$

The boundary term is $\pi/8$. Since $x^2/(1+x^2) = 1 - 1/(1+x^2)$, the remaining integral is $1 - \pi/4$. Therefore

$$I = \frac{\pi}{8} - \frac{1}{2}\left(1 - \frac{\pi}{4}\right) = \frac{\pi}{4} - \frac{1}{2}.$$

(b) Choose instead $v = (x^2+1)/2$.

This still has derivative x. Now the denominator cancels immediately:

$$I = \left[\frac{x^2+1}{2}\arctan x\right]_0^1 - \int_0^1 \frac{x^2+1}{2(1+x^2)}\,dx = \frac{\pi}{4} - \frac{1}{2}.$$

Both calculations are valid because all functions and derivatives used are continuous on $[0,1]$.

Conclusion. The common answer is $\pi/4 - 1/2$. A useful constant in v can simplify the remaining integral.

34.9 Exercise 34.9: Integrating the inverse sine

What we must show. Evaluate $\int_0^{1/2} \arcsin x\,dx$. The inverse-function integration identity of Example 3, or integration by parts with $u = \arcsin x$ and $v = x$, gives

$$\int_0^{1/2} \arcsin x\,dx = [x\arcsin x]_0^{1/2} - \int_0^{1/2} \frac{x}{\sqrt{1-x^2}}\,dx.$$

Here $\arcsin(1/2) = \pi/6$, so the boundary term is $\pi/12$. Since $d(\sqrt{1-x^2})/dx = -x/\sqrt{1-x^2}$,

$$\int_0^{1/2} \frac{x}{\sqrt{1-x^2}}\,dx = \left[-\sqrt{1-x^2}\right]_0^{1/2} = 1 - \frac{\sqrt{3}}{2}.$$

All derivatives in this calculation are continuous on $[0,1/2]$, which stays away from the singular endpoints ± 1.

Conclusion. $\displaystyle\int_0^{1/2} \arcsin x\,dx = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1.$

34.10 Exercise 34.10: The areas under a function and its inverse

What we must show. Give a geometric proof that $\int_0^1 g(x)\,dx + \int_0^1 g^{-1}(u)\,du = 1$. Because g is a continuous strictly increasing bijection of $[0,1]$ onto itself, $g(0) = 0$, $g(1) = 1$, and its inverse is continuous. Consider the unit square with horizontal coordinate x and vertical coordinate u. The graph $u = g(x)$ divides it into two regions. The region below the graph has area $\int_0^1 g(x)\,dx$, by vertical slices. The complementary region is described by $g(x) \le u$; strict increase makes this equivalent to $x \le g^{-1}(u)$. Its horizontal slice at height u therefore has length $g^{-1}(u)$, so its area is $\int_0^1 g^{-1}(u)\,du$. Together the two regions fill the square, sharing only their boundary curve, so their areas sum to its area 1.

Conclusion. The two integrals are the complementary areas in the unit square.

34.11 Exercise 34.11: A continuous square with zero integral

What we must show. Show that $\int_a^b f(x)^2\,dx = 0$ forces continuous f to vanish everywhere. Suppose instead that $f(c) \neq 0$ for some $c \in [a,b]$. Continuity at c gives a relative neighborhood on which $|f(x) - f(c)| < |f(c)|/2$. The reverse triangle inequality gives $|f(x)| \ge |f(c)| - |f(x) - f(c)| > |f(c)|/2$ there. Even if c is an endpoint, this relative neighborhood contains a closed subinterval $J \subseteq [a,b]$ of positive length. Thus

$$\int_a^b f(x)^2\,dx \ge \int_J f(x)^2\,dx \ge \frac{|f(c)|^2}{4}\,|J| > 0,$$

a contradiction. The comparison is justified because f^2 is continuous and nonnegative throughout the interval.

Conclusion. No such c exists, so $f(x) = 0$ for every $x \in [a,b]$.

34.12 Exercise 34.12: Testing against every continuous function

What we must show. If $\int_a^b f(x)g(x)\,dx = 0$ for every continuous g, prove that the continuous function f is identically zero. The assumption permits *any* continuous test function g; in particular, choose $g = f$, which is continuous by hypothesis. The assumed identity becomes

$$\int_a^b f(x)^2\,dx = 0.$$

Exercise 34.11 applies directly and yields $f = 0$ everywhere. Conversely, the zero function clearly has zero integral against every continuous test function, although only the forward implication was requested.

Conclusion. f must vanish identically on $[a,b]$.

35 Riemann–Stieltjes integrals

Read this before calculating. An ordinary integral weights an interval by its length. A Stieltjes integral weights it by the increase of an increasing function F. Here "increasing" permits equality: $s < t$ implies $F(s) \le F(t)$. Ross uses a modified definition that handles jumps explicitly. This matters: a formula from another book's traditional Stieltjes theory may not apply to this section.

Write $F(t-)$ and $F(t+)$ for the one-sided limits. On the particular interval $[a, b]$ being integrated over, the endpoint conventions are $F(a-) = F(a)$ and $F(b+) = F(b)$. Thus the mass at t is $\Delta F(t) = F(t+) - F(t-)$. For a partition $P = \{a = t_0 < \cdots < t_n = b\}$ set

$$w_k = F(t_k-) - F(t_{k-1}+), \qquad J_F(f, P) = \sum_{k=0}^{n} f(t_k)\Delta F(t_k).$$

The upper sum is $J_F(f, P) + \sum_{k=1}^{n} (\sup_{(t_{k-1}, t_k)} f) w_k$; the lower sum uses infima instead. The numbers w_k and $\Delta F(t_k)$ are nonnegative, and together they add to $W = F(b) - F(a)$. In particular, if $|h| \le B$, then every upper and lower sum for h lies between $-BW$ and BW.

The essential test is Theorem 35.6: a bounded function is F-integrable precisely when, for every $\varepsilon > 0$, some partition satisfies $U_F(f, P) - L_F(f, P) < \varepsilon$. The jump terms cancel in this difference. We will often use the equivalent expression

$$U_F(f, P) - L_F(f, P) = \sum_{k=1}^{n} \operatorname{osc}(f, (t_{k-1}, t_k)) w_k,$$

where $\operatorname{osc}(f, I) = \sup_I f - \inf_I f$ is the oscillation of f on I.

35.1 Exercise 35.1: Why an increasing function has one-sided limits

What we must show. Identify both one-sided limits of F by a supremum or an infimum, including the indicated endpoints.

Plan. Increasing values cannot move back and forth. Their supremum on the left, or infimum on the right, is therefore the only possible limiting value. We verify this using the definition of a limit.

(a) Approach $t \in (a, b]$ from the left.

The set $A = \{F(x) : a < x < t\}$ is nonempty and bounded above by $F(t)$. Consequently $L = \sup A$ is a finite real number. Given $\varepsilon > 0$, the definition of supremum supplies $y \in (a, t)$ such that $L - \varepsilon < F(y) \le L$. For every x with $y < x < t$, monotonicity gives

$$L - \varepsilon < F(y) \le F(x) \le L.$$

Take $\delta = t - y > 0$. If $x \in [a, b]$ and $0 < t - x < \delta$, then $y < x < t$, so $|F(x) - L| < \varepsilon$. This is exactly $\lim_{x \to t-} F(x) = L$. The proof works when $t = b$, since it only uses points to the left of t.

(b) Approach $t \in [a, b)$ from the right.

The nonempty set $B = \{F(x) : t < x < b\}$ is bounded below by $F(t)$. Put $R = \inf B$. Given $\varepsilon > 0$, choose $y \in (t, b)$ with $R \le F(y) < R + \varepsilon$. Whenever $t < x < y$,

$$R \le F(x) \le F(y) < R + \varepsilon.$$

Now $\delta = y - t$ proves $\lim_{x\to t+} F(x) = R$. At $t = a$ the same argument is valid using only points to the right.

Conclusion. The left limit is $\sup\{F(x) : a < x < t\}$ and the right limit is $\inf\{F(x) : t < x < b\}$. Both are finite.

35.2 Exercise 35.2: Separate a continuous increase from a jump

What we must show. Evaluate $\int_0^3 x^2\, dF(x)$ for the piecewise function in Example 4.

Record all three pieces, especially the jump.

The function is $F(t) = 0$ for $t < 0$, $F(t) = t^2$ for $0 \le t < 2$, and $F(t) = t + 5$ for $t \ge 2$. At 2 its left limit is 4 and its right limit is 7, so the jump has size 3. Write $F = F_1 + 3J_2$, where on $[0, 3]$

$$F_1(t) = \begin{cases} t^2, & 0 \le t \le 2, \\ t + 2, & 2 \le t \le 3. \end{cases}$$

This F_1 is continuous. Its derivative is $2t$ on the first piece and 1 on the second. We apply Theorem 35.13 separately on the two pieces; no derivative at the joining point is needed.

Calculate each contribution.

Linearity in the integrator, interval additivity, and the jump rule $\int f\, dJ_2 = f(2)$ give

$$\begin{aligned} \int_0^3 x^2\, dF(x) &= \int_0^2 x^2(2x)\, dx + \int_2^3 x^2\, dx + 3(2^2) \\ &= \left[\frac{x^4}{2}\right]_0^2 + \left[\frac{x^3}{3}\right]_2^3 + 12 \\ &= 8 + \frac{27 - 8}{3} + 12 = \frac{79}{3}. \end{aligned}$$

The extra 12 is essential: differentiating the smooth pieces alone would lose the mass concentrated at 2.

Conclusion. $\displaystyle\int_0^3 x^2\, dF(x) = 79/3.$

35.3 Exercise 35.3: Integrating against the integer step function

What we must show. Evaluate the three requested integrals against $F(t) = \lfloor t \rfloor$.

Plan. This function has jumps of size 1 at the integers. On a specified integration interval we include only the jumps belonging to the restricted function, with Ross's endpoint convention.

(a) Integrate x on $[0, 6]$.

On this interval $F(0-)$ means $F(0) = 0$, rather than the limit of the original function from negative numbers. Also $F(0+) = 0$. There is therefore no mass at 0. There is a mass of 1 at each of $1, 2, 3, 4, 5, 6$. The last one is included because $F(6+) = F(6) = 6$ and $F(6-) = 5$. Example 1 gives

$$\int_0^6 x\, dF(x) = \sum_{j=1}^{6} j = \frac{6 \cdot 7}{2} = 21.$$

(b) Integrate x^2 on $[0, 3]$.

The contributing jumps are at $1, 2, 3$, with the same endpoint reasoning. Thus

$$\int_0^3 x^2\, dF(x) = 1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14.$$

(c) Integrate x^2 on $[1/4, \pi/4]$.

Both endpoints lie strictly between 0 and 1. The integrator is identically 0 on the whole interval; every interval increment and every endpoint jump is zero. Its upper and lower sums are both zero.

Conclusion. The three values are 21, 14, and 0, respectively.

35.4 Exercise 35.4: A smooth trigonometric integrator

What we must show. Integrate x against $F(t) = \sin t$ on the two specified intervals.

Check the conversion to an ordinary integral.

On $[-\pi/2, \pi/2]$, $F'(t) = \cos t \geq 0$, so F is increasing. Both F' and $f(t) = t$ are continuous. Theorem 35.13 therefore permits the replacement $dF(x) = \cos x\, dx$ on each interval.

(a) The interval $[0, \pi/2]$.

Integration by parts with $u = x$ and $dv = \cos x\, dx$ gives

$$\int x \cos x\, dx = x \sin x - \int \sin x\, dx = x \sin x + \cos x.$$

Consequently

$$\int_0^{\pi/2} x\, dF(x) = [x \sin x + \cos x]_0^{\pi/2} = \frac{\pi}{2} - 1.$$

(b) The interval $[-\pi/2, \pi/2]$.

The function $x \cos x$ is odd: replacing x by $-x$ changes its sign. Its integrals over the two halves cancel, so the answer is 0. Alternatively the displayed antiderivative has value $\pi/2$ at both endpoints, giving difference 0.

Conclusion. The values are $\pi/2 - 1$ and 0.

35.5 Exercise 35.5: The rational indicator sees continuous and atomic mass differently

What we must show. For $f = \mathbf{1}_{\mathbb{Q}}$, contrast continuous integrators with the purely discrete integrators in Examples 1 and 3.

(a) A continuous nonconstant integrator.

Every nonempty open interval contains both a rational number and an irrational number. Hence on every partition interval (t_{k-1}, t_k), the supremum of f is 1 and the infimum is 0. Continuity of F

makes all jump terms zero and gives $w_k = F(t_k) - F(t_{k-1})$. Therefore, for every partition,

$$L_F(f, P) = 0, \qquad U_F(f, P) = \sum_{k=1}^{n} [F(t_k) - F(t_{k-1})] = F(b) - F(a).$$

Taking the supremum of the lower sums and infimum of the upper sums changes neither number. Since $F(b) > F(a)$, the lower and upper integrals are different. Thus f is not F-integrable.

(b) A finite or summable collection of point masses.

For $F = \sum_j c_j J_{u_j}$ as in those examples, every bounded function is F-integrable, including this f. Its integral is

$$\int_a^b f \, dF = \sum_j c_j f(u_j) = \sum_{\{j : u_j \in \mathbb{Q}\}} c_j.$$

The formula includes the finite case. In the infinite case it converges absolutely, because $0 \le c_j f(u_j) \le c_j$ and $\sum_j c_j < \infty$. These integrators assign their mass to actual points; the value of f at each mass point is well defined.

Conclusion. For continuous F of positive total increase, f is not integrable. For the discrete F of Examples 1 and 3, it is integrable, with integral equal to the total mass at rational points.

35.6 Exercise 35.6: Uniform limits preserve Stieltjes integrability

What we must show. If each f_n is F-integrable and $f_n \to f$ uniformly on $[a, b]$, prove that f is F-integrable and that the integrals converge.

Plan. Uniform convergence controls every point by the same error. Multiplying that error by the total mass $W = F(b) - F(a)$ controls every upper sum, lower sum, and ultimately the integral.

First establish boundedness.

Take one index N for which $|f(x) - f_N(x)| < 1$ for every $x \in [a, b]$. Such an index exists by uniform convergence. Since f_N is integrable, it is bounded, say $|f_N(x)| \le B$. Then $|f(x)| \le B + 1$ everywhere. This verifies the boundedness required before applying the Darboux criterion.

Make the upper–lower gap small.

Assume $W > 0$; if $W = 0$, every sum and integral is zero and the conclusions are immediate. Fix $\varepsilon > 0$ and select N so that $|f - f_N| \le \eta$ everywhere, where $\eta = \varepsilon/(4W)$. On any nonempty interval I,

$$\sup_I f \le \sup_I f_N + \eta, \qquad \inf_I f \ge \inf_I f_N - \eta.$$

Subtracting gives $\operatorname{osc}(f, I) \le \operatorname{osc}(f_N, I) + 2\eta$. Because f_N is integrable, choose P with $U_F(f_N, P) - L_F(f_N, P) < \varepsilon/2$. Since $\sum_k w_k \le W$,

$$\begin{aligned} U_F(f, P) - L_F(f, P) &\le U_F(f_N, P) - L_F(f_N, P) + 2\eta \sum_k w_k \\ &< \frac{\varepsilon}{2} + 2\eta W = \varepsilon. \end{aligned}$$

Theorem 35.6 proves that f is integrable.

Now estimate the integrals.

Put $\eta_n = \sup_{x\in[a,b]} |f_n(x) - f(x)|$; uniform convergence says $\eta_n \to 0$. The integrable difference $f_n - f$ satisfies $-\eta_n \le f_n - f \le \eta_n$. Order preservation and integration of constants give

$$-\eta_n W \le \int_a^b (f_n - f)\, dF \le \eta_n W.$$

Thus $\left|\int_a^b f_n\, dF - \int_a^b f\, dF\right| \le \eta_n W \to 0$.

Conclusion. Uniform convergence permits passage to the limit under this integral, with error at most the uniform function error times the total mass.

35.7 Exercise 35.7: Products, squares, maxima, and minima

What we must show. Show that all four indicated constructions preserve F-integrability.

(a) Squaring an integrable function.

Choose $B \ge 1$ such that $|f(x)| \le B$ on $[a, b]$. For any two points x, y in the same interval I,

$$\left|f(x)^2 - f(y)^2\right| = |f(x) - f(y)|\,|f(x) + f(y)| \le 2B\,|f(x) - f(y)|\,.$$

The supremum of pairwise differences of a bounded real function is its oscillation. It follows that $\operatorname{osc}(f^2, I) \le 2B \operatorname{osc}(f, I)$. Given $\varepsilon > 0$, take a partition making the gap for f less than $\varepsilon/(2B)$. The gap formula then makes the gap for f^2 less than ε. Since f^2 is bounded, the criterion proves its integrability.

(b) Multiplying two integrable functions.

Linearity makes $f + g$ and $f - g$ integrable. Part (a) makes their squares integrable. The identity

$$fg = \frac{(f+g)^2 - (f-g)^2}{4}$$

then expresses fg as a linear combination of integrable functions. To check the identity, expand both squares: their f^2 and g^2 terms cancel, leaving $4fg$ in the numerator.

(c) Taking the pointwise maximum and minimum.

Theorem 35.10 makes $|f - g|$ integrable. At each point,

$$\max(f, g) = \frac{f + g + |f - g|}{2}, \qquad \min(f, g) = \frac{f + g - |f - g|}{2}.$$

For example, if $f(x) \ge g(x)$ then $|f(x) - g(x)| = f(x) - g(x)$, so the first expression becomes $f(x)$ and the second becomes $g(x)$. The other ordering gives the reversed roles. The formulas and linearity prove both integrability assertions.

Conclusion. f^2, fg, $\max(f, g)$, and $\min(f, g)$ are all F-integrable.

35.8 Exercise 35.8: A continuous density defines the weight

What we must show. For continuous $g \geq 0$ and $F(t) = \int_a^t g(x)\,dx$, show that $\int_a^b f\,dF = \int_a^b f(x)g(x)\,dx$ for continuous f.

Check that F is an admissible increasing integrator.

If $a \leq s < t \leq b$, interval additivity gives

$$F(t) - F(s) = \int_s^t g(x)\,dx \geq 0,$$

because g is nonnegative. Thus F is increasing. The Fundamental Theorem of Calculus, applied to continuous g, gives $F' = g$ on the interior, with the corresponding one-sided derivatives at the endpoints. In particular F' is continuous on the closed interval.

Apply the smooth-integrator formula.

All the hypotheses of Theorem 35.13 are now verified, so

$$\int_a^b f(x)\,dF(x) = \int_a^b f(x)F'(x)\,dx = \int_a^b f(x)g(x)\,dx.$$

If g is identically zero, F is constant. Although the section usually excludes a constant integrator to avoid trivial cases, its sums are still defined and all equal zero; the formula is then also true. If g is nonzero somewhere, continuity and nonnegativity give positive integral, so $F(b) > F(a)$ as in the standing convention.

Conclusion. The density g converts a Stieltjes integral into the usual weighted Riemann integral.

35.9 Exercise 35.9: A weighted mean value is attained

What we must show. Prove the mean-value assertion for continuous f, then recover the weighted integral theorem in Exercise 33.14.

(a) Trap the average between a minimum and a maximum.

The Extreme Value Theorem gives real numbers $m = \min_{[a,b]} f$ and $M = \max_{[a,b]} f$. Put $W = F(b) - F(a) > 0$. The continuous function f is F-integrable by Theorem 35.7. Integrating $m \leq f \leq M$ with order preservation gives

$$mW \leq \int_a^b f\,dF \leq MW.$$

Divide by the positive number W and call the quotient A. We have $m \leq A \leq M$. By the Intermediate Value Theorem, f takes every value between its attained minimum and maximum. Hence some $x \in [a, b]$ satisfies $f(x) = A$. Multiplication by W proves

$$\int_a^b f\,dF = f(x)[F(b) - F(a)].$$

The conclusion naturally allows endpoints: all the mass could be concentrated at a, for example.

(b) Recover the continuous-density theorem.

For continuous $g \geq 0$ set $F(t) = \int_a^t g(s)\,ds$, as in Exercise 35.8. If $g \equiv 0$, both integrals in Exercise 33.14(a) are zero and any $x \in (a,b)$ works. Otherwise $W = \int_a^b g > 0$. Part (a) and Exercise 35.8 give

$$\int_a^b f(t)g(t)\,dt = f(x)\int_a^b g(t)\,dt$$

initially for some $x \in [a,b]$.

Explain why the point can be chosen in the interior here.

Exercise 33.14(a) asks for $x \in (a,b)$, so this small extra point deserves a proof. Let $A = (\int_a^b fg)/(\int_a^b g)$. Suppose no interior point satisfies $f(x) = A$. By continuity and the Intermediate Value Theorem, $f - A$ must then have the same strict sign on all of (a,b). Also some $t_0 \in (a,b)$ has $g(t_0) > 0$: a positive endpoint value of g would imply positive nearby interior values as well. If $f - A > 0$, continuity gives a short interval around t_0 on which both $f - A$ and g have positive lower bounds. Thus $(f - A)g \geq 0$ everywhere and has strictly positive integral, contrary to

$$\int_a^b (f - A)g = \int_a^b fg - A\int_a^b g = 0.$$

If $f - A < 0$, apply the same argument to $(A - f)g$ and obtain the same contradiction. Therefore an interior point does attain A. Taking $g \equiv 1$ recovers the ordinary integral mean-value theorem, as requested in Exercise 33.14(b).

Conclusion. Part (a) is the Stieltjes mean-value formula. With a continuous nonnegative density it gives Exercise 33.14(a), including its interior-point conclusion.

35.10 Exercise 35.10: A polynomial integrated against itself

What we must show. For a positive odd integer n and $F(x) = f(x) = x^n$, evaluate $\int_a^b f\,dF$.

Explain the parity hypothesis.

The map $x \mapsto x^n$ is increasing on all of $\mathbb{R}$ when n is odd. Its derivative is $nx^{n-1} \geq 0$ since $n - 1$ is even. Both f and F' are continuous, so Theorem 35.13 applies on any closed interval $[a,b]$.

Convert and integrate.

We have

$$\begin{aligned}\int_a^b x^n\,d(x^n) = \int_a^b x^n\,nx^{n-1}\,dx &= n\int_a^b x^{2n-1}\,dx \\ &= n\left[\frac{x^{2n}}{2n}\right]_a^b = \frac{1}{2}(b^{2n} - a^{2n}).\end{aligned}$$

The result can be negative if the interval lies on the negative half-line. That is consistent: the weight is nonnegative, but $f(x) = x^n$ is negative there.

Conclusion. $\displaystyle\int_a^b f\,dF = \tfrac{1}{2}[b^{2n} - a^{2n}]$.

35.11 Exercise 35.11: Change variables without differentiating the substitution

What we must show. For a continuous strictly increasing map $\phi : [c, d] \to [a, b]$ with the stated endpoint values, prove integrability and equality of the two Stieltjes integrals for $g = f \circ \phi$ and $G = F \circ \phi$.

Plan. A partition and its image under ϕ describe exactly the same intervals and mass points. Match their sums term by term. There is no assumption that ϕ is differentiable.

The substitution maps intervals onto intervals.

Continuity, strict increase, and the endpoint values imply that ϕ is a bijection from $[c, d]$ onto $[a, b]$. Its inverse is continuous: for an interior u, points just below and above u have image values strictly below and above $\phi(u)$, which trap the inverse of every sufficiently nearby image point. The same one-sided trapping works at the endpoints. In particular,

$$\phi((u, v)) = (\phi(u), \phi(v)) \qquad (c \le u < v \le d).$$

The composition G is increasing, and g is bounded because f is bounded on the image $[a, b]$.

Match one-sided limits and jumps.

For $u \in (c, d)$, as $v \to u-$ the numbers $\phi(v)$ approach $\phi(u)$ from below. They cover a whole one-sided interval near $\phi(u)$. Therefore

$$G(u-) = F(\phi(u)-), \qquad G(u+) = F(\phi(u)+).$$

At $u = c, d$ the endpoint conventions give these same identities for the relevant decreed limits. Subtracting proves $\Delta G(u) = \Delta F(\phi(u))$.

Match the upper and lower sums.

Take any $Q = \{c = u_0 < \cdots < u_n = d\}$ and its image partition $P = \{a = \phi(u_0) < \cdots < \phi(u_n) = b\}$. Bijection on the open subintervals gives

$$\sup_{(u_{k-1}, u_k)} g = \sup_{(\phi(u_{k-1}), \phi(u_k))} f,$$

and the analogous identity for infima. The one-sided-limit identities give matching interval weights, and $g(u_k) = f(\phi(u_k))$ gives matching jump terms. Thus

$$U_G(g, Q) = U_F(f, P), \qquad L_G(g, Q) = L_F(f, P).$$

Every partition of $[a, b]$ arises this way, by applying ϕ^{-1} to its points. Consequently the infima of the upper sums agree, and the suprema of the lower sums agree. Since f is F-integrable, those two common numbers coincide.

Conclusion. g is G-integrable, and $\displaystyle\int_c^d g\, dG = \int_a^b f\, dF$.

35.12 Exercise 35.12: A strictly increasing function with dense jumps

What we must show. Analyze the sum of positive jumps placed at every rational point of $[a, b]$.

First address the wording in the supplied edition.

The printed exercise says that the c_j are "positive integers" and that $\sum_j c_j < \infty$. These conditions are incompatible: each partial sum is at least its number of terms, so the series would diverge to $+\infty$. Thus there is no example satisfying the literal hypotheses. The intended, nonvacuous version uses *positive real numbers* with finite sum, just as in Example 3. We solve that version explicitly; for instance $c_j = 2^{-j}$ is an allowed choice. Let (u_j) enumerate the rationals in $[a, b]$ without repetition, and retain Ross's special convention for J_a.

(a) Show that the sum exists and increases strictly.

Since $0 \le J_{u_j}(t) \le 1$,

$$0 \le \sum_{j=1}^{N} c_j J_{u_j}(t) \le \sum_{j=1}^{\infty} c_j < \infty.$$

So the nondecreasing partial sums have a finite limit $F(t)$ at every point. Also their tails are at most $\sum_{j>N} c_j$, independently of t, so the convergence is uniform. For $a \le s < t \le b$, density of $\mathbb{Q}$ supplies a rational $u_m \in (s, t)$. The function J_{u_m} is 0 at s and 1 at t. Every other summand is increasing, hence has nonnegative increment. Subtracting the convergent series gives

$$F(t) - F(s) = \sum_{j=1}^{\infty} c_j [J_{u_j}(t) - J_{u_j}(s)] \ge c_m > 0.$$

This proves strict increase, not merely nondecrease.

(b) Prove continuity at every irrational point.

Let $t \in [a, b] \setminus \mathbb{Q}$ and $\varepsilon > 0$. Choose N with $\sum_{j>N} c_j < \varepsilon$. Because t differs from each of the finitely many $u_1, \ldots, u_N$, there is $\delta > 0$ such that $|x - t| < \delta$ crosses none of these points. Each of the first N step functions has the same value at x and t. For the remaining terms, $\left|J_{u_j}(x) - J_{u_j}(t)\right| \le 1$. Thus for $x \in [a, b]$ with $|x - t| < \delta$,

$$|F(x) - F(t)| \le \sum_{j>N} c_j \left|J_{u_j}(x) - J_{u_j}(t)\right| \le \sum_{j>N} c_j < \varepsilon.$$

This proves relative continuity at t, including an irrational endpoint if one occurs.

Show discontinuity at every rational point.

If $u_m \in (a, b]$, then $J_{u_m}(u_m) = 1$, whereas $J_{u_m}(x) = 0$ for every $x < u_m$. Since every summand is increasing, $F(u_m) - F(x) \ge c_m$ for $x < u_m$ in $[a, b]$. Points of the domain can approach u_m from the left while this gap stays at least $c_m > 0$, so F is discontinuous there. If the rational point is $u_m = a$, Ross defines $J_a(a) = 0$ and $J_a(x) = 1$ for $x > a$. Thus $F(x) - F(a) \ge c_m$ for all $x > a$, which proves right discontinuity at a. In fact Example 3 shows that the jump at u_m is exactly c_m.

Conclusion. Under the intended summable positive-real-weight hypothesis, F is strictly increasing, continuous exactly at irrational points of $[a, b]$, and discontinuous at every rational point. Under the printed positive-integer hypothesis, no such weight sequence exists.

36 Improper integrals

An improper integral is a *limit of ordinary integrals*; it is not an instruction to substitute an infinite number into an antiderivative. For example,

$$\int_a^b f = \lim_{d\to b^-} \int_a^d f$$

when the left endpoint is harmless. If both endpoints are troublesome, choose an interior point α and take the two limits separately. A finite limit means convergence. A limit $+\infty$ or $-\infty$ is an extended value, not convergence. An oscillating limit has no value. In particular, a symmetric limit on $[-R, R]$ cannot replace the two one-sided limits in the definition.

For Stieltjes integrals, F is increasing. Its increment $F(v) - F(u)$ replaces the interval length $v - u$. We use Ross's endpoint convention from Section 35, so that adjacent integrals add without counting a jump twice. A distribution function has total increment $F(\infty) - F(-\infty) = 1$.

36.1 Exercise 36.1: Agreement with an ordinary integral

What we must show. If f is Riemann integrable on the finite interval $[a, b]$, prove that $\lim_{d\to b^-} \int_a^d f = \int_a^b f$.

Plan. The only omitted contribution is over the short interval $[d, b]$.

Bound that contribution.

Riemann integrability includes boundedness. Choose $B \geq 0$ with $|f(x)| \leq B$ on $[a, b]$. Restrictions of an integrable function are integrable, and additivity and the integral inequality give

$$\left| \int_a^b f - \int_a^d f \right| = \left| \int_d^b f \right| \leq \int_d^b |f| \leq B(b - d).$$

If $B = 0$, the difference is already zero. If $B > 0$, given $\varepsilon > 0$, choosing $0 < b - d < \varepsilon/B$ makes the difference less than ε.

Conclusion. The difference tends to zero, so the two definitions agree.

36.2 Exercise 36.2: The splitting point does not change the value

What we must show. Show that the definition of an integral on (a, b) is independent of the chosen interior point.

Compare two splitting points by a finite integral.

Suppose $a < \alpha < \beta < b$ and set $C = \int_\alpha^\beta f$. This is a finite ordinary integral by the local integrability hypothesis. For $a < c < \alpha$ and $\beta < d < b$,

$$\int_c^\beta f = \int_c^\alpha f + C, \qquad \int_\beta^d f = \int_\alpha^d f - C.$$

Pass to the endpoint limits. Adding or subtracting a finite number preserves the existence and sign of an infinite limit. Thus, if $A = \int_a^\alpha f$ and $B = \int_\alpha^b f$ exist, then

$$\int_a^\beta f = A + C, \qquad \int_\beta^b f = B - C.$$

Check that the permitted sum is unchanged.

If A, B are finite, their sum is plainly unchanged. If one is infinite and their sum is permitted, the finite changes do not alter either infinite sign. An excluded sum $+\infty + (-\infty)$ remains excluded. The same argument in reverse proves the converse, and interchanging α, β covers either order.

Conclusion. The existence and the value of the improper integral do not depend on the split.

36.3 Exercise 36.3: Power singularities at zero and at infinity

What we must show. Analyze $\int_0^1 x^{-p}\,dx$ and $\int_0^\infty x^{-p}\,dx$ for $p > 0$.

(a) Cut off the singular endpoint before integrating.

For $0 < c < 1$ and $p \neq 1$, the power rule gives

$$\int_c^1 x^{-p}\,dx = \frac{1 - c^{1-p}}{1-p}.$$

If $0 < p < 1$, the exponent $1 - p$ is positive, so $c^{1-p} \to 0$; the answer is $1/(1-p)$. If $p > 1$, then $c^{1-p} = 1/c^{p-1} \to +\infty$. Rewriting the last display as $(c^{1-p} - 1)/(p-1)$ shows that the integral tends to $+\infty$. For completeness, the boundary case $p = 1$ has $\int_c^1 dx/x = -\log c \to +\infty$.

(b) Treat both ends separately.

For $p \neq 1$,

$$\int_1^d x^{-p}\,dx = \frac{d^{1-p} - 1}{1-p}.$$

This tends to $1/(p-1)$ if $p > 1$, and to $+\infty$ if $0 < p < 1$. For $p = 1$ it equals $\log d \to +\infty$. Thus $p > 1$ fails at zero, $0 < p < 1$ fails at infinity, and $p = 1$ fails at both ends. All integrands are nonnegative, so there is no cancellation issue.

Conclusion. The integral on $(0, 1)$ converges exactly for $0 < p < 1$; the integral on $(0, \infty)$ is $+\infty$ for every $p > 0$.

36.4 Exercise 36.4: Three explicit improper integrals

(a) The logarithm near zero.

Integration by parts on $[c, 1]$, where $c > 0$, yields

$$\int_c^1 \log x\,dx = [x \log x - x]_c^1 = -1 - c\log c + c.$$

To justify $c \log c \to 0$, put $t = 1/c$. The elementary estimate $\log t = \int_1^t du/u \le \int_1^t du/\sqrt{u} = 2(\sqrt{t} - 1)$ for $t \ge 1$ gives $0 \le (\log t)/t \le 2/\sqrt{t} \to 0$. Consequently $\int_0^1 \log x\,dx = -1$.

(b) A logarithm divided by its argument.

For $d > 2$, substitution $u = \log x$, $du = dx/x$, gives

$$\int_2^d \frac{\log x}{x}\,dx = \frac{(\log d)^2 - (\log 2)^2}{2} \longrightarrow +\infty.$$

(c) The reciprocal quadratic.

Since $(\arctan x)' = 1/(1+x^2)$,

$$\int_0^d \frac{dx}{1+x^2} = \arctan d \longrightarrow \frac{\pi}{2}.$$

Conclusion. The values, in order, are -1, $+\infty$, and $\pi/2$.

36.5 Exercise 36.5: A nonnegative integral as a supremum

What we must show. For continuous $f \geq 0$ on (a, b), identify its improper integral with the supremum of all integrals over compact subintervals.

Establish the two one-sided limits.

Choose $\alpha \in (a, b)$. Define

$$A(c) = \int_c^\alpha f \quad (a < c < \alpha), \qquad B(d) = \int_\alpha^d f \quad (\alpha < d < b).$$

Because $f \geq 0$, $A(c)$ increases as c moves toward a, and $B(d)$ increases as d moves toward b. The monotone-limit property therefore gives limits $A, B \in [0, \infty]$, equal to their respective suprema. The improper integral exists and equals $A + B$; this sum can never have the forbidden form $+\infty + (-\infty)$.

Compare with all compact subintervals.

Let $M = \sup\{\int_c^d f : [c, d] \subset (a, b)\}$. Enlarging any $[c, d]$ so that it contains α can only increase its integral. An enlarged interval has integral $A(c') + B(d') \leq A + B$; thus $M \leq A + B$. Conversely, all intervals with $c < \alpha < d$ are allowed in the supremum, so $M \geq A(c) + B(d)$. If A, B are finite, choose c, d making these terms within $\varepsilon/2$ of their suprema, and obtain $M \geq A + B - \varepsilon$. If either supremum is infinite, these same integrals exceed every prescribed bound, so $M = +\infty$.

Conclusion. $\displaystyle\int_a^b f = \sup_{[c,d] \subset (a,b)} \int_c^d f$, with $+\infty$ allowed.

36.6 Exercise 36.6: Comparison tests

What we must show. Suppose $0 \leq f \leq g$ and both functions are continuous on (a, b). Prove the convergence and divergence comparisons.

Compare finite intervals first.

For every compact $[c, d] \subset (a, b)$, ordinary integral monotonicity gives $0 \leq \int_c^d f \leq \int_c^d g$. Exercise 36.5 allows us to take suprema, giving

$$0 \leq \int_a^b f \leq \int_a^b g \leq +\infty.$$

(a) Finite upper bound.

If $\int_a^b g$ is finite, the middle quantity is bounded above by that finite number. The nonnegative integral of f therefore converges.

(b) Infinite lower bound.

If $\int_a^b f = +\infty$, no finite number can bound the integral of g from above. Therefore $\int_a^b g = +\infty$.

Conclusion. A smaller nonnegative function inherits convergence; a larger one inherits divergence to $+\infty$.

36.7 Exercise 36.7: The Gaussian integral

(a) Prove finiteness before attempting evaluation.

The integrand is even. On $[0,1]$ it is at most 1; on $[1,\infty)$, $x^2 \geq x$, so $e^{-x^2} \leq e^{-x}$. The comparison test gives

$$0 < I := \int_{-\infty}^{\infty} e^{-x^2}\,dx \leq 2\left(1 + \int_1^{\infty} e^{-x}\,dx\right) = 2(1+e^{-1}) < \infty.$$

(b) Explain the double-integral calculation.

This part uses polar coordinates for a double integral, as requested by the exercise. The following bounded-region argument avoids assuming in advance that improper iterated integrals may be interchanged. Set $I_R = \int_{-R}^{R} e^{-x^2}\,dx$. On the square $Q_R = [-R,R]^2$, integration of the continuous product gives

$$\iint_{Q_R} e^{-x^2-y^2}\,dx\,dy = I_R^2.$$

Let D_R be the disk of radius R. The inclusions $D_R \subset Q_R \subset D_{\sqrt{2}R}$ and positivity give a lower and an upper bound for I_R^2. In polar coordinates,

$$\iint_{D_R} e^{-x^2-y^2}\,dx\,dy = \int_0^{2\pi}\int_0^R e^{-r^2} r\,dr\,d\theta = \pi(1-e^{-R^2}).$$

Here the extra factor r is the polar area factor, and $\int_0^R e^{-r^2} r\,dr = (1-e^{-R^2})/2$. Hence

$$\pi(1-e^{-R^2}) \leq I_R^2 \leq \pi(1-e^{-2R^2}).$$

Both outer expressions tend to π. Since part (a) shows $I_R \to I$, we obtain $I^2 = \pi$; and $I > 0$ selects the positive root.

Conclusion. $\displaystyle\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$.

36.8 Exercise 36.8: Absolute convergence implies convergence

What we must show. If f is continuous on (a,b) and $\int_a^b |f| < \infty$, prove that $\int_a^b f$ exists and is finite.

Plan. Write f as the difference of two nonnegative functions whose integrals are bounded above by integrals already known to converge.

Use positive and negative parts.

Define

$$f_+(x) = \max\{f(x), 0\}, \qquad f_-(x) = \max\{-f(x), 0\}.$$

These are continuous because, for example, $f_+ = (|f| + f)/2$. Pointwise, $0 \le f_+, f_- \le |f|$ and $f = f_+ - f_-$. Exercise 36.6 proves that both nonnegative improper integrals are finite.

Pass to limits without subtracting infinities.

On each compact subinterval, ordinary linearity gives $\int_c^d f = \int_c^d f_+ - \int_c^d f_-$. Fix an interior split, and take the left and right endpoint limits separately. Each term has a finite limit, so the limit law for subtraction applies.

Conclusion. $\int_a^b f$ converges and $\left|\int_a^b f\right| \le \int_a^b |f|$.

36.9 Exercise 36.9: Integrals for the normal distribution

Write $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ and $F(t) = \int_{-\infty}^t \varphi(x)\,dx$. Substitution $x = \sqrt{2}u$ in Exercise 36.7 gives $\int_{\mathbb{R}} \varphi = 1$.

(a) Replace the Stieltjes integrator by its density.

The Fundamental Theorem of Calculus gives $F' = \varphi$, a continuous function. For continuous f, Theorem 35.13 on each finite $[a, b]$ therefore yields

$$\int_a^b f\,dF = \int_a^b f(x)F'(x)\,dx = \frac{1}{\sqrt{2\pi}} \int_a^b f(x)e^{-x^2/2}\,dx.$$

Taking the two endpoint limits separately proves the same identity on $\mathbb{R}$ whenever the integral specified in the question exists; positive constant multiplication preserves finite and infinite limits.

(b) The second moment.

Since $\varphi'(x) = -x\varphi(x)$, integration by parts on $[a, b]$ gives

$$\int_a^b x^2\varphi(x)\,dx = [-x\varphi(x)]_a^b + \int_a^b \varphi(x)\,dx.$$

The boundary term tends to zero at either infinity. For instance, $e^t \ge t^2/2$ for $t \ge 0$ implies $xe^{-x^2/2} \le 8/x^3$ for $x > 0$, and negative x are handled by absolute values. Thus $\int_{\mathbb{R}} x^2\,dF = 1$.

(c) The exponential of the square.

The integrand here is e^{x^2}, not e^x. Its density product is

$$e^{x^2}\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{x^2/2} \ge \frac{1}{\sqrt{2\pi}}.$$

The integral of this positive constant over an infinite half-line is infinite, so comparison proves $\int_{\mathbb{R}} e^{x^2}\,dF = +\infty$.

(d) The absolute first moment.

Evenness and the antiderivative of $xe^{-x^2/2}$ give

$$\int_{\mathbb{R}} |x|\,dF = \frac{2}{\sqrt{2\pi}} \int_0^\infty xe^{-x^2/2}\,dx = \frac{2}{\sqrt{2\pi}}[-e^{-x^2/2}]_0^\infty = \sqrt{\frac{2}{\pi}}.$$

(e) The first moment and a necessary convergence check.

Part (d) shows absolute convergence. Each half-integral is therefore finite. Because $x\varphi(x)$ is odd, substitution $u = -x$ in the negative half-integral makes it the negative of the positive half. Their sum is zero. Oddness alone would not suffice without the existence of both halves.

Conclusion. The values in (b)–(e) are 1, $+\infty$, $\sqrt{2/\pi}$, and 0.

36.10 Exercise 36.10: Linearity on the whole real line

What we must show. For F-integrable f, g on $\mathbb{R}$, prove integrability and the sum rule for $f + g$.

Apply finite-interval linearity.

For every finite $a < 0 < b$, Theorem 35.8 gives

$$\int_a^0 (f+g)\,dF = \int_a^0 f\,dF + \int_a^0 g\,dF, \qquad \int_0^b (f+g)\,dF = \int_0^b f\,dF + \int_0^b g\,dF.$$

Since f, g are F-integrable on $\mathbb{R}$, all four half-line limits are finite. The limit law for addition proves that the two limits for $f + g$ exist and are finite. Add them and regroup finite real numbers.

Conclusion. $f + g$ is F-integrable and $\int_{\mathbb{R}}(f+g)\,dF = \int_{\mathbb{R}} f\,dF + \int_{\mathbb{R}} g\,dF$. The same argument proves scalar multiplication, including multiplication by -1.

36.11 Exercise 36.11: Monotonicity on the whole real line

What we must show. If f, g are F-integrable and $f \le g$, prove the integral inequality.

Preserve the inequality through finite limits.

Theorem 35.9 gives $\int_a^0 f\,dF \le \int_a^0 g\,dF$ for each $a < 0$, and $\int_0^b f\,dF \le \int_0^b g\,dF$ for each $b > 0$. Both sides have finite limits by hypothesis. If real-valued quantities satisfy an inequality before taking a limit, the limits satisfy the same weak inequality. Pass to $a \to -\infty$ and $b \to +\infty$ separately, and add.

Conclusion. $\displaystyle\int_{\mathbb{R}} f\,dF \le \int_{\mathbb{R}} g\,dF.$

36.12 Exercise 36.12: Comparison for improper Stieltjes integrals

What we must show. State and prove the analogue of Exercise 36.6. Assume F is increasing and f, g are F-integrable on every finite interval, with $0 \le f \le g$. Continuity of f, g is one sufficient condition for local integrability.

Nonnegativity supplies the endpoint limits.

For $b > 0$, $\int_0^b f\,dF$ is increasing in b, since its increment over an added interval is nonnegative. Likewise, $\int_a^0 f\,dF$ increases as $a \to -\infty$. Thus both limits exist in $[0, \infty]$. The same holds for g.

Compare before and after the limits.

Finite-interval monotonicity gives

$$0 \le \int_a^0 f\,dF \le \int_a^0 g\,dF, \qquad 0 \le \int_0^b f\,dF \le \int_0^b g\,dF.$$

Taking limits and adding nonnegative extended values proves $0 \le \int_{\mathbb{R}} f\,dF \le \int_{\mathbb{R}} g\,dF$.

Conclusion. If $\int_{\mathbb{R}} g\,dF < \infty$, then $\int_{\mathbb{R}} f\,dF < \infty$. If $\int_{\mathbb{R}} f\,dF = +\infty$, then $\int_{\mathbb{R}} g\,dF = +\infty$.

36.13 Exercise 36.13: Absolute convergence for Stieltjes integrals

What we must show. If f is continuous on $\mathbb{R}$ and $\int_{\mathbb{R}} |f|\,dF < \infty$, prove f is F-integrable on $\mathbb{R}$.

Obtain finite nonnegative integrals.

Continuity makes f, $|f|$, and $h = f + |f|$ locally F-integrable. Pointwise, $0 \le h \le 2|f|$. Exercise 36.12 shows that h has a finite integral, since $2|f|$ does.

Subtract a function with a known finite integral.

Scalar multiplication shows $-|f|$ is F-integrable. Exercise 36.10 now applies to $f = h + (-|f|)$, so f is F-integrable. Applying monotonicity to $-|f| \le f \le |f|$ also gives $|\int_{\mathbb{R}} f\,dF| \le \int_{\mathbb{R}} |f|\,dF$.

Conclusion. Absolute F-integrability implies F-integrability. The argument still works if continuity is replaced by local F-integrability of f and $|f|$.

36.14 Exercise 36.14: A countable collection of point masses

What we must show. Let the distinct points be $u_j \in \mathbb{R}$, let $c_j \ge 0$, and suppose $C = \sum_{j=1}^{\infty} c_j < \infty$. Analyze $F(t) = \sum_j c_j J_{u_j}(t)$, where $J_u(t) = 0$ for $t < u$ and $J_u(t) = 1$ for $t \ge u$ on $\mathbb{R}$.

(a) Existence and monotonicity.

For each t, $0 \le c_j J_{u_j}(t) \le c_j$. The series therefore converges, and its tail is at most $\sum_{j>N} c_j$, independently of t. Thus convergence is uniform. Each summand is increasing; for $s < t$, termwise comparison gives $F(s) \le F(t)$.

(b) Finite intervals and their endpoint convention.

Let $|f| \le B$ on $\mathbb{R}$. Theorem 35.12(b), or its finite-mass tail estimate, makes every bounded f F-integrable on $[a, b]$. For this right-continuous F, the formula is

$$\int_a^b f\,dF = \sum_{j:\, a < u_j \le b} c_j f(u_j).$$

Indeed a single global step contributes its mass exactly when $a < u_j \le b$; a jump at a has already occurred before the interval's increment begins. Summing the steps is justified because the absolute value of the contribution from all indices $j > N$ is at most $B \sum_{j>N} c_j$. This tends to zero.

Pass to the whole line, controlling the tails.

The numerical series $\sum_j c_j f(u_j)$ is absolutely convergent, because $\sum_j |c_j f(u_j)| \le BC$. To see directly that the finite-interval sums approach it, choose N with $B \sum_{j>N} c_j < \varepsilon$ (the case $B = 0$ is immediate), then choose $a < \min_{j \le N} u_j$ and $b > \max_{j \le N} u_j$. All first N terms are included, and the omitted terms have total absolute value below ε. The same argument on the two halves treats $u_j = 0$ in the negative half only. Hence

$$\int_{-\infty}^{\infty} f\,dF = \sum_{j=1}^{\infty} c_j f(u_j).$$

(c) The total mass condition.

The same finite-head/small-tail argument gives $F(-\infty) = 0$ and $F(\infty) = C$. Consequently F is a distribution function exactly when $\sum_j c_j = 1$. It is also right-continuous: each finite partial sum is right-continuous, and uniform convergence preserves this one-sided continuity.

(d) Dense jumps and a necessary qualification.

For $s < t$,

$$F(t) - F(s) = \sum_{j:\, s<u_j\le t} c_j.$$

If the u_j enumerate $\mathbb{Q}$ and *every* $c_j > 0$, there is a rational u_j strictly between s and t, so this difference is positive. Thus F is strictly increasing. More generally the exact condition is that $\{u_j : c_j > 0\}$ is dense in $\mathbb{R}$.

Conclusion. Parts (a)–(c) hold for nonnegative coefficients as stated. Part (d), read with merely nonnegative coefficients, needs the additional density condition: all $c_j = 0$ would give $F = 0$. This is the positive-coefficient situation of Exercise 35.12.

36.15 Exercise 36.15: Uniform convergence on spaces of infinite and finite mass

(a) Small height can be offset by large width.

Define $f_n(x) = 1/n$ on $[0, n]$, and $f_n(x) = 0$ elsewhere. Each function is piecewise constant and has a finite improper Riemann integral. Its rectangle has height $1/n$ and width n, so

$$\int_{\mathbb{R}} f_n(x)\,dx = 1, \qquad \sup_{x\in\mathbb{R}} |f_n(x) - 0| = \frac{1}{n} \longrightarrow 0.$$

Thus uniform convergence alone does not preserve an integral over a domain of infinite ordinary length.

(b) Local integrability of the uniform limit.

Now suppose F is a distribution function, all f_n are F-integrable on $\mathbb{R}$, and $f_n \to f$ uniformly on $\mathbb{R}$. On each finite $[a, b]$, the uniform-limit theorem for Stieltjes integrals (Exercise 35.6) makes f F-integrable there. For clarity, that theorem follows from

$$U_F(f, P) - L_F(f, P) \le U_F(f_n, P) - L_F(f_n, P) + 2\|f - f_n\|_\infty[F(b) - F(a)] :$$

first choose n to make the second term small, then choose P using integrability of f_n.

Prove global integrability without assuming global boundedness.

Choose one index N with $\|f - f_N\|_\infty \le 1$. The difference $h = f - f_N$ is bounded and locally F-integrable. Since F has finite total increment, Theorem 36.4 makes h F-integrable on $\mathbb{R}$. Exercise 36.10 now shows that $f = f_N + h$ is F-integrable. This argument does not require f_N, or f, to be bounded on all of $\mathbb{R}$.

Estimate the difference of the integrals.

For any sufficiently large n, put $\eta_n = \|f - f_n\|_\infty$. By monotonicity and the total-mass identity $\int_{\mathbb{R}} 1\,dF = F(\infty) - F(-\infty) = 1$,

$$-\eta_n \le \int_{\mathbb{R}} (f - f_n)\,dF \le \eta_n.$$

Linearity therefore gives

$$\left| \int_{\mathbb{R}} f \, dF - \int_{\mathbb{R}} f_n \, dF \right| \leq \eta_n \longrightarrow 0.$$

Conclusion. For a distribution function, uniform convergence does preserve the integral. The decisive bound is the finite total mass 1, rather than the geometric length of the real line.

37 A discussion of exponents and logarithms

In this section we must avoid using familiar exponential and logarithm rules before they have been proved. Our starting point is

$$L(y) = \int_1^y \frac{dt}{t} \quad (y > 0).$$

Theorem 37.4 proves that $L'(y) = 1/y$, that L is strictly increasing onto $\mathbb{R}$, and that $L(yz) = L(y) + L(z)$. Its inverse is E, so $L(E(x)) = x$ and $E(L(y)) = y$. Theorem 37.6 proves $E' = E$ and $E(u + v) = E(u)E(v)$. Real powers are then *defined* by

$$b^x = E(xL(b)) \quad (b > 0,\ x \in \mathbb{R}).$$

These are the integral-based results permitted by the exercise instructions. No result from the informal approaches in 37.1 or 37.2 is needed below. We write $\log_e y = L(y)$ after that identification has been established.

37.1 Exercise 37.1: Prove the product law directly from the integral

What we must show. For $y, z > 0$, show directly that $\int_1^{yz} dt/t = \int_1^y dt/t + \int_1^z dt/t$.

Plan. Split at y and rescale the remaining interval by $t = yu$.

Use additivity with oriented endpoints.

All endpoints are positive, so $1/t$ is continuous on the compact interval containing $1, y, yz$. The oriented-integral convention gives

$$\int_1^{yz} \frac{dt}{t} = \int_1^y \frac{dt}{t} + \int_y^{yz} \frac{dt}{t}.$$

This equality remains true even when $y < 1$ or $yz < y$; reversing endpoints changes the sign of an integral.

Make the substitution, including its scale factor.

Set $t = yu$, so $dt = y\,du$. The lower endpoint $t = y$ becomes $u = 1$, and $t = yz$ becomes $u = z$. Therefore

$$\int_y^{yz} \frac{dt}{t} = \int_1^z \frac{y\,du}{yu} = \int_1^z \frac{du}{u}.$$

Because $y > 0$, this is a valid continuously differentiable change of variable, with oriented endpoints also handling $z < 1$.

Conclusion. Substituting this last identity into the split integral gives the desired formula, without assuming a logarithm product rule.

37.2 Exercise 37.2: The limits of the inverse logarithm

What we must show. Prove $E(x) \to +\infty$ as $x \to +\infty$ and $E(x) \to 0$ as $x \to -\infty$.

Translate a large-output condition through the inverse.

To show the first limit, let $M > 0$ be any proposed lower bound. The number $L(M)$ is real. Since E is strictly increasing,

$$x > L(M) \quad \Longrightarrow \quad E(x) > E(L(M)) = M.$$

Thus $E(x)$ eventually exceeds every positive bound, which is exactly the definition of divergence to $+\infty$.

Translate a small-output condition in the same way.

Let $\varepsilon > 0$. The range of E is $(0, \infty)$, and

$$x < L(\varepsilon) \quad \Longrightarrow \quad 0 < E(x) < E(L(\varepsilon)) = \varepsilon.$$

This is exactly the condition for $E(x) \to 0$ at $-\infty$.

Conclusion. The two claims follow from monotonicity and the inverse identities; there is no need to assume exponential limit rules.

37.3 Exercise 37.3: All four properties of a general exponential base

What we must show. Prove the assertions of Theorem 37.8 for fixed $b > 0$ and $B(x) = b^x = E(xL(b))$.

(i) Continuity and differentiability.

The inner function $x \mapsto xL(b)$ is linear; its derivative is the constant $L(b)$. Since E is differentiable on all of $\mathbb{R}$, the chain rule gives

$$B'(x) = E'(xL(b))L(b) = L(b)E(xL(b)) = L(b)b^x.$$

Differentiability also implies continuity. This includes $b = 1$: $L(1) = 0$, so $B(x) = E(0) = 1$ and $B' = 0$.

(ii) Determine the direction of monotonicity.

Because L is strictly increasing and $L(1) = 0$, $L(b) > 0$ for $b > 1$, whereas $L(b) < 0$ for $0 < b < 1$. In the first case the linear inner function is strictly increasing, so its composition with increasing E is strictly increasing. In the second case it is strictly decreasing, so the composition is strictly decreasing.

(iii) Verify the range, rather than merely stating it.

Assume $b \neq 1$, so $L(b) \neq 0$. Every value $B(x)$ is positive, because E is positive. Conversely, given $y > 0$, choose

$$x = \frac{L(y)}{L(b)}.$$

Then $B(x) = E(L(y)) = y$. Thus every positive value is reached, and the range is exactly $(0, \infty)$.

(iv) The addition rule in the exponent.

The defining formula and Theorem 37.6(ii) yield

$$b^{u+v} = E((u+v)L(b)) = E(uL(b) + vL(b)) = E(uL(b))E(vL(b)) = b^u b^v.$$

Conclusion. All assertions are proved, with the constant-base case $b = 1$ kept separate where invertibility is required.

37.4 Exercise 37.4: Change of logarithm base

What we must show. For $b > 0$, $b \neq 1$, prove $\log_b y = L(y)/L(b)$ for $y > 0$.

Use the definition of the inverse function.

Let $x = \log_b y$. By definition, $b^x = y$. Since $b^x = E(xL(b))$, applying L to both sides gives

$$xL(b) = L(E(xL(b))) = L(y).$$

The denominator $L(b)$ is nonzero because L is one-to-one and $L(1) = 0$. Division therefore gives $x = L(y)/L(b)$.

Conclusion. $\log_b y = \dfrac{L(y)}{L(b)}$. In particular, since $L(e) = 1$, $\log_e y = L(y)$.

37.5 Exercise 37.5: Differentiate a real power

What we must show. For fixed $p \in \mathbb{R}$, prove that $f(x) = x^p$ on $(0, \infty)$ has derivative px^{p-1}.

Start with the definition of a real power.

For a positive base x, the definition says $f(x) = E(pL(x))$. Both functions in this composition are differentiable on the relevant domains. The chain rule gives

$$f'(x) = E'(pL(x))\frac{p}{x} = \frac{p}{x}E(pL(x)) = \frac{p}{x}x^p.$$

Justify the exponent simplification.

The product rule for E gives $E(t)E(-t) = E(0) = 1$, so $E(-L(x)) = 1/x$. Consequently

$$\frac{x^p}{x} = E(pL(x))E(-L(x)) = E((p-1)L(x)) = x^{p-1}.$$

This also works for $p = 0$, since then the derivative is zero.

Conclusion. $(x^p)' = px^{p-1}$ for every real p and every $x > 0$.

37.6 Exercise 37.6: A common real power of a product

What we must show. Show $x^p y^p = (xy)^p$ when $x, y > 0$ and $p \in \mathbb{R}$.

Rewrite both powers using E and L.

The product law for L and the sum law for E give

$$\begin{aligned}(xy)^p &= E(pL(xy)) \\ &= E(p[L(x) + L(y)]) \\ &= E(pL(x) + pL(y)) \\ &= E(pL(x))E(pL(y)) = x^p y^p.\end{aligned}$$

Each logarithm is defined because the two bases and their product are positive. Multiplication by a negative p causes no problem; no inequality has been multiplied by p.

Conclusion. The identity holds for all real exponents under the stated positive-base assumption.

37.7 Exercise 37.7: Derivatives of exponential and logarithmic functions

(a) Differentiate $B(x) = b^x$.

For $b > 0$, the chain-rule calculation from Exercise 37.3 gives

$$B'(x) = E'(xL(b))L(b) = L(b)b^x = (\log_e b)b^x.$$

When $b = 1$ the formula correctly reads $B' = 0$.

(b) Differentiate the inverse, checking the base restriction.

The logarithm $\log_b$ exists only for $b > 0$, $b \neq 1$. Exercise 37.4 expresses it as a constant multiple of L:

$$\frac{d}{dy}\log_b y = \frac{1}{L(b)}L'(y) = \frac{1}{yL(b)} = \frac{1}{y\log_e b}, \qquad y > 0.$$

For $0 < b < 1$ this derivative is negative, agreeing with the decreasing character of that logarithm.

Conclusion. $(b^x)' = (\log_e b)b^x$ and $(\log_b y)' = 1/(y\log_e b)$ on their stated domains.

37.8 Exercise 37.8: A variable base and a variable exponent

What we must show. Differentiate $f(x) = x^x$ for $x > 0$.

Plan. The fixed-exponent power rule cannot be applied directly, because the exponent now changes with x.

Rewrite before differentiating.

By the definition of real powers, $f(x) = E(xL(x))$. The product rule gives

$$\frac{d}{dx}[xL(x)] = L(x) + x\frac{1}{x} = L(x) + 1.$$

Applying the chain rule and $E' = E$ now yields

$$f'(x) = E(xL(x))[L(x) + 1] = x^x[1 + \log_e x].$$

Conclusion. $(x^x)' = (1 + \log_e x)x^x$ for $x > 0$.

37.9 Exercise 37.9: An elementary growth comparison for the logarithm

What we must show. Prove the requested inequalities and then $\lim_{y\to\infty}(\log_e y)/y = 0$ without L'Hospital's rule.

(a) Bound the defining integral.

For $y > 1$, we have $0 < 1/t \leq 1$ on $[1, y]$. Monotonicity of the ordinary integral gives

$$0 < \log_e y = \int_1^y \frac{dt}{t} \leq \int_1^y 1\,dt = y - 1 < y.$$

(b) Apply the first estimate at a square root.

Since $\sqrt{y} > 1$, part (a) gives $\log_e \sqrt{y} < \sqrt{y}$. The product identity for L gives $\log_e y = \log_e(\sqrt{y}\sqrt{y}) = 2\log_e \sqrt{y}$. Dividing by positive y preserves the inequality:

$$0 < \frac{\log_e y}{y} < \frac{2\sqrt{y}}{y} = \frac{2}{\sqrt{y}}.$$

(c) Turn the estimate into a limit.

The rightmost term tends to zero. More explicitly, if $\varepsilon > 0$ and $y > \max\{1, 4/\varepsilon^2\}$, then $0 < (\log_e y)/y < \varepsilon$. This is precisely the required limit definition.

Conclusion. The logarithm grows more slowly than y: $\lim_{y\to\infty} \dfrac{\log_e y}{y} = 0$.

38 Continuous nowhere-differentiable functions

No point of a nowhere differentiable function has a finite derivative. Such a function can nevertheless be continuous. Section 38 constructs examples with both properties. When an exercise needs such a function, Theorem 38.1 supplies a bounded continuous example on $\mathbb{R}$.

One useful strategy is to suppose a derivative exists and use derivative rules to force a derivative of the known nowhere-differentiable function. This is a contradiction argument: the rules are applied only to functions that are differentiable under the temporary assumption, never directly to a function already known to be nondifferentiable.

In the metric-space notation used in this section, $C(\mathbb{R})$ consists of *bounded* continuous functions, and $d(f,g) = \sup_{x\in\mathbb{R}} |f(x) - g(x)|$. The notation C^∞ instead means that derivatives of every finite order exist and are continuous. These are two different uses of the letter C.

38.1 Exercise 38.1: Why a Taylor series cannot produce the desired example

What we must show. Explain the obstruction to constructing a continuous nowhere-differentiable function as a convergent Taylor series.

Recall what a positive-radius power series guarantees.

Suppose

$$f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$$

has radius of convergence $R > 0$. On the open interval $(c-R, c+R)$, the power-series differentiation theorem gives

$$f'(x) = \sum_{n=1}^{\infty} n a_n (x-c)^{n-1}.$$

In particular f is differentiable at every interior point of its convergence interval, however complicated the coefficients might be. Repeating the theorem gives derivatives of every order there.

Check the possible loopholes.

Behavior at the two endpoints of a finite convergence interval can be different, but those endpoints do not remove differentiability at all the interior points. A radius of convergence $R = 0$ gives only a value at the center, not a function on an interval. Moreover, a literal Taylor series of a function requires derivatives at its center in order to define the coefficients in the first place.

Conclusion. A positive-radius Taylor series is necessarily differentiable on its interior, so it cannot be a nowhere-differentiable function.

38.2 Exercise 38.2: A polynomial derivative forces a polynomial function

What we must show. If $f^{(n)}$ is a polynomial on $\mathbb{R}$ for some nonnegative integer n, prove that f is a polynomial.

Build a polynomial with the prescribed nth derivative.

The case $n = 0$ already says that f is a polynomial. For $n \geq 1$, write

$$f^{(n)}(x) = p(x) = \sum_{k=0}^{m} a_k x^k, \qquad P(x) = \sum_{k=0}^{m} a_k \frac{k!}{(k+n)!} x^{k+n}.$$

Differentiating x^{k+n} exactly n times multiplies it by $(k+n)!/k!$ and leaves x^k. Hence $P^{(n)} = p$.

Subtract and use the zero-derivative principle.

Let $h = f - P$. Then $h^{(n)} = 0$ everywhere. Proposition 38.4 implies that h is a polynomial of degree at most $n - 1$. To recall why: $h^{(n-1)}$ has derivative zero, so the Mean Value Theorem makes it constant. Subtract an antiderivative of that constant from $h^{(n-2)}$; the resulting derivative is zero, so that difference is constant as well. Repeating gives a polynomial expression for h after n steps.

Conclusion. $f = P + h$ is a sum of polynomials and is therefore a polynomial.

38.3 Exercise 38.3: A differentiable function with a nowhere-differentiable derivative

What we must show. Construct a differentiable function whose first derivative is nowhere differentiable.

Integrate a continuous nowhere-differentiable function.

Let w be a continuous nowhere-differentiable function on $\mathbb{R}$, whose existence is proved in this section. Define

$$H(x) = \int_0^x w(t)\, dt.$$

For every real x, this is an ordinary integral over a finite interval (with reversed orientation if $x < 0$), so it exists.

Use the hypothesis that makes the Fundamental Theorem applicable.

Since w is continuous at every x, the Fundamental Theorem of Calculus gives $H'(x) = w(x)$ at every real x. Thus H is differentiable everywhere, and H' is nowhere differentiable. Equivalently, H has no second derivative at any point.

Conclusion. An antiderivative of a continuous nowhere-differentiable function is the required example.

38.4 Exercise 38.4: Adding a differentiable function cannot repair a derivative

What we must show. If f is nowhere differentiable and g is differentiable on $\mathbb{R}$, decide whether $f + g$ must be nowhere differentiable.

Suppose there were an exceptional point.

Assume, for contradiction, that $f + g$ is differentiable at some a. By hypothesis g is differentiable at that same a. The difference of two functions differentiable at a point is differentiable there. Thus

$$f = (f + g) - g$$

would be differentiable at a, with derivative $(f+g)'(a)-g'(a)$. This contradicts nowhere differentiability of f.

Conclusion. Yes. The sum $f + g$ is nowhere differentiable.

38.5 Exercise 38.5: Two rough functions can cancel in a sum or a product

(a) A sum need not remain nowhere differentiable.

Take any continuous nowhere-differentiable function w and let $f = w$, $g = -w$. The function g is also nowhere differentiable: if $g'(a)$ existed, multiplying by -1 would give $w'(a)$. Both are continuous, but $f + g = 0$, which is differentiable everywhere.

(b) Arrange a nonzero function before taking its reciprocal.

Choose a bounded continuous nowhere-differentiable w and a finite bound $M \geq 0$ with $|w(x)| \leq M$. Define

$$f(x) = 1 + \frac{w(x)}{2(M+1)}.$$

This is continuous, satisfies

$$d(f, 1^*) \leq \frac{M}{2(M+1)} < \frac{1}{2} < 1, \qquad \frac{1}{2} < f(x) < \frac{3}{2},$$

and is nowhere differentiable: differentiability of f would imply differentiability of $w = 2(M+1)(f-1)$ at the same point.

Verify that the reciprocal is itself nowhere differentiable.

Let $g(x) = 1/f(x)$. It is continuous because f never vanishes. Suppose g were differentiable at a point a. We have $g(a) = 1/f(a) \neq 0$. The reciprocal differentiation rule, applied to this temporarily differentiable function g, would then make $1/g = f$ differentiable at a, with derivative $-g'(a)/g(a)^2$. This contradicts the established property of f. Thus g is nowhere differentiable too, yet $f(x)g(x) = 1$ everywhere.

Conclusion. Neither a sum nor a product of two continuous nowhere-differentiable functions must be nowhere differentiable.

38.6 Exercise 38.6: A quotient of smooth functions is smooth

What we must show. Suppose $f, g \in C^\infty(\mathbb{R})$ and $g(x) \neq 0$ for every x. Prove $f/g \in C^\infty(\mathbb{R})$.

Plan. Inductively describe each derivative so that its denominator is a power of the nonvanishing function g.

State a formula suited to induction.

We claim that, for every integer $n \geq 0$, there is $h_n \in C^\infty(\mathbb{R})$ for which

$$\left(\frac{f}{g}\right)^{(n)} = \frac{h_n}{g^{2^n}}.$$

For $n = 0$, choose $h_0 = f$; the denominator is $g^{2^0} = g$, so the formula is correct.

Differentiate the inductive expression.

Suppose the formula holds for some n, and write $m = 2^n$. The function g^m is smooth, by the product rule applied a finite number of times, and it never vanishes. Thus the ordinary quotient rule, requiring only first derivatives at this step, gives

$$\left(\frac{h_n}{g^m}\right)' = \frac{h_n' g^m - h_n\, m g^{m-1} g'}{g^{2m}}.$$

Define

$$h_{n+1} = h_n' g^m - m h_n g^{m-1} g'.$$

All factors are smooth, so finite sums and products show that h_{n+1} is smooth. Since $2m = 2^{n+1}$, this is the desired formula at the next order. The induction also proves the existence of that next derivative.

Check continuity, not only existence.

For each n, the formula expresses the nth derivative as a quotient of continuous functions with a denominator that is nowhere zero. That derivative is therefore continuous. We have proved the existence and continuity of derivatives of every order.

Conclusion. f/g is infinitely differentiable. The nonvanishing assumption is what allows every quotient-rule step.